RPSC · RAS & RTS PRELIMINARY EXAMINATION 2026

Science and Technology

685 practice questions across 13 chapters, each chapter opening with a must-know fact sheet. Answers and explanations are at the back, so you can work through the paper first.

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685 questions · **389** fact-sheet points · **77** tagged as previously asked
Paper pattern: 150 questions · 200 marks · 3 hours · 1/3 negative marking

ST01

Physics in Everyday Life

55 questions · 30 fact-sheet points

Must-know facts

1. Seven SI base units: metre (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (amount of substance), candela (luminous intensity).
2. Derived SI units: newton = kg m/s2 (force); pascal = N/m2 (pressure); joule = N m (work, energy); watt = J/s (power); hertz (frequency); coulomb (charge); volt, ohm, farad, tesla, weber, henry.
3. Units of pressure: pascal, bar, torr, atmosphere, mm of mercury (1 atm = 1.013 x 10^5 Pa = 760 torr = 76 cm of Hg = 1.013 bar).
4. Unit traps: light year and parsec measure DISTANCE (1 parsec = 3.26 light years); barn = area; knot = speed; decibel = sound level; dioptre = power of a lens; poise = viscosity; Chandrasekhar limit = mass.
5. 1 horsepower = 746 watt. 1 kilowatt-hour (Board of Trade unit) = 3.6 x 10^6 joule. Angstrom = 10^-10 m; nanometre = 10^-9 m; fermi = 10^-15 m.
6. Scalars: mass, speed, distance, work, energy, power, temperature, density, pressure. Vectors: displacement, velocity, acceleration, force, momentum, impulse, torque, weight.
7. Newton's laws: first = law of inertia (seat belts, dusting a carpet, passenger lurching in a bus); second = F = ma = rate of change of momentum; third = action-reaction (rocket, recoil of a gun, swimming, walking).
8. Conservation of linear momentum explains the recoil of a gun and jet or rocket propulsion. Impulse = force x time = change in momentum, which is why air bags, sand pits and a cricketer's follow-through reduce injury.
9. g = 9.8 m/s2 at sea level; maximum at the poles, minimum at the equator; ZERO at the centre of the earth; g on the Moon = 1/6 of that on Earth. Mass is the same everywhere, weight is not.
10. Escape velocity of Earth = 11.2 km/s; of the Moon = 2.38 km/s (why the Moon has no atmosphere); orbital velocity of a near-earth satellite is about 7.9 km/s. Weightlessness in an orbiting spacecraft or a freely falling lift occurs because body and vehicle fall together, not because gravity is absent.
11. Geostationary satellite: circular equatorial orbit at about 36,000 km (35,786 km), period 24 hours, used for communication and meteorology. Polar and sun-synchronous orbits (about 800-900 km) are used for remote sensing.
12. Pascal's law (pressure transmitted equally in an enclosed fluid) - hydraulic press, hydraulic brakes, hydraulic jack. Archimedes' principle (upthrust) - ship, submarine, hydrometer, lactometer. Bernoulli's principle (fast flow means low pressure) - aerofoil lift, atomiser, chimney draught, two boats drawn together.
13. Atmospheric pressure at sea level = 76 cm of mercury = 1.013 x 10^5 Pa; measured by a barometer (Torricelli). It falls with altitude, so water boils below 100 C on hills - hence the pressure cooker.
14. Surface tension: spherical raindrops, floating needle, insects walking on water, capillary rise in a wick and in soil; soap and detergent LOWER surface tension. Surface tension decreases as temperature rises.
15. Viscosity of a LIQUID decreases with rise in temperature; viscosity of a GAS increases with temperature. Units: poise (CGS), pascal-second (SI).
16. Water: specific heat 4.2 J/g/K, the highest among common liquids - used as a coolant and it moderates coastal climate. Latent heat of fusion of ice = 80 cal/g (336 J/g); latent heat of vaporisation of water = 540 cal/g (2260 J/g), which is why steam burns worse than boiling water.

17. Anomalous expansion of water: density is maximum at 4 C, so ice floats and lakes freeze from the top downwards, allowing aquatic life to survive.
18. Heat transfer: conduction (solids), convection (fluids - land and sea breeze), radiation (needs no medium - Sun to Earth). A thermos flask blocks all three: vacuum stops conduction and convection, silvering stops radiation, cork stopper reduces conduction.
19. Refrigeration and sweating cool by evaporation - the refrigerant absorbs latent heat as it vaporises. A bimetallic strip (unequal expansion of two metals) is the thermostat in irons, geysers and fire alarms.
20. Sound needs a material medium; there is no sound in vacuum. Speed of sound is greatest in solids, least in gases; about 332 m/s in air at 0 C. Audible range 20 Hz to 20,000 Hz; above this is ultrasonic, below it infrasonic.
21. Ultrasound uses: SONAR (sea depth, submarines), sonography, echocardiography, lithotripsy (breaking kidney stones), detecting flaws in metal castings, cleaning. Bats and dolphins use echolocation. Minimum distance for a distinct echo = 17.2 m. A stethoscope works by MULTIPLE REFLECTION of sound.
22. Doppler effect = apparent change of frequency due to relative motion (radar speed guns; red shift of receding galaxies). Loudness depends on amplitude, pitch on frequency, quality (timbre) on waveform.
23. Optics shortcuts: blue sky and red sunset = scattering (short wavelengths scatter most); mirage, sparkle of diamond and optical fibre = total internal reflection; twinkling of stars and seeing the Sun before actual sunrise = atmospheric refraction; rainbow = refraction + dispersion + internal reflection in raindrops.
24. Vision defects: myopia - concave lens; hypermetropia - convex lens; presbyopia - bifocal lens; astigmatism - cylindrical lens; cataract - the eye lens turns cloudy owing to changes in its protein, corrected surgically. Power of a lens (dioptre) = 1 / focal length in metres; a concave lens has negative power.
25. Convex mirror - rear-view mirror (erect, diminished, wide field). Concave mirror - vehicle headlight, torch, shaving mirror, dentist's mirror, solar cooker. Periscope uses plane mirrors or total-internal-reflection prisms. LASER = Light Amplification by Stimulated Emission of Radiation (monochromatic, coherent).
26. Electromagnetic spectrum in increasing order of wavelength: gamma rays, X-rays, ultraviolet, visible, infrared, microwaves, radio waves. TV remote controls use INFRARED; microwave ovens use microwaves; UHF radio waves travel as space waves.
27. Electricity: Ohm's law V = IR; Indian household supply is 220 V, 50 Hz AC; appliances are wired in PARALLEL; ammeter in series, voltmeter in parallel. A fuse is a thin low-melting-point tin-lead wire connected in SERIES in the LIVE wire.
28. Faraday's electromagnetic induction runs the generator or dynamo; Lenz's law gives the direction of the induced current. A transformer works on MUTUAL INDUCTION and only with AC. Fleming's left-hand rule - motor; right-hand rule - generator. A photovoltaic cell converts solar energy directly into electrical energy.
29. Modern physics: photoelectric effect explained by Einstein (Nobel 1921); X-rays discovered by W. C. Roentgen (first Physics Nobel, 1901); radioactivity by Henri Becquerel; neutron by James Chadwick. Alpha = helium nucleus, beta = electron, gamma = electromagnetic wave (most penetrating). Superconductivity (zero resistance, Meissner effect) was found by Kamerlingh Onnes in mercury, 1911.
30. Nuclear reactors use controlled FISSION of uranium-235 with heavy water or graphite as MODERATOR and cadmium or boron rods as CONTROL rods; the Sun's energy comes from nuclear FUSION of hydrogen into helium.

1. Which one of the following is a base unit in the International System of Units (SI)?

(A) Candela
(B) Newton
(C) Pascal
(D) Joule

2. Consider the following units:

I. bar

II. pascal

III. torr

IV. atmosphere

Which of the units given above are units of pressure? RAS PRE 2024

(A) I and II only
(B) II and IV only
(C) I, II and III only
(D) I, II, III and IV

3. One horsepower is approximately equal to

(A) 546 watt
(B) 646 watt
(C) 746 watt
(D) 846 watt

4. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Unit) List-II (Physical quantity)

A. Dioptre 1. Viscosity

B. Poise 2. Power of a lens

C. Becquerel 3. Radioactivity

D. Tesla 4. Magnetic flux density

(A) A-1, B-2, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-2, B-1, C-3, D-4
(D) A-2, B-1, C-4, D-3

5. Which one of the following pairs of a unit and the physical quantity it measures is NOT correctly matched?

(A) Knot — Speed
(B) Barn — Area
(C) Chandrasekhar limit — Mass
(D) Light year — Time

6. Which one of the following is a scalar quantity?

(A) Impulse
(B) Torque
(C) Momentum
(D) Speed

7. Consider the following statements about Newton's first law of motion:

I. It is also known as the law of inertia.

II. A standing passenger in a moving bus tends to fall forward when the bus stops suddenly.

III. The mass of a body is a measure of its inertia.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

8. The propulsion of a rocket is based on

(A) Newton's first law of motion
(B) Archimedes' principle
(C) Newton's third law of motion
(D) Bernoulli's principle

9. When a bullet is fired from a gun, the gun recoils backwards. This is a direct consequence of

(A) conservation of energy
(B) conservation of angular momentum
(C) conservation of linear momentum
(D) Newton's law of gravitation

10. Consider the following statements:
I. The value of acceleration due to gravity is zero at the centre of the earth.
II. The escape velocity from the surface of the earth is about 11.2 km per second.
III. The value of g is greater at the equator than at the poles.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

11. Consider the following statements about a geostationary satellite:
I. Its orbital period is 24 hours.
II. It is placed in a circular orbit above the equator at a height of about 36,000 km.
III. It is the preferred orbit for remote sensing satellites.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

12. An astronaut inside an orbiting spacecraft experiences weightlessness because

(A) there is no gravitational force at that height
(B) the astronaut and the spacecraft are both in a state of free fall
(C) the mass of the astronaut becomes zero in orbit
(D) there is no air resistance in space

13. A coolie carrying a load on his head walks on a level platform. The work done by him against gravity is

(A) maximum
(B) equal to the weight multiplied by the distance walked
(C) zero, because the force and the displacement are mutually perpendicular
(D) negative

14. One kilowatt-hour, the commercial unit of electrical energy, is equal to HARD

(A) 3.6 x 10^3 joule
(B) 3.6 x 10^5 joule
(C) 3.6 x 10^6 joule
(D) 3.6 x 10^9 joule

15. The working of a hydraulic brake in an automobile is based on

(A) Archimedes' principle
(B) Bernoulli's principle
(C) Boyle's law
(D) Pascal's law

16. Which one of the following pairs of a device and the principle on which it works is NOT correctly matched?

(A) Hydrometer — Archimedes' principle
(B) Hydraulic lift — Pascal's law
(C) Atomiser (spray pump) — Bernoulli's principle
(D) Pressure cooker — Surface tension

17. Food is cooked faster in a pressure cooker because

(A) the cooker lowers the boiling point of water
(B) the increased pressure inside raises the boiling point of water
(C) the specific heat of water increases inside the cooker
(D) loss of heat by radiation is completely stopped

18. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Instrument) List-II (Measures)

A. Lactometer 1. Purity of milk

B. Hydrometer 2. Atmospheric pressure

C. Barometer 3. Relative density of a liquid

D. Anemometer 4. Wind speed

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

19. Two boats moving side by side in the same direction tend to be drawn towards each other. This is explained by

(A) Pascal's law
(B) Archimedes' principle
(C) Newton's law of cooling
(D) Bernoulli's principle

20. Which one of the following is NOT an effect of surface tension?

(A) The nearly spherical shape of a freely falling raindrop
(B) The floating of a greased steel needle on water
(C) The rise of kerosene in the wick of a lamp
(D) The operation of a hydraulic jack

21. Consider the following statements:

I. Mercury is depressed in a fine glass capillary tube instead of rising in it.

II. Oil rises in the wick of a lamp by capillary action.

III. The surface tension of a liquid increases with rise in temperature.

Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

22. With a rise in temperature, the viscosity of a liquid

(A) decreases
(B) increases
(C) first increases and then decreases
(D) remains unchanged

23. Water is used as a coolant in automobile radiators mainly because it has

(A) a low boiling point
(B) a very high specific heat capacity
(C) a low density
(D) a high viscosity

24. Consider the following statements:

I. The latent heat of fusion of ice is about 80 calorie per gram.

II. The latent heat of vaporisation of water is about 540 calorie per gram.

III. The temperature of ice rises steadily while it is melting at atmospheric pressure.

Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

25. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Everyday observation) List-II (Explanation)

A. Land and sea breeze 1. Convection

B. The handle of a metal spoon kept in hot tea becomes warm 2. Conduction

C. Warmth felt while sitting in front of a fire 3. Radiation

D. Vacuum between the double walls of a thermos flask 4. Prevents conduction and convection

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-3, C-2, D-4
(D) A-3, B-2, C-4, D-1

26. The cooling produced in a domestic refrigerator is essentially due to

(A) compression of the refrigerant
(B) conduction through the metal walls
(C) radiation from the cooling coils
(D) absorption of latent heat when the liquid refrigerant evaporates

27. Ice floats on water and a lake freezes from the surface downwards because water

(A) has a very high latent heat of fusion
(B) attains its maximum density at 4 degrees Celsius
(C) is a poor conductor of heat
(D) has a very high surface tension

28. Which one of the following statements relating to heat and thermometry is NOT correct?

(A) A thermos flask has a vacuum between its double walls to check conduction and convection
(B) The walls of a thermos flask are silvered to reduce loss of heat by radiation
(C) A bimetallic strip works as a thermostat because two metals expand unequally on heating
(D) Mercury is used in thermometers because it has a very high specific heat

29. Consider the following statements about sound:

I. Sound cannot travel through vacuum.

II. The speed of sound is greater in solids than in gases.

III. The loudness of a sound depends on its frequency.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

30. Consider the following statements about ultrasonic waves:
I. Their frequency is above 20,000 hertz.
II. They are used in SONAR to measure the depth of the sea.
III. They are used to break kidney stones in lithotripsy.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

31. A stethoscope works on the principle of RAS PRE 2021

(A) refraction of sound
(B) multiple reflection of sound
(C) diffraction of sound
(D) resonance of the chest cavity

32. The apparent change in the frequency of the whistle of a train as it approaches and then recedes from an observer is called

(A) reverberation
(B) the Doppler effect
(C) resonance
(D) diffraction

33. A distinct echo is heard only if the reflecting surface is at a minimum distance of about HARD

(A) 7.2 metre
(B) 12.2 metre
(C) 17.2 metre
(D) 34.4 metre

34. The blue colour of a clear sky is due to

(A) scattering of sunlight by the molecules of the atmosphere
(B) reflection of light by air molecules
(C) total internal reflection in the atmosphere
(D) dispersion of sunlight by dust particles

35. Arrange the following electromagnetic radiations in increasing order of wavelength:
I. Gamma rays
II. X-rays
III. Visible light
IV. Radio waves

(A) I, II, III, IV
(B) IV, III, II, I
(C) II, I, III, IV
(D) III, II, I, IV

36. Arrange the following colours of the visible spectrum in increasing order of their wavelength:
I. Red
II. Yellow
III. Green
IV. Violet

(A) I, II, III, IV
(B) IV, III, II, I
(C) IV, II, III, I
(D) III, IV, I, II

37. The waves used in a common television remote control are RAS PRE 2018

(A) ultraviolet rays
(B) infrared rays
(C) microwaves
(D) X-rays

38. Stars appear to twinkle whereas planets generally do not. The twinkling of stars is caused by

(A) dispersion of starlight
(B) total internal reflection in the atmosphere
(C) atmospheric refraction of starlight
(D) diffraction at the pupil of the eye

39. Consider the following phenomena:
I. Formation of a mirage on a hot road.
II. Sparkling of a cut diamond.
III. Transmission of light through an optical fibre.
Which of the above are examples of total internal reflection?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. Optical fibres, widely used in telecommunication and in endoscopy, work on the principle of

(A) diffraction
(B) total internal reflection
(C) polarisation
(D) dispersion

41. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Defect of vision) List-II (Corrective lens)
A. Myopia 1. Convex lens
B. Hypermetropia 2. Concave lens
C. Presbyopia 3. Cylindrical lens
D. Astigmatism 4. Bifocal lens RAS PRE 2023

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

42. A convex mirror is preferred as the rear-view mirror of a vehicle because it always forms

(A) a real, inverted and magnified image
(B) a virtual, erect and diminished image with a wide field of view
(C) a real, erect image of the same size
(D) a virtual, inverted and magnified image

43. Which one of the following pairs of an optical device and the mirror or lens used in it is NOT correctly matched?

(A) Vehicle headlight reflector — Concave mirror
(B) Solar cooker reflector — Concave mirror
(C) Simple microscope — Convex lens
(D) Correction of short-sightedness — Convex lens

44. Consider the following statements about the formation of a rainbow:
I. It involves refraction, dispersion and internal reflection of sunlight in water droplets.
II. It is always seen in the part of the sky opposite to the sun.
III. In a primary rainbow the red colour appears on the outer edge.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

45. Consider the following statements about domestic electric supply in India:
I. The supply is alternating current at 220 volt and 50 hertz.
II. Domestic appliances are connected in parallel.
III. A voltmeter is connected in series with the appliance whose voltage is to be measured.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

46. An electric fuse is a safety device that is

(A) connected in parallel with the appliance in the neutral wire
(B) a thick wire of high melting point placed in the earth wire
(C) a thin wire of low melting point connected in series in the live wire
(D) a device that cuts off supply by creating an earth connection

47. The metal body of an electrical appliance is earthed in order to

(A) reduce the electricity bill
(B) increase the current supplied to the appliance
(C) provide a low-resistance path to the ground so that a person touching it is not shocked
(D) convert alternating current into direct current

48. Consider the following statements about a transformer:
I. It works on the principle of mutual induction.
II. It can be used with direct current as well as with alternating current.
III. A step-up transformer increases the voltage and decreases the current.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

49. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Device) List-II (Energy conversion)
A. Electric motor 1. Electrical energy into mechanical energy
B. Dynamo 2. Mechanical energy into electrical energy
C. Microphone 3. Sound energy into electrical energy
D. Electric bulb 4. Electrical energy into light and heat energy

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-1, B-3, C-2, D-4

50. A photovoltaic (solar) cell directly converts RAS PRE 2016

(A) heat energy into electrical energy
(B) solar energy into electrical energy
(C) solar energy into chemical energy
(D) electrical energy into light energy

51. A material that offers zero electrical resistance below a certain critical temperature is called HARD

(A) a semiconductor
(B) a superconductor
(C) a dielectric
(D) a ferromagnet

52. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Scientist) List-II (Discovery / explanation)
A. W. C. Roentgen 1. X-rays
B. Henri Becquerel 2. Radioactivity
C. James Chadwick 3. Photoelectric effect
D. Albert Einstein 4. Neutron

(A) A-1, B-2, C-4, D-3
(B) A-1, B-2, C-3, D-4
(C) A-2, B-1, C-3, D-4
(D) A-2, B-1, C-4, D-3

53. The energy radiated by the sun is produced by

(A) controlled nuclear fission of uranium
(B) nuclear fusion of hydrogen nuclei into helium
(C) chemical combustion of hydrogen
(D) radioactive decay of heavy elements

54. Which one of the following statements about a nuclear reactor is NOT correct?

(A) It produces energy by the controlled fission of uranium-235
(B) Heavy water and graphite are used as moderators
(C) Cadmium and boron rods are used as moderators to slow down neutrons
(D) The moderator slows down fast neutrons so that they can cause further fission

55. Among the alpha, beta and gamma radiations emitted by radioactive substances, the most penetrating is

(A) alpha radiation
(B) beta radiation
(C) gamma radiation
(D) all three are equally penetrating

ST02

Chemistry in Everyday Life

55 questions · 29 fact-sheet points

Must-know facts

1. Common name - chemical name - formula: baking soda = sodium hydrogencarbonate = $NaHCO_3$; washing soda = sodium carbonate decahydrate = $Na_2CO_3.10H_2O$; caustic soda = sodium hydroxide = NaOH; caustic potash = KOH.
2. Quicklime = calcium oxide = CaO; slaked lime = calcium hydroxide = $Ca(OH)_2$; limestone / marble / chalk = calcium carbonate = $CaCO_3$; gypsum = $CaSO_4.2H_2O$; plaster of Paris = $CaSO_4.(1/2)H_2O$ (gypsum heated to about 100 C).
3. Bleaching powder = calcium oxychloride = $CaOCl_2$, made by passing chlorine over dry slaked lime; used for bleaching and for disinfecting drinking water.
4. Vitriols: blue vitriol = $CuSO_4.5H_2O$; green vitriol = $FeSO_4.7H_2O$; white vitriol = $ZnSO_4.7H_2O$; oil of vitriol = concentrated H_2SO_4. Epsom salt = $MgSO_4.7H_2O$; Glauber's salt = $Na_2SO_4.10H_2O$; hypo (photographic fixer) = sodium thiosulphate $Na_2S_2O_3.5H_2O$.
5. Other everyday names: dry ice = solid carbon dioxide; laughing gas = nitrous oxide N_2O; marsh gas = methane CH_4; sal ammoniac = NH_4Cl; saltpetre = KNO_3; borax = $Na_2B_4O_7.10H_2O$; potash alum = $K_2SO_4.Al_2(SO_4)_3.24H_2O$; milk of magnesia = $Mg(OH)_2$; heavy water = D_2O.
6. Aqua regia = 3 parts concentrated hydrochloric acid to 1 part concentrated nitric acid; it dissolves gold and platinum.
7. Acids in daily life: formic (methanoic) acid - ant and nettle sting; acetic acid - vinegar; citric acid - lemon and orange; lactic acid - curd and sour milk; oxalic acid - tomato and spinach; tartaric acid - tamarind; malic acid - apple; hydrochloric acid - gastric juice.
8. pH: 7 is neutral at 25 C; below 7 acidic, above 7 basic. Tooth decay starts when mouth pH falls below 5.5; human blood pH is 7.35-7.45. Antacids such as milk of magnesia and baking soda neutralise excess stomach acid; a bee sting (acidic) is treated with baking soda and a wasp sting (alkaline) with vinegar.
9. Rusting = hydrated ferric oxide $Fe_2O_3.xH_2O$; it needs BOTH oxygen and moisture. Prevention: painting, greasing, galvanisation (zinc coating), tinning (kalai - a tin coating on brass and copper utensils), electroplating, alloying, cathodic protection.
10. Alloys: brass = copper + zinc; bronze = copper + tin; german silver = copper + zinc + nickel (contains NO silver); duralumin = aluminium + copper + magnesium + manganese; solder = lead + tin; stainless steel = iron + chromium (about 18%) + nickel (about 8%) + carbon; gunmetal = copper + tin + zinc; bell metal = copper + tin (about 78:22); nichrome = nickel + chromium + iron; magnalium = aluminium + magnesium; invar = iron + nickel (low expansion); amalgam = any alloy containing mercury.
11. Ores: bauxite - aluminium; haematite (Fe_2O_3) and magnetite (Fe_3O_4) - iron; galena (PbS) - lead; cinnabar (HgS) - mercury; zinc blende (ZnS) and calamine ($ZnCO_3$) - zinc; copper pyrites ($CuFeS_2$) and malachite - copper; pitchblende - uranium; monazite - thorium; argentite - silver; rock salt - sodium.
12. Atomic structure: electron - J. J. Thomson; proton - Goldstein/Rutherford; neutron - James Chadwick. Isotopes have the same atomic number but different mass numbers; isobars have the same mass number but different atomic numbers.
13. Periodic table: Mendeleev arranged elements by increasing atomic MASS; the modern periodic law (Moseley) uses ATOMIC NUMBER. Group 1 = alkali metals, group 2 = alkaline earth metals, group 17 = halogens, group 18 = noble gases. Fluorine is the most electronegative element; caesium is among the most electropositive.

14. Mercury is the only metal liquid at room temperature; bromine is the only non-metal liquid at room temperature. Sodium and potassium are stored under kerosene. Graphite is a non-metal that conducts electricity.
15. Carbon allotropes: diamond (hardest natural substance, each carbon bonded to FOUR others, non-conductor), graphite (layered, soft, lubricant, conducts electricity because of one free electron per atom), fullerene C60 (buckminsterfullerene, football-shaped), graphene (single layer), carbon nanotubes.
16. Catalysts: Haber process for ammonia - finely divided IRON; contact process for sulphuric acid - vanadium pentoxide V2O5; hydrogenation of vegetable oil into vanaspati - NICKEL; Ostwald process for nitric acid - platinum.
17. Gases: air is about 78% nitrogen and 21% oxygen by volume. Argon (with nitrogen) fills incandescent bulbs; helium fills balloons and airships and is used in diving mixtures; neon glows red in advertising signs; ozone (O3) in the stratosphere absorbs ultraviolet radiation; carbon dioxide is used in fire extinguishers and as dry ice.
18. Fuel gases: CNG - mainly METHANE; LPG - mainly BUTANE and propane, with ethyl mercaptan added as an odorant; biogas / gobar gas - mainly methane; water gas = CO + H2; producer gas = CO + N2; coal gas = H2 + CH4 + CO. Hydrogen has the highest calorific value of all fuels.
19. Calcium carbide with water gives ACETYLENE (ethyne), used in oxy-acetylene welding and misused for artificial ripening of fruit; ethylene (ethene) is the natural ripening hormone.
20. Polymers: thermoplastics soften on heating and can be remoulded (polythene, PVC, polystyrene, nylon, PET, teflon); thermosetting plastics set permanently (bakelite, melamine). Nylon is a synthetic NON-cellulosic fibre; rayon is regenerated cellulose. Vulcanisation = heating natural rubber with SULPHUR (Charles Goodyear).
21. Soaps are sodium or potassium salts of long-chain fatty acids made by saponification of oils with alkali; they do not lather in hard water (Ca and Mg salts). Synthetic detergents are alkyl benzene sulphonates and work even in hard water. Cleaning works through micelle formation and lowering of surface tension.
22. Cement: made from limestone and clay; gypsum (2-3 per cent) is added to SLOW DOWN the setting. Concrete = cement + sand + gravel + water; reinforced cement concrete uses steel bars. Glass is mainly silica (SiO2); borosilicate (Pyrex) glass resists heat; cobalt oxide gives blue glass.
23. Fertilisers: urea CO(NH2)2 has the highest nitrogen content among solid fertilisers, about 46 per cent; DAP supplies nitrogen and phosphorus; muriate of potash is potassium chloride; superphosphate supplies phosphorus.
24. Explosives: gunpowder = potassium nitrate + charcoal + sulphur; TNT = trinitrotoluene; RDX = cyclonite; dynamite = nitroglycerine absorbed in kieselguhr, invented by Alfred Nobel.
25. Batteries: dry cell (zinc-carbon) and lead-acid storage battery (lead, lead dioxide, sulphuric acid) in vehicles; lithium-ion cells power mobile phones, laptops and electric vehicles; a fuel cell converts the chemical energy of hydrogen and oxygen directly into electricity, giving water as the only product.
26. Food adulteration: argemone seed oil in mustard oil (causes epidemic dropsy); metanil yellow in turmeric and pulses; brick powder in chilli powder; chicory in coffee; melamine in milk; papaya seeds in black pepper. Common preservatives: sodium benzoate, sodium metabisulphite, salt, sugar, vinegar (acetic acid).
27. Drugs: analgesic relieves pain (aspirin, paracetamol; morphine is a narcotic analgesic); antipyretic reduces fever (paracetamol); antibiotic kills or checks micro-organisms (penicillin, discovered by Alexander Fleming in 1928); antiseptic is applied to LIVING tissue (Dettol, tincture of iodine) while a disinfectant is used on NON-LIVING objects (phenol, chlorine); antacid neutralises stomach acid.
28. Radioisotopes: carbon-14 - dating of fossils and archaeological material; cobalt-60 - cancer radiotherapy; iodine-131 - thyroid disorders; phosphorus-32 - agricultural and leukaemia studies; sodium-24 - tracing blood circulation and detecting leaks in underground pipelines; uranium-235 - nuclear fission.

29. Sublimation (solid straight to vapour): camphor, naphthalene, iodine, ammonium chloride, dry ice. Plasma is the fourth state of matter; the Bose-Einstein condensate is the fifth.

1. The chemical formula of baking soda is RAS PRE 2016

(A) $NaHCO_3$ (B) Na_2CO_3
(C) NaOH (D) Na_2SO_4

2. Plaster of Paris is obtained by

(A) treating limestone with dilute hydrochloric acid
(B) heating gypsum to about 100 degrees Celsius
(C) passing chlorine over dry slaked lime
(D) burning limestone in a kiln

3. Bleaching powder, used for disinfecting drinking water, is prepared by

(A) dissolving chlorine in water (B) heating limestone in a lime kiln
(C) passing chlorine over dry slaked lime (D) adding sulphur dioxide to quicklime

4. Aqua regia, which can dissolve gold, is a mixture of

(A) concentrated sulphuric acid and concentrated nitric acid in the ratio 3 : 1
(B) concentrated nitric acid and concentrated hydrochloric acid in the ratio 3 : 1
(C) dilute hydrochloric acid and dilute nitric acid in equal volumes
(D) concentrated hydrochloric acid and concentrated nitric acid in the ratio 3 : 1

5. Heavy water, used as a moderator in nuclear reactors, is chemically HARD

(A) deuterium oxide
(B) water containing dissolved calcium and magnesium salts
(C) water enriched with oxygen-18
(D) tritiated hydrogen peroxide

6. Vinegar, which is used as a food preservative, is a dilute aqueous solution of RAS PRE 2016

(A) citric acid (B) acetic acid
(C) lactic acid (D) tartaric acid

7. Calcium carbide, sometimes misused for the artificial ripening of fruit, reacts with water to produce RAS PRE 2016

(A) ethylene (B) methane
(C) acetylene (D) carbon dioxide

8. Fullerene, an allotrope of carbon shaped like a football, contains

(A) 20 carbon atoms (B) 30 carbon atoms
(C) 50 carbon atoms (D) 60 carbon atoms

9. Graphite can be used as a lubricant as well as an electrode because

(A) its layers slide over one another easily and it has free electrons that conduct electricity
(B) it is the hardest allotrope of carbon
(C) each of its carbon atoms is bonded to four other carbon atoms
(D) it is chemically inert to all acids and alkalis

10. The hardest naturally occurring substance known is

(A) quartz (B) diamond
(C) corundum (D) topaz

11. The most electronegative element in the periodic table is HARD

(A) chlorine
(B) oxygen
(C) fluorine
(D) nitrogen

12. An ordinary incandescent electric bulb is filled with

(A) oxygen
(B) hydrogen
(C) carbon dioxide
(D) argon along with some nitrogen

13. The ozone layer in the stratosphere is important because it

(A) absorbs harmful ultraviolet radiation from the sun
(B) absorbs infrared radiation emitted by the earth
(C) supplies oxygen to the upper atmosphere
(D) reflects radio waves back to the earth

14. Which one of the following gases is commonly used in fire extinguishers?

(A) Oxygen
(B) Nitrogen
(C) Carbon dioxide
(D) Hydrogen

15. Which one of the following is a synthetic, non-cellulosic fibre? RAS PRE 2018

(A) Cotton
(B) Jute
(C) Nylon
(D) Rayon

16. Brass utensils are coated with a thin layer of which metal, a process traditionally called kalai, to prevent contamination of food by copper? RAS PRE 2018

(A) Zinc
(B) Aluminium
(C) Nickel
(D) Tin

17. An amalgam is an alloy in which one of the constituents is always

(A) mercury
(B) zinc
(C) tin
(D) lead

18. Stainless steel is an alloy of iron with

(A) copper and tin
(B) chromium and nickel
(C) zinc and nickel
(D) aluminium and magnesium

19. German silver, widely used for utensils and ornamental ware, is an alloy of

(A) silver, copper and nickel
(B) silver, zinc and nickel
(C) copper, zinc and nickel
(D) copper, tin and silver

20. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Alloy) List-II (Chief constituents)

A. Brass 1. Copper and tin

B. Bronze 2. Lead and tin

C. Solder 3. Copper and zinc

D. Duralumin 4. Aluminium, copper, magnesium and manganese

(A) A-1, B-2, C-3, D-4
(B) A-3, B-1, C-2, D-4
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

21. Which one of the following pairs of an alloy and its constituents is NOT correctly matched? HARD

(A) German silver — Copper, silver and nickel
(B) Gunmetal — Copper, tin and zinc
(C) Nichrome — Nickel, chromium and iron
(D) Bell metal — Copper and tin

22. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Ore) List-II (Metal obtained)

A. Bauxite 1. Aluminium

B. Cinnabar 2. Mercury

C. Galena 3. Lead

D. Zinc blende 4. Zinc

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-3, B-2, C-1, D-4

23. Consider the following statements:

I. Mercury is the only metal that is liquid at ordinary room temperature.

II. Bromine is the only non-metal that is liquid at ordinary room temperature.

III. Sodium is stored under water to prevent it from being oxidised.

Which of the statements given above are correct?

(A) II and III only
(B) I and III only
(C) I and II only
(D) I, II and III

24. Rust formed on iron articles is chemically

(A) ferrous sulphide
(B) ferric chloride
(C) ferrous carbonate
(D) hydrated ferric oxide

25. Galvanisation is the process of coating iron or steel with a thin layer of

(A) tin
(B) chromium
(C) nickel
(D) zinc

26. Which one of the following statements about the rusting of iron is NOT correct?

(A) Rust is chemically hydrated ferric oxide
(B) Rusting of iron requires both oxygen and moisture
(C) Galvanisation is the coating of iron with a layer of zinc
(D) Rusting is a physical change that can be reversed simply by heating the article

27. Assertion (A): Stainless steel does not rust in moist air.

Reason (R): The chromium present in stainless steel forms a thin, adherent oxide film that protects the metal underneath.

Select the correct answer using the codes given below:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

28. Which one of the following is used in mobile phones, laptops and electric vehicles as a rechargeable source of electricity?

(A) Lithium-ion cell
(B) Zinc-carbon dry cell
(C) Lead-acid storage battery
(D) Hydrogen-oxygen fuel cell

29. Consider the following statements:
I. The pH of a neutral solution at 25 degrees Celsius is 7.
II. Tooth decay begins when the pH of the mouth falls below about 5.5.
III. An acidic solution has a pH greater than 7.
Which of the statements given above are correct?

(A) II and III only
(B) I and II only
(C) I and III only
(D) I, II and III

30. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Acid) List-II (Natural source)
A. Formic acid 1. Sting of an ant
B. Lactic acid 2. Tamarind
C. Citric acid 3. Curd
D. Tartaric acid 4. Lemon

(A) A-1, B-3, C-4, D-2
(B) A-1, B-2, C-3, D-4
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

31. Arrange the following in increasing order of their pH value:
I. Milk of magnesia
II. Gastric juice
III. Sodium hydroxide solution
IV. Pure water

(A) I, II, IV, III
(B) II, I, IV, III
(C) II, IV, I, III
(D) IV, II, I, III

32. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Common name) List-II (Chemical name)
A. Quicklime 1. Calcium oxide
B. Slaked lime 2. Calcium hydroxide
C. Marble 3. Calcium carbonate
D. Plaster of Paris 4. Calcium sulphate hemihydrate

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-3, B-2, C-1, D-4

33. Which one of the following pairs of a common name and the substance it denotes is NOT correctly matched?

(A) Dry ice — Solid carbon dioxide
(B) Laughing gas — Nitrous oxide
(C) Marsh gas — Methane
(D) Hypo — Sodium hydrogencarbonate

34. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Common name) List-II (Chemical formula)
A. Glauber's salt 1. $Na_2CO_3.10H_2O$
B. Washing soda 2. $CuSO_4.5H_2O$
C. Blue vitriol 3. $MgSO_4.7H_2O$
D. Epsom salt 4. $Na_2SO_4.10H_2O$

(A) A-4, B-1, C-2, D-3
(B) A-1, B-2, C-3, D-4
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

35. Consider the following statements:
I. Isotopes of an element have the same atomic number but different mass numbers.
II. The modern periodic law states that the properties of elements are a periodic function of their atomic number.
III. The elements of group 17 of the periodic table are called alkali metals.
Which of the statements given above are correct? HARD

(A) II and III only
(B) I and II only
(C) I and III only
(D) I, II and III

36. Consider the following statements about the allotropes of carbon:
I. Diamond and graphite are both allotropes of carbon.
II. Graphite conducts electricity whereas diamond does not.
III. In diamond each carbon atom is covalently bonded to three other carbon atoms.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

37. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Industrial process) List-II (Catalyst used)
A. Haber process for ammonia 1. Finely divided iron
B. Contact process for sulphuric acid 2. Vanadium pentoxide
C. Hydrogenation of vegetable oil 3. Nickel
D. Ostwald process for nitric acid 4. Platinum HARD

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-3, B-2, C-1, D-4

38. Consider the following statements:
I. Hydrogen is the lightest element known.
II. Nitrogen makes up about 78 per cent of the atmosphere by volume.
III. Liquid nitrogen is used as a refrigerant for preserving biological material.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

39. Consider the following statements:
I. The main constituent of compressed natural gas (CNG) is methane.
II. Liquefied petroleum gas (LPG) is mainly a mixture of butane and propane.
III. Ethyl mercaptan is added to LPG so that a leak can be detected by its smell.
Which of the statements given above are correct? RAS PRE 2021

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. Which one of the following pairs of a fuel gas and its chief constituents is NOT correctly matched?

(A) Producer gas — Carbon monoxide and nitrogen
(B) Water gas — Carbon monoxide and carbon dioxide
(C) Coal gas — Hydrogen, methane and carbon monoxide
(D) Biogas — Mainly methane

41. Among the following fuels, the highest calorific value belongs to HARD

(A) coal
(B) hydrogen
(C) petrol
(D) biogas

42. Which one of the following is NOT a thermoplastic?

(A) Polythene
(B) Polyvinyl chloride
(C) Bakelite
(D) Nylon

43. Vulcanisation, which makes natural rubber harder and more elastic, involves heating it with

(A) phosphorus
(B) sulphur
(C) carbon black
(D) nitrogen

44. Consider the following statements:
I. Soaps are sodium or potassium salts of long-chain fatty acids.
II. Soaps do not form a good lather with hard water.
III. Synthetic detergents work effectively even in hard water.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

45. A small quantity of gypsum is added to cement in order to

(A) increase its strength
(B) give it a grey colour
(C) slow down the rate of setting
(D) make it waterproof

46. The chief constituent of ordinary glass is

(A) calcium carbonate
(B) aluminium oxide
(C) sodium sulphate
(D) silica

47. Consider the following statements:
I. Urea has the highest nitrogen content among solid nitrogenous fertilisers, about 46 per cent.
II. Muriate of potash is chemically potassium chloride.
III. Di-ammonium phosphate supplies both nitrogen and phosphorus to the soil.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

48. Gunpowder is a mixture of HARD

(A) ammonium nitrate, charcoal and sulphur
(B) sodium nitrate, sulphur and phosphorus
(C) potassium chlorate, charcoal and phosphorus
(D) potassium nitrate, charcoal and sulphur

49. Consider the following statements:
I. Argemone seed oil is a common adulterant of mustard oil.
II. Metanil yellow is used to adulterate turmeric powder and pulses.
III. Sodium benzoate is used as a chemical preservative in food.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

50. Morphine, obtained from opium, is classified as RAS PRE 2023

(A) an antibiotic
(B) an antiseptic
(C) a narcotic analgesic
(D) an antipyretic

51. The antibiotic penicillin was discovered by

(A) Louis Pasteur
(B) Alexander Fleming
(C) Robert Koch
(D) Edward Jenner

52. The essential difference between an antiseptic and a disinfectant is that

(A) an antiseptic kills bacteria while a disinfectant only checks their growth
(B) an antiseptic is a natural product while a disinfectant is synthetic
(C) an antiseptic is applied to living tissue while a disinfectant is used on non-living objects
(D) an antiseptic is taken orally while a disinfectant is applied externally

53. Which one of the following pairs of a drug and its category is NOT correctly matched?

(A) Paracetamol — Antipyretic
(B) Aspirin — Antibiotic
(C) Penicillin — Antibiotic
(D) Dettol — Antiseptic

54. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Radioisotope) List-II (Use)

A. Carbon-14 1. Detection of thyroid disorders

B. Cobalt-60 2. Determining the age of fossils

C. Iodine-131 3. Treatment of cancer

D. Sodium-24 4. Detection of leaks in underground pipelines HARD

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-2, B-3, C-1, D-4
(D) A-1, B-2, C-4, D-3

55. Which one of the following substances sublimes, that is, changes directly from the solid to the vapour state?

(A) Common salt
(B) Camphor
(C) Sugar
(D) Copper sulphate

ST03

Biology: Cell, Tissues and the Human Body

55 questions · 30 fact-sheet points

Must-know facts

1. Cell theory: M.J. Schleiden (1838, plants) and Theodor Schwann (1839, animals); 'Omnis cellula e cellula' added by Rudolf Virchow (1855).
2. Robert Hooke coined the term 'cell' in 1665 from cork; Anton van Leeuwenhoek first saw living cells; Robert Brown discovered the nucleus (1831).
3. Mitochondrion - 'powerhouse of the cell'; double membrane, inner membrane folded into cristae; own circular DNA and 70S ribosomes; Krebs cycle + oxidative phosphorylation.
4. Ribosome - site of protein synthesis; 70S in prokaryotes (50S + 30S), 80S in eukaryotes (60S + 40S); mitochondria and chloroplasts carry 70S.
5. Lysosome - 'suicide bag' (Christian de Duve); hydrolytic enzymes at acidic pH. Golgi apparatus (Camillo Golgi) - packaging, secretion, lysosome formation.
6. Plastids: chloroplast (photosynthesis, grana + stroma), chromoplast (colour), leucoplast (storage). Chlorophyll contains magnesium; haemoglobin contains iron.
7. Plant cell exclusive: cellulose cell wall, large central vacuole, plastids. Animal cell exclusive: centriole/centrosome (absent in higher plants), prominent lysosomes.
8. Mitosis - equational, somatic, 2 identical diploid cells. Meiosis - reductional, gamete-forming, 4 haploid cells; crossing over occurs in pachytene of prophase-I.
9. Energy value: carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g. Enzymes are proteins (except ribozymes) and lower activation energy.
10. DNA double helix - Watson and Crick, 1953; A=T two hydrogen bonds, G(triple bond)C three; RNA has uracil in place of thymine and ribose sugar.
11. Five-kingdom classification - R.H. Whittaker, 1969 (Monera, Protista, Fungi, Plantae, Animalia); binomial nomenclature - Carolus Linnaeus (genus + species).
12. Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$; light reaction in grana/thylakoid, Calvin (dark) reaction in stroma; evolved O_2 comes from photolysis of water.
13. Plant hormones: auxin - apical dominance; gibberellin - stem elongation/bolting; cytokinin - cell division; abscisic acid - 'stress hormone', stomatal closure; ethylene - gaseous, fruit ripening.
14. Digestion: ptyalin (salivary amylase) on starch; pepsin + HCl in stomach; rennin in infants; bile from liver has no enzyme but emulsifies fat; trypsin, amylase and lipase from pancreas; maximum absorption in small intestine (villi).
15. Liver - largest gland (about 1.5 kg); makes bile and urea, stores glycogen. Pancreas is a mixed (heterocrine) gland. Appendix is a vestigial organ.
16. Respiration: 12 pairs of ribs; diaphragm + intercostal muscles; tidal volume about 500 ml; O_2 carried as oxyhaemoglobin, most CO_2 as bicarbonate in plasma.
17. Heart: 4 chambers, cardiac muscle; SA node in right atrium is the pacemaker; 'lub' = closure of atrioventricular valves, 'dub' = closure of semilunar valves; normal BP 120/80 mm Hg; resting rate about 72/min.
18. William Harvey described the circulation of blood. Blood pressure is measured by sphygmomanometer; ECG records electrical activity of the heart.
19. Blood: plasma about 55%; RBC 4.5-5.5 million/microlitre, anucleate in mammals, lifespan 120 days, formed in red bone marrow, destroyed in spleen ('graveyard of RBCs'); platelets 1.5-4.5 lakh/microlitre.

20. Blood groups discovered by Karl Landsteiner (1901; Nobel 1930). O = universal donor, AB = universal recipient; AB plasma has no anti-A/anti-B antibody, O plasma has both.
21. Rh factor - named after the Rhesus monkey (Landsteiner and Wiener, 1940). Rh-negative mother carrying an Rh-positive foetus risks erythroblastosis foetalis, usually from the second pregnancy; prevented by anti-D immunoglobulin.
22. Excretion: nephron is the structural and functional unit (about 10 lakh per kidney); Bowman's capsule + glomerulus = Malpighian corpuscle; humans are ureotelic; urea is made in the liver (ornithine cycle); ADH (vasopressin) controls water reabsorption.
23. Nervous system: neuron is the longest cell; adult brain about 1300-1400 g; cerebrum largest (intelligence, memory, will); cerebellum - balance and coordination; medulla oblongata - heartbeat, breathing, blood pressure; hypothalamus - temperature, hunger, thirst.
24. Humans have 12 pairs of cranial nerves and 31 pairs of spinal nerves; reflex action is mediated by the spinal cord through a reflex arc.
25. Endocrine: pituitary is the 'master gland'; GH deficiency - dwarfism, excess in childhood - gigantism, in adults - acromegaly; thyroxine needs iodine (goitre, cretinism, myxoedema); parathormone regulates calcium; calcitonin is from the thyroid.
26. Adrenal cortex - cortisol and aldosterone (deficiency: Addison's disease); adrenal medulla - adrenaline, the 'emergency/fight-or-flight hormone'; islets of Langerhans - insulin (beta cells), glucagon (alpha cells); ADH deficiency causes diabetes insipidus.
27. Skeleton: 206 bones in an adult (about 300 at birth); femur is the longest bone, stapes of the middle ear the smallest; skull 22 bones; gluteus maximus is the largest muscle, stapedius the smallest.
28. Muscle types: skeletal (striated, voluntary), smooth (unstriated, involuntary), cardiac (striated, involuntary, non-fatiguing).
29. Eye: rods contain rhodopsin and work in dim light, cones give colour vision; yellow spot (fovea) has maximum visual acuity; blind spot has no photoreceptors. Ear ossicles - malleus, incus, stapes; cochlea for hearing, semicircular canals for balance.
30. Human evolution: Australopithecus -> Homo habilis ('handy man', first tool maker) -> Homo erectus -> Homo neanderthalensis -> Homo sapiens; humans have 46 chromosomes (23 pairs), XX female and XY male.

1. The term 'cell' was coined by which scientist?

(A) Robert Hooke (B) Robert Brown
(C) Anton van Leeuwenhoek (D) Rudolf Virchow

2. The dictum 'Omnis cellula e cellula' (every cell arises from a pre-existing cell) was given by

(A) M.J. Schleiden (B) Theodor Schwann
(C) Rudolf Virchow (D) J.E. Purkinje

3. Which cell organelle is known as the 'powerhouse of the cell'?

(A) Ribosome (B) Mitochondrion
(C) Golgi apparatus (D) Lysosome

4. Which of the following cell organelles is called the 'suicide bag' of the cell?

(A) Peroxisome (B) Vacuole
(C) Centrosome (D) Lysosome

5. Ribosomes found in prokaryotic cells are of which type?

(A) 80S (B) 70S
(C) 60S (D) 55S

6. Cellulose is the chief structural constituent of which of the following?

(A) Plasma membrane of animal cells
(B) Nuclear membrane
(C) Mitochondrial matrix
(D) Cell wall of plants

7. Meiosis in a diploid cell results in the formation of

(A) Two diploid cells
(B) Two haploid cells
(C) Four haploid cells
(D) Four diploid cells

8. Crossing over between homologous chromosomes takes place during which sub-stage of meiosis? HARD

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

9. The approximate energy value of one gram of fat is

(A) 4 kilocalories
(B) 7 kilocalories
(C) 9 kilocalories
(D) 11 kilocalories

10. Which nitrogenous base is present in RNA but absent in DNA?

(A) Thymine
(B) Uracil
(C) Cytosine
(D) Adenine

11. How many hydrogen bonds are present between guanine and cytosine in a DNA double helix? HARD

(A) One
(B) Two
(C) Three
(D) Four

12. The five-kingdom system of classification of living organisms was proposed by

(A) Carolus Linnaeus
(B) R.H. Whittaker
(C) Ernst Haeckel
(D) Charles Darwin

13. In the system of binomial nomenclature, the first word of a scientific name denotes the

(A) Species
(B) Family
(C) Order
(D) Genus

14. The oxygen released during photosynthesis in green plants comes from

(A) Carbon dioxide
(B) Water
(C) Glucose
(D) Atmospheric oxides

15. The dark reaction (Calvin cycle) of photosynthesis takes place in which part of the chloroplast?

(A) Thylakoid membrane
(B) Grana
(C) Stroma
(D) Outer membrane

16. Which plant hormone is chiefly responsible for the ripening of fruits?

(A) Auxin
(B) Gibberellin
(C) Ethylene
(D) Cytokinin

17. Which enzyme present in human saliva begins the digestion of starch?

(A) Pepsin
(B) Ptyalin
(C) Trypsin
(D) Lipase

18. Bile secreted by the liver is stored in which organ?

(A) Duodenum
(B) Pancreas
(C) Spleen
(D) Gall bladder

19. The pacemaker of the human heart, the sino-atrial node, is located in the

(A) Left atrium (B) Right atrium
(C) Left ventricle (D) Inter-ventricular septum

20. The circulation of blood in the human body was first described by

(A) Karl Landsteiner (B) William Harvey
(C) Marcello Malpighi (D) Andreas Vesalius

21. Which blood group is known as the universal donor? RAS PRE 2023

(A) A (B) B
(C) AB (D) O

22. The Rh factor present in human blood derives its name from RAS PRE 2021

(A) The chimpanzee (B) The Rhesus monkey
(C) The guinea pig (D) The rabbit

23. The average lifespan of a human red blood corpuscle is about

(A) 20 days (B) 60 days
(C) 120 days (D) 180 days

24. The structural and functional unit of the human kidney is the

(A) Nephron (B) Neuron
(C) Alveolus (D) Villus

25. In the human body, urea is chiefly synthesised in the

(A) Kidney (B) Spleen
(C) Pancreas (D) Liver

26. Which endocrine gland is called the 'master gland' of the human body?

(A) Pituitary gland (B) Thyroid gland
(C) Adrenal gland (D) Thymus

27. Insulin is secreted by which cells of the human pancreas?

(A) Acinar cells (B) Alpha cells of the islets of Langerhans
(C) Beta cells of the islets of Langerhans (D) Delta cells of the islets of Langerhans

28. The number of bones in the body of a normal adult human being is

(A) 186 (B) 206
(C) 230 (D) 300

29. Which is the smallest bone in the human body?

(A) Malleus (B) Incus
(C) Stapes (D) Nasal bone

30. Vision in dim light in the human eye is due to which photoreceptors?

(A) Rods (B) Cones
(C) Ganglion cells (D) Bipolar cells

31. Consider the following statements about the mitochondrion:
I. It is bounded by a double membrane whose inner membrane is folded into cristae.
II. It possesses its own circular DNA and 70S ribosomes.
III. The Calvin cycle of photosynthesis takes place in it.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

32. Consider the following statements about the human heart:
I. It is a four-chambered organ made of cardiac muscle.
II. The sino-atrial node is situated in the left atrium.
III. The first heart sound 'lub' is produced by the closure of the atrio-ventricular valves.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

33. Consider the following statements about human blood groups:
I. The plasma of a person with blood group AB contains neither anti-A nor anti-B antibody.
II. The plasma of a person with blood group O contains both anti-A and anti-B antibodies.
III. The ABO blood group system was discovered by Karl Landsteiner.
Which of the statements given above are correct? RAS PRE 2023

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

34. Consider the following statements about the human liver:
I. It is the largest gland in the human body.
II. The bile it secretes contains no digestive enzyme.
III. It is the principal site of urea formation.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

35. Consider the following statements regarding cell division:
I. Meiosis takes place in the somatic cells of the body.
II. Mitosis produces two genetically identical diploid daughter cells.
III. Crossing over occurs during the first meiotic division.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

36. Consider the following statements about enzymes:
I. All enzymes are without exception proteins.
II. Enzymes act by lowering the activation energy of a reaction.
III. Enzyme activity is affected by temperature and pH.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) I, II and III
(D) II and III only

37. Consider the following statements about pancreatic hormones:
I. Insulin is secreted by the beta cells of the islets of Langerhans.
II. Glucagon lowers the level of glucose in the blood.
III. Deficiency of insulin causes diabetes mellitus.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

38. Consider the following statements about the human nervous system:
I. The cerebrum is the largest part of the human brain.
II. Balance and muscular coordination are controlled by the medulla oblongata.
III. Human beings have 31 pairs of spinal nerves.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

39. Consider the following statements about human excretion:
I. The nephron is the structural and functional unit of the kidney.
II. Human beings are uricotelic animals.
III. Anti-diuretic hormone increases the reabsorption of water from the kidney tubules.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

40. Consider the following statements about photosynthesis:
I. The light reaction takes place in the grana of the chloroplast.
II. Chlorophyll contains magnesium as its central metal atom.
III. Oxygen is evolved during the dark reaction.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

41. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Cell organelle) List-II (Chief function)
A. Ribosome 1. Photosynthesis
B. Lysosome 2. Protein synthesis
C. Chloroplast 3. Packaging and secretion
D. Golgi body 4. Intracellular digestion

(A) A-2, B-4, C-1, D-3
(B) A-2, B-1, C-4, D-3
(C) A-3, B-4, C-1, D-2
(D) A-2, B-4, C-3, D-1

42. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Endocrine gland) List-II (Hormone)
A. Thyroid 1. Adrenaline
B. Adrenal medulla 2. Insulin
C. Pancreas 3. Thyroxine
D. Anterior pituitary 4. Growth hormone

(A) A-1, B-3, C-2, D-4
(B) A-3, B-2, C-1, D-4
(C) A-3, B-1, C-2, D-4
(D) A-4, B-1, C-2, D-3

43. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Hormonal condition) List-II (Disorder)
A. Deficiency of insulin 1. Diabetes insipidus
B. Deficiency of thyroxine in infancy 2. Acromegaly
C. Excess of growth hormone in an adult 3. Diabetes mellitus
D. Deficiency of anti-diuretic hormone 4. Cretinism

(A) A-3, B-4, C-1, D-2
(B) A-3, B-4, C-2, D-1
(C) A-4, B-3, C-2, D-1
(D) A-3, B-2, C-4, D-1

44. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Plant hormone) List-II (Principal effect)
A. Auxin 1. Closure of stomata under water stress
B. Gibberellin 2. Apical dominance
C. Cytokinin 3. Cell division and delay of senescence
D. Abscisic acid 4. Stem elongation and bolting HARD

(A) A-2, B-4, C-1, D-3
(B) A-4, B-2, C-3, D-1
(C) A-2, B-3, C-4, D-1
(D) A-2, B-4, C-3, D-1

45. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Secretion) List-II (Source)
A. Ptyalin 1. Gastric gland
B. Pepsin 2. Liver
C. Trypsin 3. Salivary gland
D. Bile 4. Pancreas

(A) A-3, B-1, C-4, D-2
(B) A-3, B-4, C-1, D-2
(C) A-1, B-3, C-4, D-2
(D) A-3, B-1, C-2, D-4

46. Match List-I with List-II and select the correct answer using the codes given below:
List-I List-II
A. Femur 1. Smallest bone in the human body
B. Stapes 2. Largest muscle in the human body
C. Gluteus maximus 3. Longest bone in the human body
D. Stapedius 4. Smallest muscle in the human body

(A) A-1, B-3, C-2, D-4
(B) A-3, B-1, C-4, D-2
(C) A-3, B-1, C-2, D-4
(D) A-3, B-4, C-2, D-1

47. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Scientist) List-II (Discovery)
A. Karl Landsteiner 1. Circulation of blood
B. William Harvey 2. Nucleus of the cell
C. Robert Brown 3. Human blood groups
D. Camillo Golgi 4. Golgi apparatus

(A) A-3, B-2, C-1, D-4
(B) A-1, B-3, C-2, D-4
(C) A-3, B-1, C-4, D-2
(D) A-3, B-1, C-2, D-4

48. Which one of the following pairs of cell organelle and its function is NOT correctly matched?

(A) Nucleolus - Synthesis of ribosomal RNA
(B) Smooth endoplasmic reticulum - Lipid synthesis
(C) Ribosome - Synthesis of lipids
(D) Centriole - Formation of the spindle apparatus

49. Which one of the following is NOT a function of the human liver?

(A) Secretion of bile
(B) Storage of glycogen
(C) Secretion of insulin
(D) Formation of urea

50. Which one of the following gland-hormone pairs is NOT correctly matched? HARD

(A) Thyroid - Calcitonin
(B) Parathyroid - Calcitonin
(C) Adrenal cortex - Cortisol
(D) Adrenal medulla - Adrenaline

51. Which one of the following statements about human blood is NOT correct?

(A) Plasma constitutes about 55 per cent of the volume of blood
(B) Mature mammalian red blood corpuscles possess a nucleus
(C) Platelets help in the clotting of blood
(D) White blood corpuscles provide immunity against infection

52. Which one of the following is NOT a part of the human brain?

(A) Cerebrum
(B) Cerebellum
(C) Medulla oblongata
(D) Cochlea

53. Which one of the following pairs of blood group and the antibodies present in its plasma is NOT correctly matched? RAS PRE 2013

(A) Blood group A - Anti-B antibody
(B) Blood group B - Anti-A antibody
(C) Blood group AB - Anti-A and anti-B antibodies
(D) Blood group O - Anti-A and anti-B antibodies

54. Arrange the following stages of mitosis in their correct sequence:
I. Anaphase
II. Metaphase
III. Telophase
IV. Prophase

(A) IV, II, I, III
(B) IV, I, II, III
(C) II, IV, I, III
(D) IV, II, III, I

55. Arrange the following parts of the human alimentary canal in the order in which swallowed food passes through them:
I. Duodenum
II. Oesophagus
III. Ileum
IV. Stomach

(A) II, IV, III, I
(B) II, I, IV, III
(C) IV, II, I, III
(D) II, IV, I, III

ST04

Human Health, Diseases and Public Health Programmes

60 questions · 30 fact-sheet points

Must-know facts

1. Malaria - protozoan Plasmodium; vector female Anopheles mosquito. P. falciparum causes malignant/cerebral malaria (most fatal); P. vivax is the commonest in India; P. malariae causes quartan malaria.
2. Ronald Ross demonstrated the malaria parasite in the Anopheles mosquito at Secunderabad in 1897 and won the Nobel Prize in 1902.
3. Dengue, chikungunya, Zika and yellow fever are transmitted by Aedes aegypti, a daytime biter; dengue is a flavivirus and causes thrombocytopenia (fall in platelet count).
4. Lymphatic filariasis (elephantiasis) - Wuchereria bancrofti; vector Culex quinquefasciatus. Japanese encephalitis - JE virus; vector Culex tritaeniorhynchus; the pig is the amplifying host.
5. Kala-azar (visceral leishmaniasis) - Leishmania donovani; vector the female sandfly Phlebotomus argentipes. Plague - Yersinia pestis; vector the rat flea Xenopsylla cheopis.
6. Scrub typhus - Orientia tsutsugamushi, spread by the larval trombiculid mite (chigger). African sleeping sickness - Trypanosoma, spread by the tsetse fly.
7. Tuberculosis - Mycobacterium tuberculosis, identified by Robert Koch in 1882; air-borne; BCG vaccine; diagnosis by sputum microscopy and CBNAAT; DOTS therapy; Ni-kshay portal.
8. Leprosy (Hansen's disease) - Mycobacterium leprae, discovered by G.H.A. Hansen. Cholera - Vibrio cholerae; typhoid - Salmonella typhi, diagnosed by the Widal test.
9. Hepatitis A and E spread by the faeco-oral route (contaminated food and water); hepatitis B, C and D spread through blood and body fluids. Hepatitis B vaccine is part of the UIP.
10. Polio - poliovirus; OPV is the live attenuated oral vaccine of Albert Sabin, IPV the killed injectable vaccine of Jonas Salk. India was certified polio-free by WHO in March 2014.
11. Rabies - a lyssavirus transmitted by the bite of an infected dog; the first vaccine was made by Louis Pasteur; it is virtually 100 per cent fatal once symptoms appear.
12. COVID-19 - SARS-CoV-2; declared a pandemic by WHO on 11 March 2020. Swine flu is influenza A H1N1 and bird flu is avian influenza H5N1.
13. Nipah virus - natural reservoir is the fruit bat (Pteropus); Zika virus is spread by Aedes and can cause microcephaly in newborns. Mpox is a viral zoonosis of the orthopoxvirus group.
14. AIDS - Human Immunodeficiency Virus, a retrovirus that destroys CD4 helper T lymphocytes; screened by ELISA and confirmed by Western blot; NACO was set up in 1992 and the HIV and AIDS (Prevention and Control) Act was passed in 2017.
15. Deficiency diseases: vitamin A - night blindness and xerophthalmia; B1 - beriberi; B3 - pellagra; B12 - pernicious anaemia; C - scurvy; D - rickets in children and osteomalacia in adults; K - delayed blood clotting; iodine - goitre; iron - anaemia.
16. Genetic disorders: Down syndrome - trisomy of chromosome 21 (47 chromosomes); Turner syndrome - 45, XO (female); Klinefelter syndrome - 47, XXY (male); haemophilia and colour blindness are X-linked recessive; sickle cell anaemia and thalassaemia are autosomal recessive.
17. Immunity: innate versus acquired; active immunity follows infection or vaccination, passive immunity is transferred ready-made (colostrum, antiserum). Edward Jenner developed smallpox vaccination in 1796; smallpox was declared eradicated by WHO in 1980 (last Indian case 1975).
18. The Universal Immunisation Programme was launched in 1985 (from the Expanded Programme on Immunisation, 1978); it now protects against 12 vaccine-preventable diseases. Most vaccines are stored in the cold chain at 2-8 degrees Celsius.

19. Mission Indradhanush was launched on 25 December 2014 to raise full immunisation coverage to at least 90 per cent; Intensified Mission Indradhanush began in October 2017 and IMI 5.0 (August 2023) covered children up to five years of age with a focus on measles-rubella.
20. National Health Mission (2013) = National Rural Health Mission (2005) + National Urban Health Mission (2013). The ASHA (Accredited Social Health Activist) is the community link worker created under NRHM.
21. Janani Suraksha Yojana (2005) gives cash assistance for institutional delivery; Janani Shishu Suraksha Karyakram (2011) gives free delivery entitlements; LaQshya (2017) targets labour-room quality.
22. Ayushman Bharat (2018) has two components - Health and Wellness Centres, renamed Ayushman Arogya Mandir in November 2023 with the tagline 'Arogyam Parmam Dhanam', and PM-JAY.
23. PM-JAY was launched on 23 September 2018 from Ranchi; it gives health cover of Rs 5 lakh per family per year for secondary and tertiary hospitalisation, and the Ayushman Vay Vandana card (announced October 2024) extends cover to all citizens aged 70 years and above.
24. Ayushman Bharat Digital Mission was launched nationwide on 27 September 2021 (piloted from 15 August 2020); ABHA is a 14-digit health account number; its registries are the Healthcare Professionals Registry and the Health Facility Registry.
25. The Revised National TB Control Programme was renamed the National TB Elimination Programme in 2020; India's TB elimination target is 2025, five years ahead of the SDG target of 2030. Ni-kshay Poshan Yojana (April 2018) gives DBT nutrition support, raised from Rs 500 to Rs 1,000 per month in November 2024.
26. The NVBDCP directorate is now the National Center for Vector Borne Diseases Control (NCVBDC); it covers six diseases - malaria, dengue, chikungunya, Japanese encephalitis, kala-azar and lymphatic filariasis - of which malaria, kala-azar and lymphatic filariasis are targeted for elimination; the malaria elimination target year is 2030.
27. Anaemia Mukt Bharat was launched in 2018 on a 6x6x6 strategy - six beneficiary groups, six interventions and six institutional mechanisms.
28. NOTTO, the National Organ and Tissue Transplant Organisation, functions at New Delhi under the DGHS; ROTTOs and SOTTOs work at regional and state level under the Transplantation of Human Organs Act, 1994, amended in 2011. Indian Organ Donation Day is observed on 3 August.
29. AYUSH covers Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa and Homoeopathy; the Ministry of AYUSH was created in November 2014 and the WHO Global Centre for Traditional Medicine is at Jamnagar, Gujarat.
30. As per the SRS Report 2021 released in May 2025, India's maternal mortality ratio is 93 per lakh live births (2019-21), infant mortality rate 27, under-five mortality rate 31 and neonatal mortality rate 19 per 1,000 live births.

1. Malaria is transmitted to human beings by the bite of `RAS PRE 2024`

(A) Female Culex mosquito (B) Female Anopheles mosquito
(C) Female Aedes mosquito (D) Sandfly

2. Which species of Plasmodium causes the most severe, malignant form of malaria in human beings? `RAS PRE 2023`

(A) Plasmodium vivax (B) Plasmodium malariae
(C) Plasmodium falciparum (D) Plasmodium ovale

3. The transmission of the malaria parasite by the mosquito was demonstrated by

(A) Robert Koch (B) Ronald Ross
(C) Louis Pasteur (D) Charles Laveran

4. Dengue fever is transmitted by which of the following mosquitoes? RAS PRE 2021

(A) Anopheles stephensi
(B) Culex quinquefasciatus
(C) Aedes aegypti
(D) Mansonia annulifera

5. Kala-azar, or visceral leishmaniasis, is caused by

(A) Trypanosoma gambiense
(B) Entamoeba histolytica
(C) Giardia lamblia
(D) Leishmania donovani

6. Lymphatic filariasis, popularly called elephantiasis, is caused by

(A) Wuchereria bancrofti
(B) Ascaris lumbricoides
(C) Ancylostoma duodenale
(D) Taenia solium

7. In the transmission cycle of Japanese encephalitis, which animal acts as the amplifying host? HARD

(A) Dog
(B) Cattle
(C) Pig
(D) Goat

8. Plague in human beings is caused by which of the following organisms?

(A) Yersinia pestis
(B) Vibrio cholerae
(C) Rickettsia prowazekii
(D) Bacillus anthracis

9. Scrub typhus is transmitted to human beings by the bite of HARD

(A) A larval trombiculid mite
(B) A tick
(C) A body louse
(D) A sandfly

10. The bacterium causing tuberculosis was discovered by

(A) Louis Pasteur
(B) Robert Koch
(C) Joseph Lister
(D) Alexander Fleming

11. BCG vaccine given to newborns provides protection primarily against

(A) Tuberculosis
(B) Tetanus
(C) Typhoid
(D) Diphtheria

12. Leprosy, also known as Hansen's disease, is caused by

(A) Mycobacterium leprae
(B) Mycobacterium tuberculosis
(C) Treponema pallidum
(D) Corynebacterium diphtheriae

13. The Widal test is used for the diagnosis of which disease?

(A) Malaria
(B) Tuberculosis
(C) Dengue
(D) Typhoid

14. Cholera is caused by which of the following organisms?

(A) Shigella dysenteriae
(B) Vibrio cholerae
(C) Escherichia coli
(D) Salmonella typhi

15. Which type of hepatitis spreads chiefly through contaminated food and water?

(A) Hepatitis A
(B) Hepatitis B
(C) Hepatitis C
(D) Hepatitis D

16. The oral polio vaccine used in mass immunisation campaigns is a

(A) Killed vaccine developed by Jonas Salk
(B) Live attenuated vaccine developed by Albert Sabin
(C) Toxoid vaccine
(D) Recombinant subunit vaccine

17. India was certified polio-free by the World Health Organization in which year?

(A) 2011 (B) 2012
(C) 2014 (D) 2016

18. The first vaccine against rabies was developed by

(A) Edward Jenner (B) Jonas Salk
(C) Robert Koch (D) Louis Pasteur

19. Swine flu in human beings is caused by which strain of influenza virus?

(A) H5N1 (B) H1N1
(C) H7N9 (D) H3N2

20. The natural reservoir of the Nipah virus is HARD

(A) The fruit bat (B) The domestic pig
(C) The rodent (D) The camel

21. The Human Immunodeficiency Virus principally destroys which cells of the human body?

(A) B lymphocytes (B) CD4 helper T lymphocytes
(C) Platelets (D) Erythrocytes

22. The technique of vaccination against smallpox was introduced by

(A) Louis Pasteur (B) Edward Jenner
(C) Robert Koch (D) Joseph Lister

23. Down syndrome in human beings is caused by RAS PRE 2021

(A) Trisomy of chromosome 21 (B) Monosomy of chromosome 21
(C) Trisomy of chromosome 18 (D) Deletion in chromosome 5

24. Haemophilia in human beings is an example of which pattern of inheritance?

(A) Autosomal dominant (B) Autosomal recessive
(C) X-linked recessive (D) Y-linked

25. Turner syndrome in human beings is associated with which chromosomal constitution? HARD

(A) 47, XXY (B) 47, XYY
(C) 47, XXX (D) 45, XO

26. Scurvy in human beings is caused by the deficiency of

(A) Vitamin A (B) Vitamin B12
(C) Vitamin C (D) Vitamin K

27. Enlargement of the thyroid gland known as goitre is caused by the dietary deficiency of

(A) Iron (B) Iodine
(C) Calcium (D) Zinc

28. Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana provides a health cover of how much per family per year?

(A) Rs 1 lakh
(B) Rs 2 lakh
(C) Rs 5 lakh
(D) Rs 10 lakh

29. Ayushman Bharat Health and Wellness Centres were renamed as which of the following in 2023?

(A) Jan Arogya Kendra
(B) Ayushman Arogya Mandir
(C) Swasthya Suraksha Kendra
(D) Arogya Setu Kendra

30. Mission Indradhanush, the special immunisation drive of the Government of India, was launched in which year?

(A) 2012
(B) 2014
(C) 2016
(D) 2018

31. Under the National TB Elimination Programme, India has set which year as its target for the elimination of tuberculosis?

(A) 2025
(B) 2027
(C) 2030
(D) 2035

32. The Ayushman Bharat Health Account (ABHA) number created under the Ayushman Bharat Digital Mission consists of how many digits? HARD

(A) 10
(B) 12
(C) 14
(D) 16

33. Consider the following statements about malaria:
I. It is caused by a protozoan parasite of the genus Plasmodium.
II. It is transmitted by the bite of the female Anopheles mosquito.
III. Plasmodium falciparum causes the benign tertian form of malaria.
Which of the statements given above are correct? RAS PRE 2023

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

34. Consider the following statements about dengue:
I. It is caused by a virus.
II. Its vector Aedes aegypti bites mainly during the daytime.
III. A fall in the platelet count is a characteristic feature of the disease.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

35. Consider the following statements about immunity:
I. Immunity acquired through vaccination is a form of active immunity.
II. The antibodies received by an infant through the mother's colostrum give passive immunity.
III. BCG is a killed vaccine.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

36. Consider the following statements about Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana:
I. It provides a health cover of Rs 5 lakh per family per year.
II. It was launched in September 2018 from Ranchi.
III. Beneficiary families must pay an annual premium to enrol in the scheme.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

37. Consider the following statements about the Ayushman Bharat Digital Mission:
I. It was launched nationwide in September 2021.
II. ABHA is a unique health account number issued under it.
III. The Health Facility Registry is one of its building blocks.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

38. Consider the following statements about the National TB Elimination Programme:
I. The Revised National TB Control Programme was renamed the National TB Elimination Programme in 2020.
II. India's target year for the elimination of tuberculosis is 2030.
III. Ni-kshay Poshan Yojana provides direct benefit transfer for the nutritional support of TB patients.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

39. Consider the following statements about genetic disorders:
I. Down syndrome results from the trisomy of chromosome 21.
II. Colour blindness is an X-linked recessive disorder.
III. Thalassaemia is a sex-linked disorder.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. Consider the following statements about antibiotics and antimicrobial resistance:
I. Antibiotics are ineffective against viral infections such as the common cold.
II. Penicillin, the first antibiotic, was discovered by Alexander Fleming.
III. India adopted a National Action Plan on Antimicrobial Resistance in 2017.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

41. Consider the following statements about the Rh factor and blood transfusion:
I. An Rh-negative person can safely receive Rh-positive blood repeatedly without any risk.
II. Erythroblastosis foetalis may occur when an Rh-negative mother carries an Rh-positive foetus.
III. Anti-D immunoglobulin given to the mother helps prevent this condition.
Which of the statements given above are correct? RAS PRE 2023 HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

42. Consider the following statements about the National Center for Vector Borne Diseases Control:
I. It covers six vector-borne diseases including dengue and chikungunya.
II. India's target year for the elimination of malaria is 2030.
III. Chikungunya is one of the diseases targeted for elimination under the programme.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

43. Consider the following statements about poliomyelitis:
I. The oral polio vaccine contains live attenuated virus.
II. The injectable polio vaccine developed by Jonas Salk contains killed virus.
III. India was certified polio-free in the year 2014.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

44. Consider the following statements about deficiency diseases:
I. Beriberi is caused by the deficiency of thiamine.
II. Pellagra is caused by the deficiency of niacin.
III. Rickets is caused by the deficiency of vitamin C.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

45. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Disease) List-II (Causative organism)
A. Malaria 1. Salmonella typhi
B. Kala-azar 2. Wuchereria bancrofti
C. Filariasis 3. Plasmodium vivax
D. Typhoid 4. Leishmania donovani RAS PRE 2024

(A) A-3, B-4, C-2, D-1
(B) A-3, B-2, C-4, D-1
(C) A-4, B-3, C-2, D-1
(D) A-3, B-4, C-1, D-2

46. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Disease) List-II (Vector)
A. Malaria 1. Aedes aegypti
B. Dengue 2. Sandfly
C. Filariasis 3. Female Anopheles
D. Kala-azar 4. Culex RAS PRE 2024

(A) A-3, B-1, C-2, D-4
(B) A-1, B-3, C-4, D-2
(C) A-3, B-4, C-1, D-2
(D) A-3, B-1, C-4, D-2

47. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Disease) List-II (Type of pathogen)
A. Tuberculosis 1. Fungus
B. Dengue 2. Protozoan
C. Ringworm 3. Bacterium
D. Amoebic dysentery 4. Virus

(A) A-3, B-4, C-2, D-1
(B) A-3, B-4, C-1, D-2
(C) A-4, B-3, C-1, D-2
(D) A-3, B-1, C-4, D-2

48. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Deficiency) List-II (Disease)
A. Vitamin A 1. Rickets
B. Vitamin C 2. Goitre
C. Vitamin D 3. Night blindness
D. Iodine 4. Scurvy

(A) A-3, B-1, C-4, D-2
(B) A-4, B-3, C-1, D-2
(C) A-3, B-4, C-1, D-2
(D) A-3, B-4, C-2, D-1

49. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Programme) List-II (Year of launch)
A. National Rural Health Mission 1. 2021
B. Mission Indradhanush 2. 2018
C. Pradhan Mantri Jan Arogya Yojana 3. 2005
D. Ayushman Bharat Digital Mission (national launch) 4. 2014

(A) A-3, B-4, C-2, D-1
(B) A-3, B-2, C-4, D-1
(C) A-4, B-3, C-2, D-1
(D) A-3, B-4, C-1, D-2

50. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Disorder) List-II (Genetic basis)
A. Down syndrome 1. 47, XXY
B. Turner syndrome 2. X-linked recessive gene
C. Klinefelter syndrome 3. Trisomy of chromosome 21
D. Haemophilia 4. 45, XO HARD

(A) A-3, B-1, C-4, D-2
(B) A-3, B-4, C-2, D-1
(C) A-1, B-4, C-3, D-2
(D) A-3, B-4, C-1, D-2

51. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Scientist) List-II (Discovery)
A. Robert Koch 1. Penicillin
B. Ronald Ross 2. Smallpox vaccination
C. Edward Jenner 3. Tuberculosis bacillus
D. Alexander Fleming 4. Malaria parasite in the mosquito

(A) A-3, B-4, C-1, D-2
(B) A-4, B-3, C-2, D-1
(C) A-3, B-2, C-4, D-1
(D) A-3, B-4, C-2, D-1

52. Which one of the following pairs of disease and its vector is NOT correctly matched? RAS PRE 2024 HARD

(A) Japanese encephalitis - Culex mosquito
(B) Chikungunya - Aedes mosquito
(C) Kala-azar - Tsetse fly
(D) Plague - Rat flea

53. Which one of the following is NOT a viral disease?

(A) Measles
(B) Mumps
(C) Typhoid
(D) Chickenpox

54. Which one of the following is NOT a component of the Ayushman Bharat programme?

(A) Health and Wellness Centres
(B) Pradhan Mantri Jan Arogya Yojana
(C) Ayushman Arogya Mandir
(D) Pradhan Mantri Bhartiya Janaushadhi Pariyojana

55. Which one of the following is NOT a water-borne disease?

(A) Cholera
(B) Typhoid
(C) Hepatitis A
(D) Tetanus

56. Which one of the following pairs of vaccine and the disease it prevents is NOT correctly matched?

(A) BCG - Hepatitis B
(B) OPV - Poliomyelitis
(C) MMR - Measles, mumps and rubella
(D) DPT - Diphtheria, pertussis and tetanus

57. Which one of the following diseases is NOT covered under the National Center for Vector Borne Diseases Control?

(A) Malaria
(B) Dengue
(C) Tuberculosis
(D) Lymphatic filariasis

58. Which one of the following pairs of health programme and its year of launch is NOT correctly matched? HARD

(A) National Mental Health Programme - 1982
(B) National Programme for Control of Blindness - 1976
(C) Janani Suraksha Yojana - 2005
(D) Anaemia Mukt Bharat - 2012

59. Arrange the following health initiatives of the Government of India in the correct chronological order of their launch:

I. Pradhan Mantri Jan Arogya Yojana

II. National Rural Health Mission

III. Ayushman Bharat Digital Mission

IV. Mission Indradhanush

(A) II, IV, I, III
(B) II, I, IV, III
(C) IV, II, I, III
(D) II, IV, III, I

60. Assertion (A): Plasmodium falciparum causes the most severe form of malaria in human beings.

Reason (R): Plasmodium falciparum can infect red blood cells of all ages and cause blockage of cerebral capillaries.

Select the correct answer from the codes given below: RAS PRE 2023

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

ST05

Food and Nutrition

40 questions · 30 fact-sheet points

Must-know facts

1. Calorific values: carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g, alcohol 7 kcal/g.
2. Balanced diet energy split: carbohydrate 55-60%, fat 20-30%, protein 10-15%. Water = about 60-70% of body weight.
3. Fat-soluble vitamins: A, D, E, K (stored in liver/fat, can cause hypervitaminosis). Water-soluble: B-complex and C.
4. Vitamin A = retinol -> night blindness, Bitot's spots, xerophthalmia, keratomalacia.
5. B1 thiamine -> beriberi; B2 riboflavin -> cheilosis/angular stomatitis; B3 niacin (nicotinic acid) -> pellagra (dermatitis, diarrhoea, dementia).
6. B5 pantothenic acid -> burning feet syndrome; B6 pyridoxine -> anaemia and convulsions; B7 biotin -> dermatitis and hair loss.
7. B9 folic acid -> megaloblastic anaemia and neural tube defects; B12 cyanocobalamin (contains cobalt) -> pernicious anaemia.
8. Vitamin C = ascorbic acid -> scurvy; humans, other primates and guinea pigs cannot synthesise it.
9. Vitamin D = calciferol (D2 ergocalciferol, D3 cholecalciferol) -> rickets in children, osteomalacia in adults; made in skin from 7-dehydrocholesterol using UV-B.
10. Vitamin E = tocopherol (antioxidant, sterility in rats); Vitamin K = phylloquinone (K1)/menaquinone (K2) -> delayed blood clotting; K2 is made by gut bacteria.
11. Minerals: iodine -> goitre and cretinism; iron -> anaemia; calcium -> rickets, osteoporosis, tetany; zinc -> growth retardation; fluoride deficiency -> dental caries, excess -> fluorosis.
12. Iodised salt must carry not less than 30 ppm iodine at manufacture and 15 ppm at the consumer end (NIDDCP). Desirable fluoride limit in drinking water (BIS): 1.0 mg/L, permissible 1.5 mg/L.
13. Kwashiorkor = protein deficiency with near-adequate calories: oedema, moon face, flaky-paint dermatosis; term coined by Cicely Williams from a Ghanaian word.
14. Marasmus = simultaneous deficiency of protein AND calories: extreme wasting, no oedema, 'old man's face', usually under 1 year.
15. Stunting = low height-for-age (chronic); wasting = low weight-for-height (acute); underweight = low weight-for-age.
16. NFHS-5 (2019-21) India: stunting 35.5%, wasting 19.3%, underweight 32.1%; anaemia in children 6-59 months 67.1%, women 15-49 years 57.0%.
17. Anaemia (WHO Hb cut-off): adult men <13 g/dL, non-pregnant women <12 g/dL, pregnant women <11 g/dL, children 6-59 months <11 g/dL.
18. BMI = weight (kg) / height (m)^2. WHO: <18.5 underweight, 18.5-24.9 normal, 25-29.9 overweight, >=30 obese. Asian-Indian cut-offs: >=23 overweight, >=25 obese.
19. Nine essential amino acids for adult humans: histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine. Egg protein is the reference protein.
20. Fortification (adding nutrients during processing) carries the FSSAI '+F' logo. Salt -> iodine (+iron in DFS); wheat flour and rice -> iron, folic acid, vitamin B12; milk and edible oil -> vitamins A and D.
21. Biofortification = raising nutrient content by breeding/agronomy. Dhanashakti (iron-rich pearl millet) was India's first biofortified variety; Golden Rice carries provitamin A (beta-carotene); QPM maize is rich in lysine and tryptophan.

22. Adulterants: argemone oil in mustard oil -> epidemic dropsy; kesari dal (Lathyrus sativus, BOAA/ODAP) in arhar -> lathyrism; metanil yellow/lead chromate in turmeric; papaya seeds in black pepper; brick powder in chilli powder.
23. Preservation: pasteurisation LTLT 63 C/30 min, HTST 72 C/15 sec, UHT 135-140 C/2-3 sec. Canning was invented by Nicolas Appert. Irradiation uses Cobalt-60 gamma rays and carries the Radura logo. Lyophilisation = freeze drying.
24. Class I preservatives are natural (salt, sugar, vinegar, spices, honey, edible oil); Class II are chemical (sodium benzoate, sulphur dioxide, sorbic acid, nitrites).
25. FSSAI was set up under the Food Safety and Standards Act, 2006, under the Ministry of Health and Family Welfare, headquartered in New Delhi. FSSAI licence numbers have 14 digits.
26. FSSAI initiatives: Eat Right India (2018), Eat Right Challenge, RUCO (Repurpose Used Cooking Oil), Clean Street Food Hub, BHOG. Industrial trans fat in foods capped at 2% by weight from January 2022.
27. ICDS was launched on 2 October 1975 and delivers six services; it now runs as Saksham Anganwadi and Poshan 2.0.
28. POSHAN Abhiyaan (National Nutrition Mission) was launched on 8 March 2018 from Jhunjhunu, Rajasthan: cut stunting and underweight by 2% a year, anaemia by 3% a year.
29. PM POSHAN (Pradhan Mantri Poshan Shakti Nirman) replaced the Mid-Day Meal Scheme in September 2021 and covers Bal Vatika and Classes I-VIII. NFSA 2013 gives 5 kg per person per month (35 kg per AAY household) at Rs 3/2/1 per kg for rice/wheat/coarse grain, covering 75% rural and 50% urban population.
30. Anaemia Mukt Bharat follows a 6x6x6 strategy. 2023 was the UN International Year of Millets on India's proposal; millets were branded 'Shree Anna'. Ragi is the richest cereal source of calcium; Rajasthan is India's largest producer of bajra.

1. The chemical name of Vitamin B1 is:

(A) Riboflavin
(B) Niacin
(C) Thiamine
(D) Pyridoxine

2. Deficiency of nicotinic acid in the human diet causes:

(A) Beriberi
(B) Scurvy
(C) Pellagra
(D) Rickets

3. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Vitamin) List-II (Chemical name)

A. Vitamin B2 1. Ascorbic acid

B. Vitamin B12 2. Riboflavin

C. Vitamin C 3. Tocopherol

D. Vitamin E 4. Cyanocobalamin

(A) A-2, B-4, C-1, D-3
(B) A-2, B-1, C-4, D-3
(C) A-3, B-4, C-1, D-2
(D) A-4, B-2, C-3, D-1

4. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Vitamin) List-II (Deficiency disease)
A. Vitamin A 1. Rickets
B. Vitamin D 2. Scurvy
C. Vitamin C 3. Delayed blood clotting
D. Vitamin K 4. Night blindness

(A) A-4, B-2, C-1, D-3
(B) A-1, B-4, C-3, D-2
(C) A-4, B-1, C-2, D-3
(D) A-3, B-1, C-2, D-4

5. Which one of the following groups contains only fat-soluble vitamins?

(A) A, B, C, D
(B) A, D, E, K
(C) B, C, D, K
(D) A, C, E, K

6. Which metal is present in the Vitamin B12 molecule?

(A) Iron
(B) Magnesium
(C) Cobalt
(D) Zinc

7. The energy value of one gram of fat in the human diet is approximately:

(A) 4 kilocalories
(B) 7 kilocalories
(C) 9 kilocalories
(D) 12 kilocalories

8. Which one of the following pairs of vitamin and deficiency disease is NOT correctly matched?

(A) Thiamine - Beriberi
(B) Riboflavin - Cheilosis
(C) Folic acid - Megaloblastic anaemia
(D) Pyridoxine - Pellagra

9. Consider the following statements about kwashiorkor:
I. It results mainly from deficiency of protein while calorie intake is nearly adequate.
II. Oedema and a 'moon face' are characteristic features.
III. It usually appears in infants below six months of age.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

10. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Term) List-II (Meaning)
A. Marasmus 1. Low height for age
B. Stunting 2. Low weight for height
C. Wasting 3. Low weight for age
D. Underweight 4. Severe simultaneous deficiency of protein and calories

(A) A-4, B-1, C-2, D-3
(B) A-1, B-4, C-3, D-2
(C) A-4, B-2, C-1, D-3
(D) A-3, B-1, C-2, D-4

11. Endemic goitre in an adult population is a direct consequence of the dietary deficiency of:

(A) Iron
(B) Fluorine
(C) Iodine
(D) Calcium

12. As per the standards enforced under the National Iodine Deficiency Disorders Control Programme, iodised salt must contain iodine of not less than: HARD

(A) 15 ppm at manufacture and 30 ppm at consumer level
(B) 30 ppm at manufacture and 15 ppm at consumer level
(C) 10 ppm at both manufacture and consumer level
(D) 50 ppm at manufacture and 25 ppm at consumer level

13. Consider the following statements about fluoride in drinking water:
I. Deficiency of fluoride increases the incidence of dental caries.
II. Excess fluoride causes dental and skeletal fluorosis.
III. The permissible limit of fluoride in drinking water as per BIS is 1.5 mg per litre.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

14. Bitot's spots on the conjunctiva of the eye are a clinical sign of the deficiency of:

(A) Vitamin A
(B) Vitamin B2
(C) Vitamin C
(D) Vitamin K

15. Scurvy, characterised by bleeding gums and delayed wound healing, is caused by deficiency of:

(A) Calciferol
(B) Tocopherol
(C) Retinol
(D) Ascorbic acid

16. Deficiency of Vitamin D in adults leads to:

(A) Rickets
(B) Osteomalacia
(C) Osteoporosis due to protein loss
(D) Pernicious anaemia

17. Consider the following statements about Vitamin K:
I. It is essential for the synthesis of prothrombin in the liver.
II. It is synthesised by bacteria in the human large intestine.
III. It is a water-soluble vitamin.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

18. Consider the following statements about the Body Mass Index (BMI):
I. It is obtained by dividing body weight in kilograms by the square of height in metres.
II. As per WHO, a BMI below 18.5 indicates underweight.
III. A BMI of 30 or above is classified as obesity.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

19. How many amino acids are considered essential in the adult human diet?

(A) Eight
(B) Nine
(C) Eleven
(D) Twenty

20. The '+F' logo introduced by FSSAI on food packets indicates that the food is:

(A) Organically grown
(B) Free of trans fats
(C) Fortified with micronutrients
(D) Free of genetically modified ingredients

21. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Fortified staple) List-II (Nutrient added)

A. Salt 1. Vitamins A and D

B. Edible oil 2. Iodine

C. Rice 3. Vitamin B12, iron and folic acid

D. Double fortified salt 4. Iodine and iron HARD

(A) A-4, B-1, C-3, D-2
(B) A-2, B-3, C-1, D-4
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

22. Consider the following statements about the Food Safety and Standards Authority of India:

I. It was established under the Food Safety and Standards Act, 2006.

II. It functions under the Ministry of Health and Family Welfare.

III. The Eat Right India movement is an initiative of this authority.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

23. Which one of the following pairs of food item and its common adulterant is NOT correctly matched?

(A) Black pepper - Papaya seeds
(B) Mustard oil - Argemone oil
(C) Turmeric powder - Metanil yellow
(D) Milk - Brick powder

24. Epidemic dropsy, marked by oedema of the legs and cardiac complications, is caused by consuming mustard oil adulterated with:

(A) Argemone oil
(B) Castor oil
(C) Linseed oil
(D) Mineral oil

25. Lathyrism, a crippling paralysis of the lower limbs, results from prolonged consumption of:

(A) Horse gram
(B) Soybean
(C) Kesari dal (Lathyrus sativus)
(D) Moth bean

26. The Integrated Child Development Services (ICDS) scheme was launched in the year:

(A) 1972
(B) 1975
(C) 1985
(D) 1995

27. POSHAN Abhiyaan, the National Nutrition Mission, was launched on 8 March 2018 from which place?

(A) Jaipur
(B) Jhunjhunu
(C) Bikaner
(D) Udaipur

28. Consider the following statements about PM POSHAN:
I. It replaced the erstwhile Mid-Day Meal Scheme in 2021.
II. Its full form is Pradhan Mantri Poshan Shakti Nirman.
III. It covers children of Bal Vatika and Classes I to VIII in government and government-aided schools.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

29. Under the National Food Security Act, 2013, a person belonging to a priority household is entitled to foodgrains per month at subsidised rates of:

(A) 3 kg
(B) 5 kg
(C) 7 kg
(D) 10 kg

30. The Anaemia Mukt Bharat programme is built on a strategy described as: HARD

(A) 3x3x3
(B) 4x4x4
(C) 5x5x5
(D) 6x6x6

31. Dhanashakti, notified as India's first biofortified variety, is a high-iron variety of which crop? HARD

(A) Wheat
(B) Pearl millet
(C) Rice
(D) Maize

32. Consider the following statements about Golden Rice:
I. It has been genetically engineered to accumulate beta-carotene in the endosperm.
II. It was developed to address Vitamin A deficiency.
III. It contains an added gene that increases its iron content.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

33. In the High Temperature Short Time (HTST) method of pasteurisation, milk is heated to:

(A) 63 degrees C for 30 minutes
(B) 72 degrees C for 15 seconds
(C) 100 degrees C for 5 minutes
(D) 140 degrees C for 2 seconds

34. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Preservation technique) List-II (Description)
A. Lyophilisation 1. Use of Cobalt-60 gamma rays
B. Food irradiation 2. Freeze drying under vacuum
C. Canning 3. Brief heating followed by rapid cooling of vegetables
D. Blanching 4. Sealing in airtight containers after heating HARD

(A) A-2, B-1, C-4, D-3
(B) A-1, B-2, C-3, D-4
(C) A-2, B-4, C-1, D-3
(D) A-3, B-1, C-4, D-2

35. Which one of the following is NOT a water-soluble vitamin?

(A) Thiamine
(B) Niacin
(C) Tocopherol
(D) Folic acid

36. Which one of the following statements correctly distinguishes probiotics from prebiotics?

(A) Probiotics are live beneficial microorganisms while prebiotics are non-digestible food components that promote their growth
(B) Probiotics are non-digestible fibres while prebiotics are live bacteria
(C) Both are live microorganisms differing only in dose
(D) Probiotics are antibiotics of plant origin while prebiotics are of animal origin

37. Which one of the following is NOT a constituent of dietary fibre?

(A) Cellulose
(B) Pectin
(C) Lignin
(D) Glycogen

38. Arrange the following in the correct chronological order of their launch or enactment:

I. Integrated Child Development Services
II. National Food Security Act
III. POSHAN Abhiyaan
IV. PM POSHAN

(A) I, II, III, IV
(B) II, I, III, IV
(C) I, III, II, IV
(D) I, II, IV, III

39. Consider the following statements:

I. The United Nations observed 2023 as the International Year of Millets on a proposal moved by India.
II. Millets were given the brand name 'Shree Anna' in the Union Budget.
III. Among cereals, ragi (finger millet) is the richest source of calcium.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. FSSAI's regulation capping industrial trans fatty acids in foods relates to a limit of: HARD

(A) 1 per cent by weight
(B) 2 per cent by weight
(C) 5 per cent by weight
(D) 10 per cent by weight

ST06

Genetics: Inheritance and Variation

45 questions · 30 fact-sheet points

Must-know facts

1. Gregor Johann Mendel worked on the garden pea (Pisum sativum), took seven pairs of contrasting characters and published his paper in 1866; it was rediscovered in 1900 by de Vries, Correns and von Tschermak.
2. The term 'genetics' was coined by William Bateson and the term 'gene' by Wilhelm Johannsen. Mendel is called the Father of Genetics.
3. Mendel's laws: Law of Dominance, Law of Segregation (law of purity of gametes - the only universally valid law) and Law of Independent Assortment.
4. Monohybrid cross F2 ratio - phenotypic 3:1, genotypic 1:2:1. Dihybrid cross F2 phenotypic ratio 9:3:3:1. Test cross ratio 1:1 (monohybrid) and 1:1:1:1 (dihybrid).
5. Incomplete dominance: Mirabilis jalapa and Antirrhinum - F2 phenotypic ratio 1:2:1, same as the genotypic ratio. Codominance: AB blood group, roan cattle, MN blood group.
6. ABO blood group is a case of multiple alleles: I(A), I(B) and i - three alleles, six genotypes, four phenotypes; I(A) and I(B) are codominant and both dominant over i. The gene lies on chromosome 9.
7. AB group is the universal recipient and O group the universal donor. Rh incompatibility (Rh-negative mother, Rh-positive foetus) causes erythroblastosis foetalis; prevented by anti-D immunoglobulin.
8. Pleiotropy = one gene, many effects (phenylketonuria, sickle cell anaemia). Polygenic inheritance = many genes, one trait (human skin colour, height); wheat kernel colour studied by Kolreuter.
9. Linkage and crossing over were established by T. H. Morgan on Drosophila melanogaster; Alfred Sturtevant prepared the first genetic map using recombination frequency.
10. Sex determination: XX-XY in humans and Drosophila (male heterogametic); XX-XO in grasshoppers (male heterogametic); ZZ-ZW in birds and butterflies (FEMALE heterogametic); haplodiploidy in honeybees - drones are haploid (16) and females diploid (32).
11. In humans maleness is determined by the SRY gene on the short arm of the Y chromosome. The sperm decides the sex of the child.
12. X-linked recessive disorders: red-green colour blindness (about 8% of men, 0.4% of women), haemophilia A (factor VIII), haemophilia B or Christmas disease (factor IX), Duchenne muscular dystrophy, G6PD deficiency. Queen Victoria was a famous haemophilia carrier.
13. A colour-blind daughter is possible only if the father is colour blind AND the mother is at least a carrier - the classic criss-cross inheritance from father to grandson through the daughter.
14. Down syndrome = trisomy of chromosome 21, karyotype 47; described by Langdon Down; risk rises sharply with maternal age; features - short stature, small round head, furrowed protruding tongue, palm crease, congenital heart defects, mental retardation.
15. Other trisomies: Edwards syndrome (trisomy 18) and Patau syndrome (trisomy 13). Cri-du-chat results from deletion of the short arm of chromosome 5; the Philadelphia chromosome t(9;22) causes chronic myeloid leukaemia.
16. Klinefelter syndrome = 47, XXY - a sterile male with gynaecomastia. Turner syndrome = 45, X0 - a sterile female with short stature, webbed neck and rudimentary ovaries. Both arise from non-disjunction.
17. Number of Barr bodies = number of X chromosomes minus one (normal female 1, normal male 0, Klinefelter 1, Turner 0).
18. Sickle cell anaemia is autosomal recessive: glutamic acid is replaced by valine at the sixth position of the beta-globin chain (a missense point mutation). Thalassaemia is a quantitative defect - reduced

globin chain synthesis.

19. Humans have 46 chromosomes - 22 pairs of autosomes plus one pair of sex chromosomes. A karyotype arranges them by size and centromere position (Denver system, groups A to G).
20. DNA double helix - Watson and Crick, 1953, using Rosalind Franklin's and Maurice Wilkins' X-ray data; Nobel 1962. B-DNA is right handed, 10 base pairs per turn, pitch 3.4 nm, rise 0.34 nm per base pair. Chargaff's rule: A = T, G = C; A-T has two and G-C three hydrogen bonds.
21. Griffith (1928) discovered transformation; Avery, MacLeod and McCarty (1944) identified DNA as the transforming principle; Hershey and Chase (1952) confirmed it with bacteriophage; Meselson and Stahl (1958) proved semi-conservative replication in E. coli.
22. Replication is semi-conservative and always 5' to 3'; the lagging strand is made in Okazaki fragments joined by DNA ligase. Central dogma (Crick): DNA -> RNA -> protein; reverse transcription was found by Temin and Baltimore.
23. The genetic code has 64 codons - 61 sense codons and 3 stop codons (UAA, UAG, UGA); AUG is the start codon and codes for methionine. The code is degenerate, non-overlapping, commaless and nearly universal; methionine (AUG) and tryptophan (UGG) have a single codon each.
24. The genetic code was deciphered by Nirenberg, Har Gobind Khorana and Holley (Nobel 1968). The lac operon (inducible) was explained by Jacob and Monod.
25. Mutation types: point/substitution (transition, transversion) and frameshift (insertion or deletion); silent, missense and nonsense. Physical mutagens - UV, X-rays, gamma rays; chemical - EMS, nitrous acid, 5-bromouracil, acridine dyes.
26. Hugo de Vries proposed the mutation theory from work on Oenothera lamarckiana and used the term 'saltation'. H. J. Muller showed X-rays induce mutations in Drosophila.
27. Hardy-Weinberg principle: p + q = 1 and p^2 + 2pq + q^2 = 1. Equilibrium requires a large random-mating population with no mutation, no migration, no genetic drift and no selection. Founder effect and bottleneck are forms of genetic drift.
28. Darwin's 'On the Origin of Species' (1859) proposed natural selection after the HMS Beagle voyage and study of Galapagos finches (adaptive radiation); Wallace independently reached the same idea. Lamarck proposed inheritance of acquired characters.
29. Industrial melanism in the peppered moth Biston betularia is the classic example of directional natural selection. Homologous organs indicate divergent evolution; analogous organs indicate convergent evolution.
30. DNA fingerprinting was developed by Alec Jeffreys (1984) and is based on VNTR/STR repeats; Lalji Singh of CCMB Hyderabad is called the Father of DNA fingerprinting in India. The Human Genome Project ran from 1990 to 2003 and found about 20,000-25,000 genes in roughly 3.2 billion base pairs.

1. Mendel selected which plant for his classical experiments on inheritance?

(A) Oenothera lamarckiana (B) Pisum sativum
(C) Antirrhinum majus (D) Drosophila melanogaster

2. How many pairs of contrasting characters of the garden pea did Mendel study?

(A) Five (B) Six
(C) Seven (D) Nine

3. Which of Mendel's laws is regarded as universally applicable, without exception?

(A) Law of dominance (B) Law of segregation
(C) Law of independent assortment (D) Law of unit characters

4. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Cross) List-II (Expected ratio)
A. F2 phenotypic ratio of a monohybrid cross 1. 1:1:1:1
B. F2 genotypic ratio of a monohybrid cross 2. 9:3:3:1
C. F2 phenotypic ratio of a dihybrid cross 3. 3:1
D. Dihybrid test cross ratio 4. 1:2:1

(A) A-3, B-4, C-2, D-1
(B) A-3, B-2, C-4, D-1
(C) A-4, B-3, C-2, D-1
(D) A-1, B-4, C-2, D-3

5. In a typical dihybrid cross involving two independently assorting genes, the F2 phenotypic ratio is:

(A) 3:1
(B) 1:2:1
(C) 9:3:3:1
(D) 1:1:1:1

6. A test cross in genetics involves crossing an individual of unknown genotype with an individual that is:

(A) Homozygous dominant
(B) Heterozygous
(C) Homozygous recessive
(D) Of the F1 generation

7. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Phenomenon) List-II (Example)
A. Incomplete dominance 1. Human skin colour
B. Codominance 2. Pink flowers in Antirrhinum
C. Pleiotropy 3. AB blood group
D. Polygenic inheritance 4. Phenylketonuria

(A) A-3, B-2, C-1, D-4
(B) A-2, B-4, C-3, D-1
(C) A-2, B-3, C-4, D-1
(D) A-1, B-3, C-4, D-2

8. The AB blood group in humans is the best-known example of: RAS PRE 2023

(A) Incomplete dominance
(B) Codominance
(C) Epistasis
(D) Polygenic inheritance

9. Consider the following statements about the inheritance of ABO blood groups in humans:
I. The system is controlled by three alleles of a single gene.
II. There are six possible genotypes and four phenotypes.
III. A child of blood group O can be born to parents who are both of blood group A.
Which of the statements given above are correct? RAS PRE 2023

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

10. A man of blood group AB marries a woman of blood group O. Which blood groups can their children possibly have? RAS PRE 2021

(A) A or B only
(B) AB or O only
(C) A, B, AB or O
(D) O only

11. Erythroblastosis foetalis in a newborn arises when: RAS PRE 2023

(A) An Rh-positive mother carries an Rh-negative foetus
(B) Both mother and foetus are Rh-negative
(C) An Rh-negative mother carries an Rh-positive foetus
(D) The mother is of blood group AB

12. The condition in which a single gene influences several apparently unrelated phenotypic traits is called:

(A) Polygeny
(B) Pleiotropy
(C) Epistasis
(D) Codominance

13. Human skin colour and height are examples of:

(A) Cytoplasmic inheritance
(B) Multiple allelism
(C) Sex-linked inheritance
(D) Polygenic inheritance

14. Linkage and crossing over in Drosophila melanogaster were established by:

(A) Hugo de Vries
(B) William Bateson
(C) T. H. Morgan
(D) Wilhelm Johannsen

15. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Organism) List-II (Mechanism of sex determination)

A. Human being 1. ZZ-ZW

B. Grasshopper 2. Haplodiploidy

C. Bird 3. XX-XY

D. Honeybee 4. XX-XO

(A) A-3, B-1, C-4, D-2
(B) A-3, B-4, C-1, D-2
(C) A-4, B-3, C-2, D-1
(D) A-1, B-4, C-3, D-2

16. In which one of the following is the female the heterogametic sex?

(A) Human being
(B) Drosophila
(C) Domestic fowl
(D) Grasshopper

17. Which one of the following statements regarding sex determination in human beings is NOT correct?

(A) The mother contributes the Y chromosome to a male child
(B) The sperm determines the sex of the child
(C) Females are homogametic and produce only one type of gamete
(D) Maleness is determined by the SRY gene present on the Y chromosome

18. Consider the following statements about red-green colour blindness in human beings:

I. It is an X-linked recessive condition.

II. It is far more frequent in males than in females.

III. A colour-blind man passes the gene to all his sons.

Which of the statements given above are correct? RAS PRE 2021

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

19. Haemophilia A in human beings results from the deficiency of: HARD

(A) Factor VIII (B) Factor IX
(C) Factor XI (D) Fibrinogen

20. A woman who is a carrier of haemophilia marries a normal man. What proportion of their sons is expected to be haemophilic? RAS PRE 2021

(A) None (B) One-fourth
(C) One-half (D) All

21. Which one of the following disorders is NOT inherited as an X-linked recessive trait?

(A) Sickle cell anaemia (B) Red-green colour blindness
(C) Duchenne muscular dystrophy (D) Haemophilia

22. Down syndrome in human beings is caused by: RAS PRE 2018, 2021

(A) Trisomy of chromosome 18 (B) Monosomy of the X chromosome
(C) Trisomy of chromosome 21 (D) An extra X chromosome in males

23. The total number of chromosomes in a person affected with Down syndrome is: RAS PRE 2018, 2021

(A) 45 (B) 46
(C) 47 (D) 48

24. Consider the following statements about Down syndrome:
I. It was first described by Langdon Down.
II. Its incidence increases with advancing maternal age.
III. Affected individuals characteristically have a small round head, a furrowed protruding tongue and a single palmar crease.
Which of the statements given above are correct? RAS PRE 2018, 2021

(A) I and II only
(B) I, II and III
(C) I and III only
(D) II and III only

25. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Syndrome) List-II (Chromosomal constitution)
A. Klinefelter 1. 45, X0
B. Turner 2. 47, XXY
C. Down 3. Trisomy 18
D. Edwards 4. Trisomy 21

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-2, B-1, C-3, D-4
(D) A-4, B-1, C-2, D-3

26. The number of Barr bodies in the somatic cell of a person with Klinefelter syndrome (47, XXY) is: HARD

(A) Zero (B) One
(C) Two (D) Three

27. Consider the following statements about sickle cell anaemia:
I. It is inherited as an autosomal recessive disorder.
II. Glutamic acid is replaced by valine at the sixth position of the beta-globin chain.
III. It is caused by a frameshift mutation.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

28. Which one of the following pairs relating to chromosomal abnormalities is NOT correctly matched?

(A) Klinefelter syndrome - 45, X0
(B) Philadelphia chromosome - Chronic myeloid leukaemia
(C) Patau syndrome - Trisomy 13
(D) Cri-du-chat syndrome - Deletion in chromosome 5

29. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Discovery) List-II (Scientist)
A. Double-helical model of DNA 1. Griffith
B. Bacterial transformation 2. Hershey and Chase
C. DNA as the genetic material of bacteriophage 3. Hugo de Vries
D. Mutation theory of evolution 4. Watson and Crick

(A) A-1, B-4, C-2, D-3
(B) A-4, B-2, C-1, D-3
(C) A-4, B-1, C-2, D-3
(D) A-3, B-1, C-2, D-4

30. Consider the following statements about the B-form of the DNA double helix:
I. The two strands are antiparallel and the helix is right handed.
II. Adenine pairs with thymine through three hydrogen bonds.
III. There are about ten base pairs in one complete turn of the helix.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

31. Consider the following statements about DNA replication:
I. It is semi-conservative, as demonstrated by Meselson and Stahl in Escherichia coli.
II. DNA polymerase synthesises the new strand only in the 5' to 3' direction.
III. The lagging strand is synthesised as short pieces called Okazaki fragments.
Which of the statements given above are correct?

(A) I and II only
(B) I, II and III
(C) I and III only
(D) II and III only

32. Arrange the following genetic experiments in correct chronological order:
I. Griffith's transformation experiment
II. Avery, MacLeod and McCarty's identification of the transforming principle
III. Hershey and Chase's bacteriophage experiment
IV. Meselson and Stahl's replication experiment HARD

(A) II, I, III, IV
(B) I, II, III, IV
(C) I, III, II, IV
(D) III, I, II, IV

33. Out of the 64 codons of the genetic code, how many are stop or termination codons?

(A) One
(B) Two
(C) Three
(D) Four

34. Which of the following amino acids is coded by only one codon in the genetic code? HARD

(A) Tryptophan
(B) Serine
(C) Leucine
(D) Arginine

35. Which one of the following is NOT a property of the genetic code?

(A) It is ambiguous
(B) It is non-overlapping
(C) It is commaless
(D) It is degenerate

36. The addition or deletion of a single nucleotide in a gene, which shifts the reading of all subsequent codons, is known as:

(A) Frameshift mutation
(B) Transversion mutation
(C) Transition mutation
(D) Silent mutation

37. Hugo de Vries proposed the mutation theory of evolution on the basis of his work on:

(A) Pisum sativum
(B) Oenothera lamarckiana
(C) Drosophila melanogaster
(D) Antirrhinum majus

38. Consider the following statements about the Hardy-Weinberg principle:
I. It states that allele frequencies in a population remain constant from generation to generation in the absence of disturbing factors.
II. It is expressed as p squared plus 2pq plus q squared equals one.
III. Genetic drift and natural selection are conditions necessary for the equilibrium to hold.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

39. Industrial melanism observed in the peppered moth, Biston betularia, is a classical illustration of: HARD

(A) Stabilising selection
(B) Directional selection
(C) Disruptive selection
(D) Genetic drift

40. Assertion (A): The forelimbs of a whale, a bat and a human being have a similar skeletal plan though they perform different functions.
Reason (R): These are homologous organs and indicate divergent evolution from a common ancestor.
Select the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

41. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Scientist) List-II (Contribution)
A. William Bateson 1. Coined the term 'gene'
B. Wilhelm Johannsen 2. Coined the term 'genetics'
C. Alec Jeffreys 3. Developed DNA fingerprinting
D. Har Gobind Khorana 4. Deciphering of the genetic code

(A) A-1, B-2, C-4, D-3
(B) A-2, B-1, C-3, D-4
(C) A-2, B-3, C-1, D-4
(D) A-2, B-1, C-4, D-3

42. DNA fingerprinting as a technique of identification is based primarily on differences in: HARD

(A) The total quantity of DNA in a cell
(B) Repetitive DNA sequences such as VNTRs
(C) The number of chromosomes
(D) The proportion of coding genes

43. Consider the following statements about the Human Genome Project:
I. It was formally launched in 1990 and declared complete in 2003.
II. The human genome contains about 3.2 billion base pairs.
III. It identified about one lakh protein-coding genes in the human genome.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

44. Which one of the following statements about the Y chromosome in human beings is NOT correct?

(A) It is smaller than the X chromosome
(B) It is present in one copy in normal females
(C) It is present in one copy in normal males
(D) It carries the SRY gene

45. Which one of the following techniques is used for prenatal detection of chromosomal disorders such as Down syndrome?

(A) Amniocentesis
(B) Electrophoresis
(C) Electrocardiography
(D) Chromatography

ST07

Biotechnology and Genetic Engineering

45 questions · 30 fact-sheet points

Must-know facts

1. Recombinant DNA technology was developed by Stanley Cohen and Herbert Boyer (1972-73); Paul Berg, who made the first recombinant DNA molecule, received the Nobel Prize in Chemistry in 1980.
2. Restriction endonucleases are the 'molecular scissors' and DNA ligase the 'molecular glue'. The first restriction enzyme isolated was Hind II. Type II enzymes cut palindromic sequences and are used in genetic engineering.
3. EcoRI is named from Escherichia coli strain RY13; it cuts GAATTC between G and A, leaving sticky (cohesive) ends that allow fragments from different sources to join.
4. Cloning vectors need an origin of replication (ori), a selectable marker and unique restriction sites. pBR322 was the first widely used artificial plasmid vector; others are pUC19, bacteriophage lambda, cosmids, BACs and YACs.
5. The Ti (tumour-inducing) plasmid of Agrobacterium tumefaciens, which causes crown gall disease, is the natural genetic engineer of plants and is the commonest vector for dicot transformation.
6. Gene transfer methods: microinjection (animal cells), biolistics or gene gun (plant cells), electroporation, treatment with divalent calcium to make competent bacterial cells, liposome-mediated transfer and Agrobacterium-mediated transfer.
7. PCR was invented by Kary Mullis (Nobel Chemistry 1993). Its three steps are denaturation (about 94-95 C), annealing (about 50-55 C) and extension (about 72 C) using thermostable Taq polymerase from Thermus aquaticus. RT-PCR is the gold standard for RNA viruses including SARS-CoV-2.
8. In agarose gel electrophoresis DNA, being negatively charged, moves towards the anode; bands are stained with ethidium bromide and seen under UV light. Southern blot = DNA, Northern = RNA, Western = protein.
9. The first commercial recombinant DNA product was human insulin (Humulin, 1982-83): the A chain (21 amino acids) and B chain (30 amino acids) were made separately in E. coli and joined by disulphide bonds.
10. CRISPR-Cas9 stands for Clustered Regularly Interspaced Short Palindromic Repeats with CRISPR-associated protein 9; it is derived from the bacterial defence system against viruses and is guided to its target by a guide RNA. Emmanuelle Charpentier and Jennifer Doudna won the 2020 Nobel Prize in Chemistry for it.
11. FELUDA, a CRISPR-Cas9 based paper-strip test for COVID-19, was developed by CSIR-IGIB, New Delhi. Casgevy became the first approved CRISPR gene-editing therapy (for sickle cell disease and beta thalassaemia).
12. India released its first genome-edited rice varieties in May 2025: DRR Dhan 100 'Kamala' (ICAR-IIRR) and Pusa DST Rice 1 (ICAR-IARI), both made using SDN-1/SDN-2 editing, which is exempted from GMO rules.
13. Bt cotton carries cry genes (cry1Ac, cry2Ab) from Bacillus thuringiensis against bollworms. The Bt protein is an inactive protoxin crystal that becomes active in the alkaline gut of the insect. Bollgard-I Bt cotton was approved in India in March 2002 and remains the ONLY GM crop approved for commercial cultivation in India.
14. Bt brinjal was developed by Mahyco using cry1Ac; GEAC cleared it in 2009 but a moratorium was imposed in February 2010 by the then Environment Minister Jairam Ramesh. Bangladesh grows it commercially.
15. GM mustard DMH-11 (Dhara Mustard Hybrid-11) was developed by Deepak Pental's team at the University of Delhi using the barnase-barstar-bar gene system; GEAC recommended environmental

release in October 2022, but the Supreme Court delivered a split verdict in July 2024 and the matter went to a larger bench.

16. Golden Rice is engineered to make beta-carotene (provitamin A) in the endosperm; Flavr Savr tomato (1994) was the first genetically modified food crop to be commercialised.
17. Plant tissue culture rests on totipotency; G. Haberlandt is the Father of plant tissue culture. Explant -> callus -> plantlet on Murashige and Skoog medium; a high auxin-to-cytokinin ratio induces roots and a high cytokinin ratio induces shoots. Meristem/shoot-tip culture yields virus-free plants; anther culture yields haploids; protoplast fusion produced the 'pomato'.
18. Biofertilisers: Rhizobium (symbiotic in legume nodules), Azotobacter (free-living aerobic), Azospirillum (associative), Anabaena and Nostoc (cyanobacteria), Azolla with Anabaena azollae in paddy, Frankia with Casuarina, and mycorrhiza/VAM for phosphorus uptake. Nitrogenase fixes nitrogen and leghaemoglobin protects it from oxygen.
19. Biopesticides: Bacillus thuringiensis, Trichoderma, Nucleopolyhedrovirus (NPV, highly species specific), Trichogramma, Beauveria bassiana and neem-based azadirachtin.
20. Bioremediation: Pseudomonas putida, the oil-eating 'superbug' engineered by Ananda Mohan Chakrabarty, became the first living organism to be patented (Diamond v. Chakrabarty, 1980).
21. Fermentation products: ethanol - Saccharomyces cerevisiae; citric acid - Aspergillus niger; acetic acid - Acetobacter aceti; penicillin - Penicillium notatum/chrysogenum (Fleming, Chain and Florey, Nobel 1945); cyclosporin A (immunosuppressant) - Trichoderma polysporum; statins (cholesterol lowering) - Monascus purpureus; streptokinase (clot buster) - Streptococcus. Swiss cheese holes come from Propionibacterium shermanii.
22. Biofuel generations: 1G from edible crops (sugarcane, molasses, corn, vegetable oil); 2G from non-food lignocellulosic biomass and crop residues; 3G from algae; 4G from genetically or metabolically engineered algae and microbes designed for carbon capture, aiming at carbon-negative fuel.
23. The National Policy on Biofuels, 2018 (amended 2022) advanced the 20% ethanol blending target from 2030 to 2025-26; India achieved 20% blending in 2025. India's first 2G ethanol bio-refinery, using paddy straw, is at Panipat, Haryana. Pradhan Mantri JI-VAN Yojana supports 2G ethanol and SATAT promotes compressed biogas.
24. Dolly the sheep (born 5 July 1996, Roslin Institute, Ian Wilmut and Keith Campbell) was the first mammal cloned from an adult somatic cell by somatic cell nuclear transfer. India's cloned buffalo work was done at NDRI, Karnal.
25. Stem cells: totipotent (zygote), pluripotent (embryonic stem cells) and multipotent (adult stem cells). Shinya Yamanaka won the 2012 Nobel Prize with John Gurdon for induced pluripotent stem cells.
26. Gene therapy was first attempted in 1990 on a child with adenosine deaminase (ADA) deficiency. RNA interference, used for pest resistance in plants, earned Andrew Fire and Craig Mello the 2006 Nobel Prize. Monoclonal antibodies are made by hybridoma technology (Kohler and Milstein, Nobel 1984) and carry the suffix '-mab'.
27. COVID-19 vaccines: Covaxin (BBV152) is a whole-virion inactivated vaccine of Bharat Biotech with ICMR-NIV Pune; Covishield is the Oxford-AstraZeneca chimpanzee-adenovirus vector vaccine made by the Serum Institute of India; ZyCoV-D of Zydus is the world's first DNA plasmid vaccine approved for human use, given in three needle-free doses; GEMCOVAC-19 of Gennova is India's first indigenous mRNA vaccine; Corbevax is a recombinant protein subunit vaccine.
28. Regulation of GMOs in India runs under the 1989 Rules of the Environment (Protection) Act, 1986 through RDAC, IBSC, RCGM, GEAC, SBCC and DLC. The Genetic Engineering Appraisal Committee (GEAC), under the Ministry of Environment, Forest and Climate Change, is the apex body for approving environmental release of GMOs.
29. The Biological Diversity Act, 2002 created the National Biodiversity Authority (Chennai), State Biodiversity Boards and Biodiversity Management Committees that prepare People's Biodiversity Registers; the Biological Diversity (Amendment) Act, 2023 decriminalised offences and exempted registered AYUSH practitioners from benefit sharing. India is a party to the Cartagena Protocol on biosafety and the Nagoya Protocol on access and benefit sharing.

30. Institutional framework: Department of Biotechnology (1986) under the Ministry of Science and Technology; BIRAC (2012) for start-ups; the BioE3 Policy - Biotechnology for Economy, Environment and Employment - approved by the Union Cabinet in August 2024 to promote high-performance biomanufacturing and biofoundries.

1. The restriction enzyme EcoRI derives the letter 'R' in its name from:

(A) The bacterial strain
(B) The species name
(C) The genus name
(D) The order of discovery

2. Consider the following statements about restriction enzymes:

I. They recognise palindromic nucleotide sequences.

II. Type II restriction enzymes are the ones used in genetic engineering.

III. They join two DNA fragments to form recombinant DNA.

Which of the statements given above are correct?

(A) II and III only
(B) I and II only
(C) I and III only
(D) I, II and III

3. Which one of the following is NOT an essential requirement of a cloning vector?

(A) Origin of replication
(B) Ability to code for a human protein
(C) Recognition site for a restriction enzyme
(D) Selectable marker

4. The Ti plasmid used widely for producing transgenic plants is obtained from:

(A) Agrobacterium tumefaciens
(B) Bacillus thuringiensis
(C) Escherichia coli
(D) Thermus aquaticus

5. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Technique/Tool) List-II (Function)

A. DNA ligase 1. Amplification of a DNA segment

B. Polymerase chain reaction 2. Joining of DNA fragments

C. Gel electrophoresis 3. Introducing DNA into plant cells

D. Gene gun 4. Separation of DNA fragments by size

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-2, B-4, C-1, D-3
(D) A-3, B-1, C-4, D-2

6. Consider the following statements about the polymerase chain reaction:

I. It was invented by Kary Mullis, who received the Nobel Prize for it.

II. It uses the thermostable Taq polymerase obtained from Thermus aquaticus.

III. Reverse transcription PCR is used for detecting RNA viruses such as SARS-CoV-2.

Which of the statements given above are correct?

(A) I and II only
(B) I, II and III
(C) I and III only
(D) II and III only

7. Arrange the three steps of one cycle of the polymerase chain reaction in the correct order:
I. Annealing of primers
II. Denaturation of the DNA template
III. Extension by DNA polymerase

(A) II, III, I
(B) I, II, III
(C) II, I, III
(D) III, II, I

8. During agarose gel electrophoresis of DNA, the fragments move towards the:

(A) Anode, because DNA is negatively charged
(B) Cathode, because DNA is positively charged
(C) Cathode, because DNA is neutral
(D) Anode, because DNA is positively charged

9. The first human protein produced commercially by recombinant DNA technology was:

(A) Insulin
(B) Interferon
(C) Human growth hormone
(D) Erythropoietin

10. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Technique) List-II (Principal use)
A. CRISPR-Cas9 1. Production of monoclonal antibodies
B. Hybridoma technology 2. Silencing of a specific messenger RNA
C. Somatic cell nuclear transfer 3. Precise editing of a target gene
D. RNA interference 4. Cloning of an animal such as Dolly

(A) A-1, B-3, C-4, D-2
(B) A-3, B-4, C-1, D-2
(C) A-2, B-1, C-4, D-3
(D) A-3, B-1, C-4, D-2

11. FELUDA, a paper-strip diagnostic test developed in India for COVID-19, is based on which technology? HARD

(A) CRISPR-Cas9
(B) ELISA
(C) Monoclonal antibodies
(D) Mass spectrometry

12. India's first genome-edited rice varieties, announced in 2025, are: HARD

(A) DRR Dhan 100 (Kamala) and Pusa DST Rice 1
(B) Swarna Sub-1 and Samba Mahsuri
(C) Pusa Basmati 1121 and IR-64
(D) Golden Rice 2 and Pusa Sugandh

13. The cry genes introduced into Bt cotton are obtained from:

(A) Bacillus thuringiensis
(B) Agrobacterium tumefaciens
(C) Pseudomonas putida
(D) Escherichia coli

14. Consider the following statements about the Bt toxin:
I. In the bacterium it exists as an inactive protoxin in crystal form.
II. It is activated by the alkaline pH of the insect gut.
III. It kills the insect by creating pores in the midgut epithelial cells.
Which of the statements given above are correct?

(A) I and II only
(B) I, II and III
(C) I and III only
(D) II and III only

15. Which is the only genetically modified crop approved for commercial cultivation in India so far?

(A) Bt cotton
(B) Bt brinjal
(C) GM mustard
(D) Golden Rice

16. The moratorium on the commercial release of Bt brinjal in India was imposed in the year:

(A) 2002
(B) 2006
(C) 2010
(D) 2016

17. Consider the following statements about GM mustard DMH-11:

I. It was developed at the University of Delhi using the barnase-barstar gene system.

II. The barnase gene induces male sterility and the barstar gene restores fertility.

III. It has been approved for commercial cultivation across India.

Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

18. Which one of the following was the first genetically modified food crop to be commercialised in the world?

(A) Bt cotton
(B) Bt maize
(C) Golden Rice
(D) Flavr Savr tomato

19. Consider the following statements about Dolly the sheep:

I. She was the first mammal cloned from an adult somatic cell.

II. She was produced by the technique of somatic cell nuclear transfer.

III. She was created at the Roslin Institute in Scotland.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

20. The ability of a single plant cell to regenerate into a whole plant is called:

(A) Micropropagation
(B) Pluripotency
(C) Multipotency
(D) Totipotency

21. Consider the following statements about plant tissue culture:

I. The undifferentiated mass of cells produced from an explant is called a callus.

II. A high cytokinin to auxin ratio in the medium promotes shoot formation.

III. Meristem or shoot-tip culture is used to raise virus-free plants.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

22. Which one of the following pairs of technique and its outcome is NOT correctly matched?

(A) Anther culture - Production of haploid plants
(B) Callus culture - Production of transgenic animals
(C) Meristem culture - Virus-free plants
(D) Protoplast fusion - Somatic hybridisation

23. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Biofertiliser) List-II (Nature/association)
A. Rhizobium 1. Free-living aerobic nitrogen fixer
B. Azotobacter 2. Symbiotic with legume root nodules
C. Anabaena azollae 3. Improves phosphorus uptake
D. Mycorrhiza 4. Symbiont within the Azolla fern

(A) A-3, B-1, C-4, D-2
(B) A-1, B-2, C-3, D-4
(C) A-2, B-4, C-1, D-3
(D) A-2, B-1, C-4, D-3

24. The 'superbug' Pseudomonas putida, developed by Ananda Mohan Chakrabarty, is significant because it:

(A) Fixes atmospheric nitrogen in cereals
(B) Causes crown gall disease in plants
(C) Produces human insulin
(D) Digests hydrocarbons of oil spills and was the first patented living organism

25. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Product) List-II (Source organism)
A. Citric acid 1. Trichoderma polysporum
B. Penicillin 2. Aspergillus niger
C. Cyclosporin A 3. Monascus purpureus
D. Statins 4. Penicillium chrysogenum `HARD`

(A) A-4, B-2, C-3, D-1
(B) A-2, B-1, C-4, D-3
(C) A-3, B-4, C-1, D-2
(D) A-2, B-4, C-1, D-3

26. Which one of the following pairs of micro-organism and product is NOT correctly matched?

(A) Saccharomyces cerevisiae - Ethanol production
(B) Acetobacter aceti - Citric acid
(C) Streptococcus - Streptokinase
(D) Lactobacillus - Curd formation

27. First-generation biofuels are those produced from: `RAS PRE 2013, 2024`

(A) Edible food crops such as sugarcane, corn and vegetable oils
(B) Agricultural residues and lignocellulosic biomass
(C) Algal biomass
(D) Genetically engineered microorganisms designed to capture carbon dioxide

28. Biofuels produced from algal biomass are classified as: `RAS PRE 2013, 2024`

(A) First-generation biofuels
(B) Second-generation biofuels
(C) Fossil fuels
(D) Third-generation biofuels

29. Fourth-generation biofuels are distinguished from earlier generations mainly by the use of: `RAS PRE 2013, 2024` `HARD`

(A) Sugarcane molasses as feedstock
(B) Rice straw and bagasse as feedstock
(C) Blending of ethanol with petrol
(D) Genetically or metabolically engineered algae and microbes aimed at capturing carbon dioxide

30. Consider the following statements about biofuels in India:
I. India's first second-generation ethanol bio-refinery, based on paddy straw, is located at Panipat, Haryana.
II. The National Policy on Biofuels was amended to advance the 20 per cent ethanol blending target to 2025-26.
III. Pradhan Mantri JI-VAN Yojana provides financial support for second-generation ethanol projects.
Which of the statements given above are correct? RAS PRE 2024

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

31. Induced pluripotent stem cells (iPSCs) are:

(A) Stem cells obtained from umbilical cord blood
(B) Stem cells obtained from the inner cell mass of the blastocyst
(C) Cancer cells cultured indefinitely
(D) Adult somatic cells reprogrammed to an embryonic-like state

32. The first successful attempt at human gene therapy, carried out in 1990, was for a patient suffering from:

(A) Cystic fibrosis
(B) Haemophilia A
(C) Sickle cell anaemia
(D) Adenosine deaminase deficiency

33. Which one of the following COVID-19 vaccines is a whole-virion inactivated vaccine developed indigenously in India?

(A) Covishield
(B) Sputnik V
(C) ZyCoV-D
(D) Covaxin

34. Covishield, manufactured in India by the Serum Institute of India, is based on which vaccine platform?

(A) mRNA
(B) Protein subunit
(C) DNA plasmid
(D) Recombinant viral vector

35. Consider the following statements about the vaccine ZyCoV-D:
I. It is the world's first DNA plasmid vaccine approved for human use.
II. It is administered through a needle-free applicator.
III. It is a three-dose vaccine.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

36. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Vaccine) List-II (Platform/Developer)
A. Covaxin 1. DNA plasmid vaccine, Zydus
B. Covishield 2. Inactivated whole virion, Bharat Biotech
C. ZyCoV-D 3. Indigenous mRNA vaccine, Gennova
D. GEMCOVAC-19 4. Adenoviral vector, Serum Institute of India

(A) A-1, B-4, C-2, D-3
(B) A-2, B-1, C-4, D-3
(C) A-4, B-2, C-1, D-3
(D) A-2, B-4, C-1, D-3

37. Monoclonal antibodies are produced by a technique known as:

(A) Gene silencing
(B) Somatic hybridisation
(C) Protoplast culture
(D) Hybridoma technology

38. Consider the following statements about the regulation of genetically modified organisms in India:
I. The regulatory framework operates under rules notified in 1989 under the Environment (Protection) Act, 1986.
II. The Institutional Biosafety Committee functions at the level of the research institution.
III. The Genetic Engineering Appraisal Committee functions under the Department of Biotechnology.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

39. The National Biodiversity Authority set up under the Biological Diversity Act, 2002 has its headquarters at:

(A) New Delhi
(B) Hyderabad
(C) Bengaluru
(D) Chennai

40. Which one of the following statements about the Biological Diversity (Amendment) Act, 2023 is NOT correct? HARD

(A) It exempts registered AYUSH practitioners from sharing benefits for the use of codified traditional knowledge
(B) It encourages cultivation of medicinal plants
(C) It abolished the National Biodiversity Authority
(D) It decriminalises offences and replaces imprisonment with monetary penalties

41. Which one of the following pairs of instrument or body and its subject is NOT correctly matched?

(A) Cartagena Protocol - Transboundary movement of living modified organisms
(B) Nagoya Protocol - Access to genetic resources and benefit sharing
(C) Protection of Plant Varieties and Farmers' Rights Act, 2001 - Regulation of biofuel blending
(D) Genetic Engineering Appraisal Committee - Approval of environmental release of genetically modified organisms

42. The BioE3 Policy approved by the Union Cabinet in 2024 stands for Biotechnology for: HARD

(A) Energy, Environment and Employment
(B) Environment, Education and Employment
(C) Ecology, Economy and Education
(D) Economy, Environment and Employment

43. Arrange the following in the correct chronological order of their occurrence:
I. Approval of Bt cotton for commercial cultivation in India
II. Birth of Dolly the sheep
III. Award of the Nobel Prize for CRISPR-Cas9 genome editing
IV. GEAC recommendation for environmental release of GM mustard DMH-11

(A) I, II, IV, III
(B) I, II, III, IV
(C) II, I, IV, III
(D) II, I, III, IV

44. BIRAC, which supports biotechnology start-ups in India, functions under the:

(A) Indian Council of Medical Research
(B) Ministry of Environment, Forest and Climate Change
(C) Council of Scientific and Industrial Research
(D) Department of Biotechnology

45. RNA interference, used to develop nematode-resistant transgenic tobacco, works by:

(A) Cutting the target DNA at a specific site
(B) Inserting an extra copy of the gene
(C) Increasing the rate of transcription of a gene
(D) Silencing a specific messenger RNA so that the protein is not translated

ST08

Nanotechnology and New Materials

35 questions · 30 fact-sheet points

Must-know facts

1. Nano comes from Greek 'nanos' (dwarf); 1 nanometre = 10^-9 metre = one-billionth of a metre = 10 angstrom.
2. Nanotechnology deals with matter having at least one dimension in the 1-100 nm range — the definition RPSC has asked repeatedly.
3. Richard Feynman's lecture 'There's Plenty of Room at the Bottom' (29 December 1959, Caltech, American Physical Society) is the conceptual origin of nanotechnology.
4. The word 'nanotechnology' was coined by Norio Taniguchi (Japan) in 1974; K. Eric Drexler popularised it in 'Engines of Creation' (1986).
5. Two nanoscale effects: huge surface-area-to-volume ratio (high reactivity) and quantum confinement (size-dependent optical/electronic behaviour).
6. Confinement: quantum well = confined in 1 direction (carriers free in 2-D); quantum wire = confined in 2 (free in 1-D); quantum dot = confined in all 3 (0-D, an 'artificial atom').
7. Quantum dots are semiconductor nanocrystals of about 2-10 nm; emission colour is size-tunable. Nobel Prize in Chemistry 2023 — Bawendi, Brus and Ekimov.
8. Fullerene C60 (buckminsterfullerene, 'bucky ball') — discovered 1985 by Kroto, Curl and Smalley (Nobel Chemistry 1996); 60 carbon atoms in 20 hexagons and 12 pentagons.
9. Carbon nanotubes were reported by Sumio Iijima (NEC, Japan) in 1991; single-walled and multi-walled; armchair CNTs are metallic, zigzag/chiral ones are usually semiconducting.
10. CNTs are about 100 times stronger than steel at roughly one-sixth the weight and are excellent thermal and electrical conductors.
11. Graphene is a one-atom-thick sheet of sp2 carbon in a honeycomb lattice, isolated in 2004 by Andre Geim and Konstantin Novoselov (Manchester) by the adhesive-tape method — Nobel Physics 2010.
12. Carbon allotropes by dimensionality: fullerene 0-D, carbon nanotube 1-D, graphene 2-D, diamond and graphite 3-D bulk.
13. Generations of nanotechnology (M.C. Roco): 1st passive nanostructures, 2nd active nanostructures, 3rd systems of nanosystems, 4th molecular nanosystems.
14. Synthesis: top-down = ball milling, lithography, etching, laser ablation; bottom-up = sol-gel, chemical vapour deposition, self-assembly, molecular beam epitaxy.
15. Characterisation: SEM and TEM (electron microscopy), AFM and STM (scanning probe), XRD for crystal structure, DLS for particle size in suspension.
16. The Scanning Tunnelling Microscope was built in 1981 by Gerd Binnig and Heinrich Rohrer at IBM Zurich (Nobel Physics 1986); the Atomic Force Microscope followed in 1986.
17. Dendrimers are repetitively branched, tree-like macromolecules used as drug-delivery carriers; liposomes and nanoshells are other nanocarriers.
18. Nano-silver is antimicrobial; nano-TiO2 and ZnO are UV blockers in sunscreen; nano-clay is used in packaging; carbon black in tyres is a legacy nanomaterial.
19. Gold nanoparticles scatter and absorb light to give a ruby-red colour — seen in the Roman Lycurgus Cup and studied by Michael Faraday in 1857.
20. India's Nano Mission was launched in May 2007 under the Department of Science and Technology; it followed the Nano Science and Technology Initiative (2001), concluded on 31 March 2017 and continues as the National Programme on Nano Science and Technology.

21. Institute of Nano Science and Technology (INST), Mohali is DST's dedicated autonomous nano institute; ARCI Hyderabad and CeNS Bengaluru are other key centres.
22. Nanotoxicology: inhaled ultrafine particles can cross biological barriers; long, thin carbon nanotubes raise asbestos-like concerns — hence the need for regulation.
23. A composite = matrix + reinforcement (carbon-fibre reinforced polymer, Kevlar/aramid, fibreglass, reinforced concrete); properties differ from either constituent.
24. Superconductivity = zero electrical resistance below a critical temperature plus expulsion of magnetic flux (Meissner effect); discovered by Kamerlingh Onnes in mercury in 1911.
25. Nitinol (nickel-titanium) is a shape-memory alloy used in stents; quartz and PZT are piezoelectric; thermochromic and magnetostrictive materials are other 'smart' materials.
26. Metamaterials get their behaviour from engineered structure, can show a negative refractive index and underpin cloaking and superlens research; aerogel ('frozen smoke') is among the lightest solids.
27. Semiconductor doping: group-15 elements (P, As) give n-type; group-13 elements (B, Ga) give p-type; silicon's band gap is about 1.1 eV; SiC and GaN are wide-band-gap semiconductors.
28. Semicon India Programme was approved in 2021 with an outlay of Rs 76,000 crore; the India Semiconductor Mission runs it under MeitY. Per PIB (1 April 2026), 10 projects worth about Rs 1.6 lakh crore stand approved and ISM 2.0 was announced in Budget 2026-27.
29. Tata Electronics with PSMC is building India's first commercial semiconductor fab at Dholera (Gujarat); Micron's ATMP and Kaynes' unit are at Sanand (Gujarat) and Tata's OSAT plant is at Jagiroad (Assam).
30. Rare earths = 15 lanthanides plus scandium and yttrium (17 elements); India's monazite beach sands (Kerala, Odisha, Tamil Nadu, Andhra Pradesh) are worked by IREL (India) Ltd; the National Critical Mineral Mission was launched in January 2025.

1. One nanometre is equal to:

(A) 10^-3 metre (B) 10^-6 metre
(C) 10^-9 metre (D) 10^-12 metre

2. Nanotechnology deals with the manipulation of matter having at least one dimension in the size range of: RAS PRE 2021

(A) 1-100 nm (B) 100-500 nm
(C) 0.01-1 nm (D) 500-1000 nm

3. The lecture 'There's Plenty of Room at the Bottom', regarded as the conceptual beginning of nanotechnology, was delivered in 1959 by:

(A) Richard Feynman (B) Norio Taniguchi
(C) K. Eric Drexler (D) Gerd Binnig

4. The term 'nanotechnology' was used for the first time in 1974 by:

(A) Richard Feynman (B) Sumio Iijima
(C) Harold Kroto (D) Norio Taniguchi

5. A nanostructure in which charge carriers are confined in all three spatial dimensions is called a: RAS PRE 2018

(A) Quantum well (B) Quantum wire
(C) Quantum dot (D) Nanotube

6. Consider the following statements:
I. In a quantum well, charge carriers are confined in one direction only.
II. A quantum dot is often described as an artificial atom.
III. The colour of light emitted by a quantum dot is independent of its size.
Which of the statements given above are correct? RAS PRE 2021

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

7. Buckminsterfullerene (C60) is composed of: RAS PRE 2016

(A) 12 hexagons and 20 pentagons
(B) 60 hexagons only
(C) 32 hexagons and 12 pentagons
(D) 20 hexagons and 12 pentagons

8. Carbon nanotubes were reported in 1991 by: RAS PRE 2023

(A) Richard Smalley
(B) Sumio Iijima
(C) Andre Geim
(D) Robert Curl

9. The Nobel Prize in Physics 2010 was awarded to Andre Geim and Konstantin Novoselov for groundbreaking experiments regarding: RAS PRE 2023

(A) Fullerene
(B) Graphene
(C) Carbon nanotubes
(D) Quantum dots

10. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Generation of nanotechnology) List-II (Characteristic)
A. First generation 1. Molecular nanosystems
B. Second generation 2. Passive nanostructures
C. Third generation 3. Systems of nanosystems
D. Fourth generation 4. Active nanostructures RAS PRE 2018

(A) A-2, B-4, C-3, D-1
(B) A-1, B-2, C-3, D-4
(C) A-2, B-3, C-4, D-1
(D) A-4, B-2, C-1, D-3

11. Which one of the following is NOT an allotrope of carbon?

(A) Nitinol
(B) Graphene
(C) Fullerene
(D) Carbon nanotube

12. The Scanning Tunnelling Microscope, which first made imaging of individual atoms possible, was developed in 1981 by:

(A) Kroto and Smalley
(B) Geim and Novoselov
(C) Feynman and Drexler
(D) Binnig and Rohrer

13. Consider the following statements about the synthesis of nanomaterials:
I. Ball milling is a top-down method.
II. The sol-gel process is a bottom-up method.
III. Chemical vapour deposition is a top-down method.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

14. The Nano Mission of India, launched in 2007, was placed under which of the following?

(A) Department of Science and Technology
(B) Ministry of Electronics and Information Technology
(C) Department of Biotechnology
(D) Council of Scientific and Industrial Research

15. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Institution) List-II (Location)
A. Institute of Nano Science and Technology 1. Bengaluru
B. International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI) 2. Mohali
C. Centre for Nano and Soft Matter Sciences 3. Mumbai
D. National Centre for Nanosciences and Nanotechnology 4. Hyderabad HARD

(A) A-2, B-4, C-1, D-3
(B) A-2, B-1, C-4, D-3
(C) A-1, B-4, C-2, D-3
(D) A-2, B-4, C-3, D-1

16. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Instrument) List-II (Working principle)
A. Transmission Electron Microscope 1. Transmission of an electron beam through a very thin specimen
B. Atomic Force Microscope 2. Force between a sharp tip and the sample surface
C. Scanning Tunnelling Microscope 3. Tunnelling current between a tip and a conducting surface
D. X-Ray Diffraction 4. Diffraction of X-rays to determine crystal structure

(A) A-2, B-1, C-3, D-4
(B) A-1, B-3, C-2, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-2, C-3, D-4

17. Repetitively branched, tree-like macromolecules widely studied as nanoscale carriers for drug delivery are called:

(A) Micelles
(B) Zeolites
(C) Dendrimers
(D) Aerogels

18. Which one of the following is NOT an application of nanotechnology?

(A) Targeted drug delivery in cancer therapy
(B) Generation of electricity by nuclear fission
(C) Self-cleaning and stain-resistant textiles
(D) Nanofiltration membranes for water purification

19. The ruby-red colour seen when the Roman Lycurgus Cup is lit from behind is caused by nanoparticles of: HARD

(A) Iron oxide
(B) Gold and silver
(C) Titanium dioxide
(D) Cobalt

20. Consider the following statements about materials at the nanoscale:
I. The surface-area-to-volume ratio increases as particle size decreases.
II. Chemical reactivity generally increases at the nanoscale.
III. Optical and electronic properties become size-dependent because of quantum confinement.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

21. Arrange the following in correct chronological order:
I. Discovery of fullerene C60
II. Feynman's lecture 'There's Plenty of Room at the Bottom'
III. Isolation of graphene
IV. Report of carbon nanotubes by Iijima

(A) II, I, IV, III
(B) I, II, III, IV
(C) II, IV, I, III
(D) I, IV, II, III

22. The Nobel Prize in Chemistry 2023 was awarded for the discovery and synthesis of:

(A) Graphene
(B) Quantum dots
(C) Carbon nanotubes
(D) Metal-organic frameworks

23. Graphene is best described as:

(A) A hollow cage of sixty carbon atoms
(B) A cylindrical rolled sheet of carbon atoms
(C) A three-dimensional network of sp3 hybridised carbon atoms
(D) A single layer of carbon atoms arranged in a honeycomb lattice

24. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Structure) List-II (Number of dimensions in which carriers are free to move)
A. Quantum dot 1. Zero
B. Quantum wire 2. One
C. Quantum well 3. Two
D. Bulk crystal 4. Three RAS PRE 2021 HARD

(A) A-4, B-3, C-2, D-1
(B) A-1, B-3, C-2, D-4
(C) A-1, B-2, C-3, D-4
(D) A-2, B-1, C-3, D-4

25. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Material) List-II (Property or use)
A. Nitinol 1. Aramid fibre used in bullet-proof vests
B. Kevlar 2. Shape-memory alloy used in vascular stents
C. Quartz 3. Extremely low-density solid used as a thermal insulator
D. Aerogel 4. Piezoelectric material used in sensors and watches

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-2, B-1, C-4, D-3
(D) A-3, B-1, C-4, D-2

26. The expulsion of magnetic flux from the interior of a material when it passes into the superconducting state is known as the:

(A) Hall effect
(B) Meissner effect
(C) Josephson effect
(D) Seebeck effect

27. Consider the following statements about metamaterials:
I. Their unusual properties arise mainly from their engineered structure rather than their chemical composition.
II. They can be designed to show a negative refractive index.
III. They are being researched for cloaking devices and superlenses.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

28. Aerogel, popularly called 'frozen smoke', is chiefly valued as:

(A) A superconducting magnet material
(B) An extremely light thermal insulator
(C) A semiconductor dopant
(D) A magnetic data storage medium

29. Which one of the following pairs relating to composite materials is NOT correctly matched?

(A) Carbon-fibre reinforced polymer — aerospace structures
(B) Fibreglass — boat hulls and thermal insulation
(C) Reinforced concrete — steel rods in a cement matrix
(D) Nitinol — a ceramic-matrix composite

30. The India Semiconductor Mission functions under which ministry?

(A) Ministry of Heavy Industries
(B) Ministry of Science and Technology
(C) Ministry of Electronics and Information Technology
(D) Ministry of Commerce and Industry

31. The Semicon India Programme approved by the Union Cabinet in 2021 for development of the semiconductor and display manufacturing ecosystem has an outlay of: HARD

(A) Rs 10,000 crore
(B) Rs 76,000 crore
(C) Rs 1,50,000 crore
(D) Rs 25,000 crore

32. India's first commercial semiconductor fabrication unit is being set up by Tata Electronics in partnership with Powerchip (PSMC) at:

(A) Jagiroad, Assam
(B) Sanand, Gujarat
(C) Dholera, Gujarat
(D) Mysuru, Karnataka

33. Consider the following statements about doping of silicon:
I. Doping with a group-15 element such as phosphorus produces an n-type semiconductor.
II. Doping with a group-13 element such as boron produces a p-type semiconductor.
Which of the statements given above is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

34. Consider the following statements about rare earth elements:
I. They comprise the fifteen lanthanides along with scandium and yttrium.
II. In India, monazite occurring in beach sands is a principal source of rare earths.
III. IREL (India) Limited is the principal public sector producer of rare earths in India.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

35. Which one of the following statements about carbon nanotubes is NOT correct?

(A) They are always electrical insulators
(B) They may be single-walled or multi-walled
(C) Their tensile strength is many times that of steel
(D) Their electrical behaviour depends on the way the graphene sheet is rolled

ST09

Computers, IT and ICT

55 questions · 30 fact-sheet points

Must-know facts

1. Charles Babbage is the 'Father of the Computer' (Difference Engine 1822, Analytical Engine 1837); Ada Lovelace is regarded as the world's first programmer.
2. Generations: I (1946-59) vacuum tube; II (1959-65) transistor; III (1965-71) integrated circuit; IV (1971-80) microprocessor and VLSI; V (1980 onwards) artificial intelligence, ULSI and parallel processing.
3. ENIAC (1946) was the first general-purpose electronic digital computer; UNIVAC-I the first commercially sold computer; EDVAC embodied the stored-programme (von Neumann) concept.
4. CPU = Arithmetic Logic Unit + Control Unit + registers. The ALU computes, the control unit fetches and decodes instructions.
5. Data units: 1 nibble = 4 bits, 1 byte = 8 bits, 1 KB = 1024 bytes, then MB, GB, TB, PB, EB, ZB, YB. Number systems: binary (base 2), octal (8), decimal (10), hexadecimal (16, digits 0-9 and A-F); ASCII is a 7-bit code of 128 characters and Unicode is the universal standard.
6. Memory hierarchy by speed: registers > cache > main memory (RAM) > secondary storage; RAM is volatile, ROM is not. PROM is written once, EPROM is erased by ultraviolet light, EEPROM electrically, and flash memory is a form of EEPROM.
7. Input devices: keyboard, mouse, scanner, OMR, OCR, MICR (used on cheques), barcode reader, light pen, joystick. Output: monitor, printer, plotter, speaker. Dot-matrix printers are impact; laser and inkjet are non-impact.
8. System software (operating system, compiler, assembler, interpreter, device drivers) versus application software. Windows, Linux, Unix, macOS, Android and iOS are operating systems; Android is built on the Linux kernel.
9. A compiler translates the whole programme at once and lists all errors; an interpreter works line by line; an assembler converts assembly to machine code. Language generations: 1GL machine, 2GL assembly, 3GL high-level (FORTRAN 1957 was the first, COBOL for business, C, Java, Python), 4GL SQL, 5GL LISP and Prolog.
10. E.F. Codd proposed the relational database model in 1970; in a relation a row is a tuple/record and a column an attribute/field; the primary key uniquely identifies a row.
11. Networks by span: PAN < LAN < MAN < WAN. Topologies: bus (single backbone), star (central hub/switch), ring (closed loop), mesh (every node to every node, most fault-tolerant), tree and hybrid.
12. The OSI reference model has 7 layers (Physical, Data Link, Network, Transport, Session, Presentation, Application) against TCP/IP's 4. Devices by layer: repeater and hub = physical, switch and bridge = data link, router = network; a gateway joins dissimilar networks and a modem modulates and demodulates.
13. IPv4 addresses are 32-bit (four octets in dotted decimal); IPv6 is 128-bit. Default ports: HTTP 80, HTTPS 443, FTP 20/21, SMTP 25, DNS 53, Telnet 23, POP3 110.
14. ARPANET (1969) was the ancestor of the internet; Tim Berners-Lee invented the World Wide Web at CERN in 1989; commercial internet reached India on 15 August 1995 through VSNL; NIXI (2003) manages the .IN registry.
15. Cloud service models: IaaS (infrastructure), PaaS (platform), SaaS (software). MeghRaj is the Government of India cloud initiative.
16. Big Data's five Vs: volume, velocity, variety, veracity and value. IoT is a network of physical objects with sensors. Blockchain is a decentralised, immutable distributed ledger; Bitcoin (2009) is attributed

to 'Satoshi Nakamoto'; the RBI began Digital Rupee (CBDC) pilots in 2022.

17. 'Artificial intelligence' was coined by John McCarthy at the Dartmouth Conference of 1956; Alan Turing proposed the Turing Test in 1950. Machine learning types: supervised, unsupervised and reinforcement learning.
18. Generative AI and large language models are built on the transformer architecture introduced in 2017. NITI Aayog's National Strategy for AI (2018) is '#AIforAll'; the IndiaAI Mission was approved on 7 March 2024 with about Rs 10,372 crore under MeitY and has seven pillars; Bhashini is India's language AI platform.
19. National Quantum Mission — approved 19 April 2023, outlay Rs 6,003.65 crore, period 2023-24 to 2030-31, under DST; target of 50-1000 physical qubits in eight years; four thematic hubs at IISc Bengaluru (computing), IIT Madras (communication), IIT Bombay (sensing and metrology) and IIT Delhi (materials and devices).
20. 5G services were launched in India on 1 October 2022 at the India Mobile Congress; the Bharat 6G Vision document came in March 2023, the Bharat 6G Alliance in July 2023, with 6G leadership targeted around 2030.
21. PARAM 8000 (1991), built by C-DAC under Dr Vijay Bhatkar, was India's first indigenous supercomputer; the National Supercomputing Mission (2015) is steered by MeitY and DST through C-DAC Pune and IISc Bengaluru. PARAM Shivay at IIT-BHU (2019) was its first system, three PARAM Rudra machines were dedicated in September 2024 (Pune, Delhi, Kolkata), AIRAWAT at C-DAC Pune ranked 75th in the Top500 list announced at ISC 2023, and Pratyush and Mihir serve weather forecasting.
22. Malware: a virus needs a host programme, a worm spreads on its own, a trojan disguises itself, ransomware encrypts files for payment, spyware and keyloggers steal information, a botnet is a network of compromised machines.
23. The Information Technology Act, 2000 is based on the UNCITRAL Model Law on Electronic Commerce (1996) and came into force on 17 October 2000; it was amended in 2008. Section 66A was struck down in Shreya Singhal v. Union of India (2015).
24. CERT-In (2004, under MeitY) is the national nodal agency for cyber incidents; NCIIPC (under NTRO, Section 70A) guards critical information infrastructure; I4C under the Ministry of Home Affairs runs the cybercrime portal and helpline 1930; Cyber Swachhta Kendra cleans botnets.
25. Digital Personal Data Protection Act, 2023 — key terms Data Principal, Data Fiduciary, Consent Manager and Significant Data Fiduciary; it sets up the Data Protection Board of India; the maximum penalty is Rs 250 crore; DPDP Rules were notified on 14 November 2025.
26. Digital India was launched on 1 July 2015 with three vision areas and nine pillars. Building blocks: DigiLocker (2015), UMANG (2017), MyGov (2014), BharatNet, Common Service Centres, UPI by NPCI (2016), ONDC (2021), Aadhaar (UIDAI 2009, 12-digit, Aadhaar Act 2016).
27. Kerala was declared India's first digital state in February 2016 by the President of India — an item RPSC asked twice in the same 2016 paper.
28. Rajasthan: DoIT&C is the nodal department; e-Mitra delivers 600-plus services through 80,000-plus kiosks; Rajasthan SSO gives single sign-on to state services; Bhamashah Yojana (relaunched 2014, woman as head of family) was subsumed into Jan Aadhaar (18 December 2019, 'One Number, One Card, One Identity').
29. Rajasthan portals: Raj Kaj (e-office and employee service matters), RAJ NIVESH (single-window investment clearance), iStart (startup registration and incubation), RajNET and RajSWAN (connectivity), Rajasthan Sampark (grievances, helpline 181), R-CAT Jaipur (advanced technology training), RKCL's RS-CIT (digital literacy).
30. ICT in education: SWAYAM (online courses), SWAYAM PRABHA (DTH channels), DIKSHA, NPTEL, e-PathShala, National Digital Library, PM eVidya. Common shortcuts: Ctrl+C copy, Ctrl+X cut, Ctrl+V paste, Ctrl+Z undo, Ctrl+Y redo, Ctrl+S save, Ctrl+P print, Ctrl+A select all, Ctrl+F find, F1 help, F2 rename, F5 refresh, Alt+F4 close.

1. Who is regarded as the 'Father of the Computer' for designing the Analytical Engine?

(A) Alan Turing
(B) John von Neumann
(C) Charles Babbage
(D) Blaise Pascal

2. ENIAC, completed in 1946, is regarded as:

(A) The first commercially sold computer
(B) The first general-purpose electronic digital computer
(C) The first stored-programme computer
(D) The first microprocessor-based computer

3. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Generation of computers) List-II (Principal electronic component)

A. First generation 1. Integrated circuit

B. Second generation 2. Microprocessor and VLSI

C. Third generation 3. Vacuum tube

D. Fourth generation 4. Transistor

(A) A-3, B-4, C-1, D-2
(B) A-3, B-1, C-4, D-2
(C) A-4, B-3, C-1, D-2
(D) A-3, B-4, C-2, D-1

4. Consider the following statements about the Central Processing Unit:

I. The Arithmetic Logic Unit performs arithmetic and logical operations.

II. The Control Unit decodes instructions and directs the working of other units.

III. Registers are the slowest form of memory in a computer.

Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

5. One byte consists of:

(A) 4 bits
(B) 16 bits
(C) 1024 bits
(D) 8 bits

6. A group of four bits is known as a:

(A) Word
(B) Byte
(C) Parity bit
(D) Nibble

7. Arrange the following units of computer memory in ascending order of size:

I. Gigabyte

II. Kilobyte

III. Terabyte

IV. Megabyte

(A) II, IV, I, III
(B) II, I, IV, III
(C) IV, II, I, III
(D) I, II, IV, III

8. The hexadecimal number system has a base of:

(A) 2
(B) 8
(C) 16
(D) 10

9. The binary equivalent of the decimal number 25 is:

(A) 10011
(B) 10101
(C) 11011
(D) 11001

10. Which one of the following is NOT an input device?

(A) Optical Mark Reader
(B) Light pen
(C) Plotter
(D) Barcode scanner

11. Which of the following memories loses its contents when the power supply is switched off?

(A) ROM
(B) RAM
(C) Hard disk drive
(D) Flash drive

12. The contents of which of the following memories can be erased by exposure to ultraviolet light?

(A) PROM
(B) EEPROM
(C) SRAM
(D) EPROM

13. Consider the following statements:

I. Cache memory is faster than main memory but smaller in capacity.

II. Secondary storage is non-volatile.

III. The CPU can directly execute instructions lying on a hard disk without loading them into main memory.

Which of the statements given above are correct?

(A) II and III only
(B) I and II only
(C) I and III only
(D) I, II and III

14. Assertion (A): A ROM chip retains its contents even when the computer is switched off.

Reason (R): ROM is a non-volatile memory.

Select the correct answer using the code given below:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

15. Which one of the following is NOT an operating system?

(A) Ubuntu
(B) Android
(C) Oracle
(D) macOS

16. The first high-level programming language, developed in the 1950s, was:

(A) COBOL
(B) FORTRAN
(C) BASIC
(D) PASCAL

17. Which one of the following translates an entire source programme into machine code at one time and reports all the errors together?

(A) Interpreter
(B) Assembler
(C) Loader
(D) Compiler

18. The relational model of database management was proposed by: HARD

(A) James Gosling
(B) Dennis Ritchie
(C) E. F. Codd
(D) Tim Berners-Lee

19. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Network topology) List-II (Description)
A. Bus 1. Every node is directly connected to every other node
B. Star 2. All nodes share a single backbone cable
C. Ring 3. All nodes are connected to a central hub or switch
D. Mesh 4. Nodes form a closed loop and data moves from node to node

(A) A-2, B-3, C-4, D-1
(B) A-2, B-4, C-3, D-1
(C) A-1, B-3, C-4, D-2
(D) A-2, B-3, C-1, D-4

20. Consider the following statements:
I. The OSI reference model has seven layers.
II. The TCP/IP model has five layers.
III. A router operates at the network layer.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

21. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Protocol) List-II (Function)
A. SMTP 1. Transfer of files between hosts
B. FTP 2. Sending of electronic mail
C. DNS 3. Automatic assignment of IP addresses to hosts
D. DHCP 4. Translation of domain names into IP addresses HARD

(A) A-2, B-1, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-1, B-2, C-4, D-3
(D) A-2, B-4, C-1, D-3

22. An IPv4 address consists of:

(A) 16 bits
(B) 64 bits
(C) 128 bits
(D) 32 bits

23. The World Wide Web was invented in 1989 at CERN by:

(A) Vint Cerf
(B) Tim Berners-Lee
(C) Robert Kahn
(D) Marc Andreessen

24. Consider the following statements about cloud computing:
I. In Software as a Service the user accesses applications over the internet without managing the underlying infrastructure.
II. Infrastructure as a Service provides virtualised computing resources such as servers and storage.
III. MeghRaj is the cloud initiative of the Government of India.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

25. Which one of the following pairs is NOT correctly matched?

(A) Internet of Things — network of physical objects with sensors that exchange data
(B) Big Data — data marked by volume, velocity and variety
(C) Blockchain — a centralised database controlled by a single trusted authority
(D) Machine learning — systems that improve their performance from data without being explicitly programmed

26. Consider the following statements about blockchain technology:
I. It maintains a distributed ledger shared across a network of computers.
II. Once recorded, a block is extremely difficult to alter retrospectively.
III. Bitcoin was launched in 2009 by the Reserve Bank of India.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

27. The term 'artificial intelligence' was coined in 1956 at the Dartmouth Conference by:

(A) Alan Turing
(B) John McCarthy
(C) Marvin Minsky
(D) Herbert Simon

28. Consider the following statements about machine learning:
I. In supervised learning the algorithm is trained on labelled data.
II. Clustering is an example of supervised learning.
III. Reinforcement learning trains an agent through rewards and penalties.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

29. Most present-day large language models used in generative AI are built on which neural network architecture, introduced in 2017?

(A) Convolutional neural network
(B) Recurrent neural network
(C) Transformer
(D) Perceptron

30. The IndiaAI Mission, approved by the Union Cabinet in March 2024 with an outlay of about Rs 10,372 crore, is being implemented by:

(A) NITI Aayog
(B) Department of Science and Technology
(C) Indian Space Research Organisation
(D) Ministry of Electronics and Information Technology

31. The National Quantum Mission approved in April 2023 has a total outlay of about: HARD

(A) Rs 6,004 crore
(B) Rs 10,372 crore
(C) Rs 76,000 crore
(D) Rs 1,000 crore

32. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Thematic hub of the National Quantum Mission) List-II (Host institution)
A. Quantum computing 1. IIT Delhi
B. Quantum communication 2. IISc Bengaluru
C. Quantum sensing and metrology 3. IIT Madras
D. Quantum materials and devices 4. IIT Bombay HARD

(A) A-2, B-3, C-4, D-1
(B) A-1, B-3, C-4, D-2
(C) A-2, B-4, C-3, D-1
(D) A-3, B-2, C-1, D-4

33. Consider the following statements about quantum computing:
I. The basic unit of quantum information is the qubit.
II. A qubit can exist in a superposition of states.
III. Entanglement allows the states of two qubits to remain correlated.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

34. 5G mobile services were formally launched in India in:

(A) October 2020
(B) October 2022
(C) August 2021
(D) March 2023

35. Consider the following statements:
I. The Bharat 6G Vision document was released in March 2023.
II. The Bharat 6G Alliance was set up under the Ministry of Defence.
III. India aims to be a leading contributor to 6G technology by around 2030.
Which of the statements given above are correct? HARD

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

36. India's first indigenously developed supercomputer PARAM 8000, built by C-DAC in 1991, was developed under the leadership of:

(A) Vijay Bhatkar
(B) Raj Reddy
(C) Narendra Karmarkar
(D) Anil Kakodkar

37. PARAM Shivay, the first supercomputer installed under the National Supercomputing Mission, is located at:

(A) IIT Kanpur
(B) IIT (BHU) Varanasi
(C) IISc Bengaluru
(D) C-DAC Pune

38. Which one of the following statements about supercomputing in India is NOT correct? HARD

(A) The National Supercomputing Mission is steered jointly by MeitY and the Department of Science and Technology
(B) C-DAC Pune and IISc Bengaluru are the implementing agencies of the mission
(C) AIRAWAT, installed at C-DAC Pune, was ranked 75th in the Top500 list announced at ISC 2023
(D) The three PARAM Rudra supercomputers were dedicated to the nation in 2019

39. Consider the following statements:
I. A computer virus needs a host programme to spread, whereas a worm can spread by itself.
II. A trojan horse disguises itself as a legitimate programme.
III. Ransomware encrypts a victim's files and demands payment for the decryption key.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. The Information Technology Act, 2000 of India is based on which of the following model laws?

(A) UNCITRAL Model Law on Electronic Commerce, 1996
(B) OECD Privacy Guidelines, 1980
(C) Budapest Convention on Cybercrime, 2001
(D) General Data Protection Regulation of the European Union

41. Section 66A of the Information Technology Act, 2000, which penalised sending 'offensive' messages through communication services, was struck down by the Supreme Court in: HARD

(A) K.S. Puttaswamy v. Union of India
(B) Anuradha Bhasin v. Union of India
(C) Shreya Singhal v. Union of India
(D) Vishaka v. State of Rajasthan

42. CERT-In, the national nodal agency for responding to computer security incidents, functions under the:

(A) Ministry of Home Affairs
(B) Ministry of Defence
(C) Ministry of Electronics and Information Technology
(D) National Security Council Secretariat

43. Consider the following statements about the Digital Personal Data Protection Act, 2023:
I. The individual whose personal data is processed is called the Data Principal.
II. The Act provides for the establishment of the Data Protection Board of India.
III. The maximum financial penalty prescribed under the Act is Rs 50 crore.
Which of the statements given above are correct?

(A) II and III only
(B) I and III only
(C) I and II only
(D) I, II and III

44. The Digital India programme was launched on:

(A) 15 August 2014
(B) 26 January 2015
(C) 2 October 2014
(D) 1 July 2015

45. The Aadhaar number issued by the Unique Identification Authority of India consists of:

(A) 10 digits
(B) 11 digits
(C) 12 digits
(D) 16 digits

46. The Unified Payments Interface (UPI) was developed by:

(A) Reserve Bank of India
(B) National Payments Corporation of India
(C) State Bank of India
(D) Ministry of Finance

47. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Initiative) List-II (Description)
A. DigiLocker 1. Single mobile application giving access to many government services
B. UMANG 2. Cloud platform for storing and sharing authentic digital documents
C. ONDC 3. Open network for digital commerce
D. SWAYAM 4. Platform for free online courses from school to postgraduate level

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-2, B-1, C-4, D-3
(D) A-2, B-3, C-1, D-4

48. Which state was declared the first digital state of India in 2016? RAS PRE 2016

(A) Rajasthan
(B) Andhra Pradesh
(C) Kerala
(D) Gujarat

49. The Jan Aadhaar Yojana of Rajasthan, launched in December 2019, works on the principle of: RAS PRE 2021

(A) One nation, one ration card
(B) One state, one portal
(C) One family, one job
(D) One number, one card, one identity

50. e-Mitra kiosks in Rajasthan are primarily meant for: RAS PRE 2018

(A) Electronic delivery of government and private services to citizens
(B) Distribution of subsidised foodgrain
(C) Registration of land records only
(D) Provision of free internet in gram panchayats

51. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Portal or application of Rajasthan) List-II (Purpose)
A. Raj Kaj 1. Single-window clearance for investors
B. RAJ NIVESH 2. e-Office and service matters of government employees
C. iStart 3. Registration and support for startups
D. Rajasthan SSO 4. Single sign-on access to state online services RAS PRE 2024

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-2, B-1, C-4, D-3
(D) A-2, B-3, C-1, D-4

52. Under the Bhamashah Yojana of Rajasthan, benefits were transferred by treating which person as the head of the family?

(A) The eldest male member
(B) The adult woman of the household
(C) The eldest member irrespective of gender
(D) Any member holding a bank account

53. Which one of the following pairs of keyboard shortcuts is NOT correctly matched?

(A) Ctrl + Z — Undo
(B) Ctrl + P — Print
(C) Ctrl + X — Cut
(D) Ctrl + V — Save

54. Which one of the following pairs of file extensions is NOT correctly matched?

(A) .xlsx — Spreadsheet file
(B) .pptx — Presentation file
(C) .mp3 — Image file
(D) .docx — Word processing document

55. Arrange the following in correct chronological order:

I. Launch of the Digital India programme
II. Creation of ARPANET
III. Invention of the World Wide Web
IV. Start of commercial internet service in India

(A) II, III, IV, I
(B) II, IV, III, I
(C) III, II, IV, I
(D) II, III, I, IV

ST10

Space Technology

55 questions · 30 fact-sheet points

Must-know facts

1. INCOSPAR set up 1962 under DAE; ISRO formed 15 August 1969; Department of Space and Space Commission created 1972.
2. Vikram Sarabhai — father of the Indian space programme; Satish Dhawan chaired ISRO 1972-84; V. Narayanan took charge as Chairman in January 2025.
3. TERLS, Thumba (Thiruvananthapuram) — first sounding rocket, a Nike-Apache, launched 21 November 1963; site chosen for the magnetic equator.
4. Aryabhata — first Indian satellite, 360 kg, launched 19 April 1975 by a Soviet Kosmos-3M from Kapustin Yar.
5. Rohini RS-1, 18 July 1980 on SLV-3 — first satellite orbited by an Indian launch vehicle; SLV-3 project director A.P.J. Abdul Kalam.
6. Bhaskara-I (1979) first experimental earth-observation satellite; APPLE (1981, Ariane) first experimental communication satellite; INSAT-1A 1982; IRS-1A 1988; Kalpana-1 (2002) first dedicated meteorological satellite (originally METSAT); EDUSAT (2004, GSLV-F01) first satellite exclusively for education.
7. Launch vehicles: SLV-3 (1979), ASLV (1987), PSLV (1993), GSLV (2001), LVM3/GSLV Mk-III (2014 experimental), SSLV (2022), RLV-TD, HLVM3 for Gaganyaan.
8. PSLV variants — PSLV-G, PSLV-CA (core alone, no strap-ons), PSLV-XL (6 XL strap-ons), PSLV-DL (2), PSLV-QL (4). PSLV = ISRO's 'workhorse'.
9. PSLV-C37, 15 February 2017 — world record 104 satellites in one flight (primary Cartosat-2 series); record surpassed by SpaceX Transporter-1 (143) in January 2021.
10. PSLV-C11 → Chandrayaan-1 (2008); PSLV-C25 → Mangalyaan (2013); PSLV-C30 → AstroSat (2015); PSLV-C57 → Aditya-L1 (2023); PSLV-C58 → XPoSat (2024); PSLV-C60 → SpaDeX (2024).
11. PSLV-C55 (April 2023) launched Singapore's TeLEOS-2 as a dedicated NSIL commercial mission.
12. Chandrayaan-1 (2008) — Moon Impact Probe; NASA's Moon Mineralogy Mapper aboard it confirmed water molecules on the Moon.
13. Chandrayaan-2 (GSLV Mk III-M1, 22 July 2019) — Vikram lander crashed 7 Sept 2019; the orbiter still works. Crash site named Tiranga Point.
14. Chandrayaan-3 (LVM3-M4, 14 July 2023) — Vikram soft-landed 23 August 2023 near the lunar south pole; India 4th to soft-land, 1st near the south pole; site 'Shiv Shakti Point'; 23 August = National Space Day; cost ~Rs 615 crore.
15. Mangalyaan/MOM — launched 5 Nov 2013, Mars orbit insertion 24 Sept 2014; India first country to reach Mars orbit on its maiden attempt; cost ~Rs 450 crore.
16. Aditya-L1 — launched 2 Sept 2023, inserted into a halo orbit around Sun-Earth L1 (~1.5 million km) on 6 January 2024; main payload VELC built by the Indian Institute of Astrophysics.
17. AstroSat (2015) — India's first multi-wavelength space observatory. XPoSat (1 Jan 2024) — world's second X-ray polarimetry mission after NASA's IXPE; payloads POLIX and XSPECT.
18. NavIC/IRNSS — constellation of 7 satellites; covers India plus 1500 km beyond; GAGAN is the ISRO-AAI satellite-based augmentation system for aviation.
19. Cryogenic engines: CE-7.5 on GSLV Mk-II, CE-20 on LVM3; first successful indigenous cryogenic flight GSLV-D5, January 2014. SCE-200 semi-cryogenic engine under development.

20. SpaDeX — PSLV-C60, 30 Dec 2024; SDX01 (Chaser) and SDX02 (Target) docked 16 January 2025, making India the 4th country after the USA, Russia and China to dock in space.
21. NISAR — ISRO-NASA dual-band SAR (NASA L-band + ISRO S-band), launched by GSLV-F16 on 30 July 2025 into a sun-synchronous orbit.
22. GSLV-F15/NVS-02, 29 January 2025 — ISRO's 100th launch from Sriharikota. CMS-03 on LVM3-M5 (2 Nov 2025, ~4,410 kg) is the heaviest satellite launched from Indian soil at the time; surpassed by BlueBird Block-2 on LVM3-M6, 24 Dec 2025.
23. Gaganyaan — Rs 20,193 crore; TV-D1 crew-escape test 21 Oct 2023; HLVM3 launcher; 3 crew, ~400 km orbit, up to 7 days; uncrewed G1 carries humanoid Vyommitra; crewed flight targeted 2027 (as of 2026).
24. Shubhanshu Shukla flew the Axiom-4 mission to the ISS in June 2025 — second Indian in space after Rakesh Sharma (Soyuz T-11, 1984).
25. Cabinet approved 18 September 2024: Chandrayaan-4 lunar sample return (~Rs 2,104 crore), Venus Orbiter Mission (~Rs 1,236 crore, target 2028), and the Bharatiya Antariksh Station (first module by 2028, complete by 2035); Indian crewed Moon landing target 2040.
26. Indian Space Policy 2023 — IN-SPACe (Ahmedabad) is the single-window authorisation body; NSIL (2019) is the commercial PSU arm; Antrix (1992) the earlier marketing arm.
27. Private sector: Skyroot Aerospace's Vikram-S (Mission Prarambh, 18 Nov 2022) first privately built Indian rocket; Agnikul Cosmos' Agnibaan SOrTeD (May 2024) used the world's first single-piece 3D-printed semi-cryogenic engine.
28. Project NETRA + IS4OM (Bengaluru) handle space-debris tracking and space situational awareness; ISRO's goal is debris-free space missions by 2030. A second launch port for SSLV is coming up at Kulasekarapattinam, Thoothukudi district, Tamil Nadu; SSLV technology has been transferred to HAL.
29. Orbits: LEO 160-2,000 km; MEO up to 35,786 km; GEO 35,786 km (period ~24 h, inclination ~0 deg); sun-synchronous polar orbits (~98 deg inclination) are used for remote sensing. Earth's escape velocity 11.2 km/s, orbital velocity ~7.9 km/s.
30. Udaipur Solar Observatory — on an island in Fateh Sagar Lake, founded 1976 by Arvind Bhatnagar, run by the Physical Research Laboratory, Ahmedabad since 1981; 50 cm Multi Application Solar Telescope (MAST) operational 2015.

1. The Indian Space Research Organisation (ISRO) was established in the year:

(A) 1962 (B) 1969
(C) 1972 (D) 1975

2. Who is regarded as the father of the Indian space programme?

(A) Homi J. Bhabha (B) Satish Dhawan
(C) Vikram Sarabhai (D) A.P.J. Abdul Kalam

3. Aryabhata, India's first satellite, was launched in 1975 from:

(A) Kapustin Yar, USSR (B) Sriharikota, India
(C) Baikonur, Kazakhstan (D) Kourou, French Guiana

4. The first Indian satellite placed in orbit by an Indian-built launch vehicle was:

(A) Aryabhata (B) Bhaskara-I
(C) APPLE (D) Rohini RS-1

5. The first sounding rocket launched from the Thumba Equatorial Rocket Launching Station in November 1963 was: HARD

(A) Rohini-75 (B) Centaure
(C) Menaka (D) Nike-Apache

6. Match List-I with List-II and select the correct answer using the codes given below:
List-I (ISRO centre) List-II (Location)
A. Vikram Sarabhai Space Centre 1. Ahmedabad
B. Space Applications Centre 2. Thiruvananthapuram
C. National Remote Sensing Centre 3. Hyderabad
D. U R Rao Satellite Centre 4. Bengaluru

(A) A-2, B-1, C-3, D-4
(B) A-2, B-3, C-1, D-4
(C) A-4, B-1, C-3, D-2
(D) A-2, B-1, C-4, D-3

7. Match List-I with List-II and select the correct answer using the codes given below:
List-I (ISRO facility) List-II (Principal function)
A. Liquid Propulsion Systems Centre 1. Reception and dissemination of remote sensing data
B. National Remote Sensing Centre 2. Design of liquid propulsion stages
C. Space Applications Centre 3. Development of payloads and space applications
D. ISRO Propulsion Complex, Mahendragiri 4. Testing of liquid and cryogenic engines

(A) A-4, B-1, C-3, D-2
(B) A-2, B-3, C-1, D-4
(C) A-4, B-3, C-1, D-2
(D) A-2, B-1, C-3, D-4

8. ISRO's Master Control Facility, which controls India's geostationary satellites, is located at:

(A) Byalalu
(B) Sriharikota
(C) Gadanki
(D) Hassan

9. Satish Dhawan Space Centre (SHAR), India's principal launch port, is located at Sriharikota in:

(A) Andhra Pradesh
(B) Tamil Nadu
(C) Odisha
(D) Telangana

10. Consider the following statements about the Polar Satellite Launch Vehicle (PSLV):
I. It is a four-stage vehicle with alternating solid and liquid stages.
II. Its first fully successful flight was PSLV-D2 in 1994.
III. The PSLV-CA variant flies without strap-on motors.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

11. In February 2017 ISRO set a world record by launching how many satellites in a single mission by PSLV-C37? `RAS PRE 2018`

(A) 83
(B) 104
(C) 36
(D) 57

12. The Singaporean earth observation satellite TeLEOS-2 was placed in orbit in April 2023 by which Indian launch vehicle? `RAS PRE 2023`

(A) GSLV-F12
(B) SSLV-D2
(C) PSLV-C55
(D) LVM3-M3

13. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Mission) List-II (Launch vehicle)
A. Chandrayaan-1 1. PSLV-C11
B. Mangalyaan 2. PSLV-C25
C. Chandrayaan-3 3. LVM3-M4
D. Aditya-L1 4. PSLV-C57 RAS PRE 2023

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-2, B-1, C-4, D-3

14. Chandrayaan-1, launched in October 2008, is chiefly credited with the discovery of:

(A) Water molecules on the lunar surface
(B) Helium-3 deposits on the Moon
(C) A strong lunar magnetic field
(D) Liquid water at the lunar south pole

15. Chandrayaan-3's lander Vikram made a soft landing on the Moon on: RAS PRE 2023

(A) 14 July 2023
(B) 6 September 2019
(C) 1 January 2024
(D) 23 August 2023

16. The landing site of Chandrayaan-3 on the Moon has been named:

(A) Tiranga Point
(B) Jawahar Point
(C) Shiv Shakti Point
(D) Vikram Point

17. Consider the following statements about Chandrayaan-2:
I. It was launched by GSLV Mk III-M1 in July 2019.
II. Its lander Vikram achieved a successful soft landing.
III. Its orbiter continues to function and send back data.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

18. India's Mars Orbiter Mission (Mangalyaan) was inserted into orbit around Mars on: RAS PRE 2016

(A) 24 September 2014
(B) 5 November 2013
(C) 22 July 2019
(D) 2 September 2023

19. India's solar mission Aditya-L1 has been placed in a halo orbit around which point?

(A) Sun-Earth L2
(B) Sun-Earth L1
(C) Earth-Moon L1
(D) Sun-Earth L4

20. XPoSat, launched by ISRO on 1 January 2024, is dedicated to the study of: HARD

(A) Lunar mineralogy
(B) Atmospheric ozone
(C) X-ray polarisation of cosmic sources
(D) Ionospheric electron density

21. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Mission) List-II (Principal domain)
A. AstroSat 1. Solar corona and solar wind studies
B. Aditya-L1 2. X-ray polarimetry
C. XPoSat 3. Multi-wavelength astronomy
D. NISAR 4. Radar earth observation

(A) A-1, B-3, C-2, D-4
(B) A-3, B-1, C-2, D-4
(C) A-3, B-2, C-1, D-4
(D) A-2, B-1, C-3, D-4

22. Which one of the following pairs of an Indian satellite series and its purpose is NOT correctly matched?

(A) RISAT — Radar imaging
(B) Oceansat — Ocean colour and wind studies
(C) Cartosat — High-resolution cartographic imaging
(D) Kalpana-1 — Satellite-based navigation

23. The Indian Regional Navigation Satellite System (NavIC) was designed as a constellation of how many satellites?

(A) 7
(B) 11
(C) 24
(D) 4

24. Consider the following statements about NavIC:
I. It provides coverage over India and about 1500 km beyond its borders.
II. Its satellites are placed in geostationary and geosynchronous orbits.
III. It is a global navigation system comparable in coverage to GPS.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

25. In a cryogenic rocket engine of the type used by ISRO, the propellant combination is:

(A) Solid HTPB with ammonium perchlorate
(B) Liquid hydrogen and liquid oxygen
(C) Unsymmetrical dimethyl hydrazine and nitrogen tetroxide
(D) Kerosene and liquid oxygen

26. Consider the following statements about India's cryogenic propulsion:
I. GSLV Mk-II uses the indigenous CE-7.5 cryogenic upper-stage engine.
II. LVM3 uses the CE-20 engine in its cryogenic upper stage.
III. India's first flight with an indigenous cryogenic stage was GSLV-D3 in April 2010.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

27. Consider the following statements about the Small Satellite Launch Vehicle (SSLV):
I. It is designed to place about 500 kg in a 500 km low Earth orbit.
II. Its first developmental flight, SSLV-D1, was fully successful.
III. Its third developmental flight placed EOS-08 in orbit in August 2024.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

28. ISRO's Reusable Launch Vehicle autonomous landing experiments (Pushpak) were conducted at the Aeronautical Test Range in:

(A) Chandipur, Odisha (B) Sriharikota, Andhra Pradesh
(C) Chitradurga, Karnataka (D) Balasore, Odisha

29. The success of ISRO's SpaDeX mission in January 2025 made India the fourth country to demonstrate in-space docking, after the USA, Russia and:

(A) Japan (B) France
(C) the European Space Agency (D) China

30. NISAR, launched in July 2025, is a joint earth observation mission of ISRO and:

(A) ESA (B) NASA
(C) JAXA (D) CNES

31. Consider the following statements about the Gaganyaan programme, as of 2026:
I. The test flight TV-D1, demonstrating the crew escape system, was conducted in October 2023.
II. The human-rated launch vehicle for the mission is designated HLVM3.
III. The first crewed flight of the programme has already been completed.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

32. Group Captain Shubhanshu Shukla, who travelled to the International Space Station in 2025, became the second Indian in space after:

(A) Rakesh Sharma (B) Kalpana Chawla
(C) Ravish Malhotra (D) Sunita Williams

33. According to the plan approved by the Union Cabinet in September 2024, the first module of the Bharatiya Antariksh Station is targeted for launch by:

(A) 2026 (B) 2028
(C) 2030 (D) 2035

34. Consider the following statements about the Chandrayaan-4 mission:
I. It has been approved as a lunar sample-return mission.
II. It is planned to use two LVM3 launches with multiple modules.
III. It will carry the first Indian astronaut to the surface of the Moon.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

35. India's Venus Orbiter Mission (Shukrayaan), approved by the Union Cabinet in September 2024, is targeted for launch in:

(A) 2026
(B) 2027
(C) 2030
(D) 2028

36. Which one of the following pairs of an ISRO facility and its location is NOT correctly matched?

(A) Physical Research Laboratory — Ahmedabad
(B) Indian Institute of Space Science and Technology — Bengaluru
(C) National Atmospheric Research Laboratory — Gadanki
(D) Human Space Flight Centre — Bengaluru

37. The Udaipur Solar Observatory is situated on an island in which lake? RAS PRE 2018, 2021

(A) Fateh Sagar Lake
(B) Pichola Lake
(C) Jaisamand Lake
(D) Rajsamand Lake

38. The Multi Application Solar Telescope (MAST) commissioned at the Udaipur Solar Observatory has an aperture of: HARD

(A) 15 cm
(B) 25 cm
(C) 50 cm
(D) 100 cm

39. Consider the following statements:
I. A geostationary satellite has an orbital period of about 24 hours and an inclination of nearly zero degrees.
II. Sun-synchronous polar orbits are preferred for remote sensing because they give consistent illumination conditions.
III. A geostationary orbit lies at an altitude of about 36,000 km above the equator.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

40. The escape velocity from the surface of the Earth is approximately:

(A) 2.4 km/s
(B) 7.9 km/s
(C) 11.2 km/s
(D) 29.8 km/s

41. Rockets are built in multiple stages mainly because:

(A) A single engine cannot burn for more than two minutes
(B) Multi-stage rockets need no guidance system
(C) Shedding spent stages reduces the mass that must be accelerated further
(D) It removes the need for a cryogenic engine

42. Arrange the following ISRO missions in the correct chronological order of their launch:
I. Chandrayaan-1
II. Mangalyaan (Mars Orbiter Mission)
III. AstroSat
IV. Aditya-L1

(A) II, I, III, IV
(B) I, III, II, IV
(C) III, I, II, IV
(D) I, II, III, IV

43. Arrange the following Indian launch vehicles in the chronological order of their first flight:
I. SLV-3
II. ASLV
III. PSLV
IV. GSLV HARD

(A) II, I, III, IV
(B) I, III, II, IV
(C) I, II, III, IV
(D) I, II, IV, III

44. Under the Indian Space Policy 2023, the single-window autonomous agency that authorises and regulates space activities by non-government entities is:

(A) NSIL
(B) Antrix Corporation
(C) Department of Space
(D) IN-SPACe

45. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Indian private space firm) List-II (Associated product)
A. Skyroot Aerospace 1. Agnibaan SOrTeD
B. Agnikul Cosmos 2. Vikram-S sounding rocket
C. Pixxel 3. Hyperspectral imaging satellites
D. Dhruva Space 4. Satellite orbital deployers

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-2, B-1, C-3, D-4
(D) A-1, B-2, C-4, D-3

46. Consider the following statements about space debris:
I. Project NETRA is ISRO's initiative for tracking space objects that could threaten Indian assets.
II. The IS4OM centre for space situational awareness is located at Bengaluru.
III. ISRO has announced a goal of debris-free space missions by 2030.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

47. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Satellite series) List-II (Principal application)
A. Cartosat 1. All-weather radar imaging
B. RISAT 2. Ocean colour and wind studies
C. Oceansat 3. High-resolution cartographic mapping
D. INSAT 4. Communication and meteorology

(A) A-1, B-3, C-2, D-4
(B) A-3, B-2, C-1, D-4
(C) A-3, B-1, C-2, D-4
(D) A-4, B-1, C-2, D-3

48. CMS-03, launched by LVM3-M5 in November 2025, is notable as:

(A) India's first reusable satellite
(B) The heaviest satellite launched from Indian soil at the time of its launch
(C) The first Indian satellite in a polar orbit
(D) India's first satellite built entirely by a private firm

49. Which one of the following statements about India's launch vehicles is NOT correct?

(A) SLV-3 was India's first vehicle to use a cryogenic upper stage
(B) PSLV is often described as ISRO's workhorse launch vehicle
(C) LVM3 was the launch vehicle used for Chandrayaan-3
(D) GSLV Mk-II uses the indigenous CE-7.5 cryogenic engine

50. Consider the following statements:
I. Aryabhata, India's first satellite, was launched in 1975.
II. Bhaskara-I was India's first experimental earth observation satellite.
III. APPLE was India's first experimental communication satellite.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

51. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Satellite) List-II (Distinction)
A. Aryabhata 1. First Indian satellite
B. Rohini RS-1 2. First Indian experimental communication satellite
C. APPLE 3. First satellite orbited by an Indian launch vehicle
D. Kalpana-1 4. First Indian dedicated meteorological satellite

(A) A-1, B-2, C-3, D-4
(B) A-1, B-3, C-2, D-4
(C) A-3, B-1, C-2, D-4
(D) A-1, B-3, C-4, D-2

52. Which one of the following statements about the institutional history of India's space programme is NOT correct? HARD

(A) INCOSPAR was set up in 1962 under the Department of Atomic Energy
(B) The Department of Space was created in 1972
(C) Vikram Sarabhai is regarded as the father of the Indian space programme
(D) Antrix Corporation was set up in 2019 as the commercial arm of ISRO

53. Consider the following statements:
I. The James Webb Space Telescope operates from the Sun-Earth L2 point.
II. NASA's Artemis-I was an uncrewed mission around the Moon.
III. China's Chang'e-6 returned the first samples from the far side of the Moon.
Which of the statements given above are correct?

(A) I and II only
(B) I, II and III
(C) II and III only
(D) I and III only

54. Which one of the following pairs of a space agency and its country is NOT correctly matched?

(A) JAXA — Japan
(B) CNSA — China
(C) Roscosmos — Russia
(D) CNES — Canada

55. Assertion (A): Most Indian remote sensing satellites are placed in sun-synchronous polar orbits.
Reason (R): A sun-synchronous orbit lets a satellite cross a given latitude at nearly the same local solar time on every pass.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

ST11

Defence Technology

50 questions · 30 fact-sheet points

Must-know facts

1. DRDO founded 1 January 1958 by merging the Technical Development Establishment, the Directorate of Technical Development & Production and the Defence Science Organisation; began with 10 laboratories, now about 52.
2. DRDO motto: 'Balasya Mulam Vigyanam' (बलस्य मूलं विज्ञानम्) — 'Strength's origin is in Science', drawn from Kautilya's Arthashastra. HQ: DRDO Bhawan, New Delhi.
3. DRDO works under the Department of Defence Research and Development; its head is Secretary DD R&D and Chairman DRDO. Five DRDO Young Scientist Laboratories opened in 2020 (AI, quantum, cognitive, asymmetric and smart-materials technologies). Aero India is held at Yelahanka, Bengaluru.
4. Key DRDO labs: DRDL & RCI & ASL Hyderabad (missiles), ADE Bengaluru (UAVs), GTRE Bengaluru (Kaveri engine), LRDE Bengaluru (radars), CABS Bengaluru (Netra AEW&C), ARDE & HEMRL Pune (armaments, explosives), TBRL Chandigarh, CVRDE Avadi (Arjun), VRDE Ahmednagar, NPOL Kochi (sonar), NSTL Visakhapatnam (torpedoes), DFRL Mysuru (food), INMAS & DIPAS & DIPR Delhi, DRDE Gwalior, DIHAR Leh, SASE Chandigarh, DIBER Haldwani.
5. Defence Laboratory Jodhpur (DLJ) is DRDO's Rajasthan laboratory — camouflage, desert environment sciences, isotopes and radiation-detection kits.
6. Lukoskin — polyherbal drug for leucoderma (vitiligo) developed by DIBER, Haldwani from Himalayan herbs; technology transferred to Aimil Pharmaceuticals.
7. Other DRDO life-science products: 2-DG anti-COVID drug (INMAS with Dr Reddy's, 2021), Oxycare oxygen system (DEBEL), retort-pouch and extreme-climate rations (DFRL), seabuckthorn beverages (DIHAR, Leh).
8. IGMDP launched 1983 under A.P.J. Abdul Kalam; five missiles — Prithvi, Agni, Trishul, Akash, Nag; formally concluded 2008.
9. Ranges: Prithvi-I 150 km, Prithvi-II 250-350 km; Agni-I ~700-900 km, Agni-II 2,000+ km, Agni-III 3,000+ km, Agni-IV ~4,000 km, Agni-V 5,000+ km, Agni-Prime 1,000-2,000 km.
10. Mission Divyastra, 11 March 2024 — first flight test of Agni-5 with MIRV (Multiple Independently Targetable Re-entry Vehicles).
11. Trishul ~9 km, Akash 25-30 km (Akash-NG ~70 km), Nag ATGM ~4 km, QRSAM ~30 km, VSHORADS/MANPADS ~6 km, Astra Mk-1 BVRAAM 100+ km, Nirbhay subsonic cruise ~1,000 km.
12. BrahMos — India-Russia joint venture (1998), named after the Brahmaputra and Moskva rivers; Mach 2.8-3, world's fastest operational supersonic cruise missile; range extended beyond the original 290 km after India joined MTCR. The Philippines was the first export customer (2022 contract, delivery 2024).
13. K-series SLBMs: K-15 Sagarika ~750 km, K-4 ~3,500 km — the sea leg of the nuclear triad on Arihant-class boats.
14. Pinaka MBRL (ARDE): Mk-I ~38 km, enhanced ~45 km, guided Pinaka 75+ km; used in the 1999 Kargil conflict. Pralay is a quasi-ballistic missile of 150-500 km. HSTDV — scramjet hypersonic demonstrator, flight tested 7 September 2020 at about Mach 6; a long-range hypersonic missile (LRAShM) was tested from Dr A.P.J. Abdul Kalam Island in November 2024.
15. Ballistic Missile Defence: Phase-I has the exo-atmospheric Prithvi Air Defence and the endo-atmospheric Advanced Air Defence. Mission Shakti (27 March 2019) destroyed a satellite at ~283 km, making India the 4th ASAT power.
16. Tejas LCA designed by ADA, built by HAL; IOC 2013, FOC 2019; Mk-1A powered by GE F404. AMCA is

India's fifth-generation stealth fighter programme, approved by the CCS in March 2024.

17. HAL platforms: Dhruv ALH, Rudra, Light Combat Helicopter 'Prachand' (inducted October 2022, first unit raised at Jodhpur), HTT-40 trainer; Su-30MKI licence-built at Nashik. C-295 transports are assembled at Vadodara with Airbus-Tata.
18. UAVs: TAPAS-BH (earlier Rustom-II), Nishant and Lakshya from ADE Bengaluru; SWiFT/Ghatak stealth demonstrator. Netra AEW&C (CABS) flies on an Embraer ERJ-145; the larger Phalcon AWACS uses IL-76.
19. Rafale — 36 aircraft from Dassault, first squadron No. 17 'Golden Arrows' at Ambala (2020); 26 Rafale-M for the Navy contracted in 2025. The MiG-21 was retired from Indian Air Force service in September 2025.
20. Navy: INS Vikrant (commissioned 2 September 2022, Cochin Shipyard, ~45,000 t) is India's first indigenously designed and built aircraft carrier; INS Vikramaditya was acquired from Russia in 2013.
21. Submarines: INS Arihant (2016) and INS Arighaat (2024) are indigenous SSBNs; the six Kalvari-class (Scorpene) diesel-electric boats under Project 75 end with INS Vagsheer (January 2025).
22. Recent commissionings: INS Nilgiri (Project 17A, Jan 2025), INS Surat (Project 15B, Jan 2025), INS Arnala (ASW shallow water craft, June 2025), INS Tamal (Talwar class, July 2025 — the last warship imported from abroad), INS Nistar (first indigenous diving support vessel, July 2025), INS Himgiri and INS Udaygiri (Project 17A, 26 August 2025).
23. Army systems: Arjun MBT (CVRDE Avadi), Zorawar light tank (DRDO with L&T), ATAGS 155 mm/52 cal towed howitzer (ARDE, ~48 km), Dhanush 155 mm/45 cal 'Desi Bofors', K-9 Vajra-T tracked self-propelled gun (L&T-Hanwha), M777 ultra-light howitzer, AK-203 rifles made at Korwa (Amethi).
24. Nuclear: Pokhran-I 'Smiling Buddha' 18 May 1974 in Jaisalmer district; Pokhran-II 'Operation Shakti' 11 and 13 May 1998 — 11 May is National Technology Day. Doctrine (2003): No First Use, credible minimum deterrence, massive retaliation, no use against non-nuclear weapon states.
25. Export-control regimes: India joined MTCR (2016), the Wassenaar Arrangement (2017) and the Australia Group (2018); India is NOT in the NSG and has not signed the NPT or CTBT. The nuclear triad was completed in 2018 with INS Arihant's first deterrence patrol.
26. Ordnance Factory Board dissolved on 1 October 2021 into 7 new DPSUs — Munitions India, Armoured Vehicles Nigam (AVANI), Advanced Weapons and Equipment India, Troop Comforts, Yantra India, India Optel and Gliders India.
27. Defence industrial corridors announced in Budget 2018-19: Uttar Pradesh (Aligarh, Agra, Jhansi, Kanpur, Chitrakoot, Lucknow) and Tamil Nadu (Chennai, Coimbatore, Hosur, Salem, Tiruchirappalli).
28. iDEX (2018) engages start-ups and MSMEs through the Defence Innovation Organisation; SRIJAN portal and the Positive Indigenisation Lists support Atmanirbharta. Defence exports reached a record of about Rs 23,622 crore in 2024-25, with a target of Rs 50,000 crore by 2029.
29. Chief of Defence Staff created 2019; Gen. Bipin Rawat first CDS (1 Jan 2020), Gen. Anil Chauhan since Sept 2022; CDS heads the Department of Military Affairs and is Permanent Chairman COSC. Tri-service formations: Andaman and Nicobar Command (2001, Port Blair) — India's first tri-service command; Integrated Defence Staff (2001); Strategic Forces Command (2003). Agnipath launched June 2022 — four-year Agniveer tenure, up to 25 per cent retention.
30. Rajasthan and defence: Pokhran Field Firing Range (nuclear tests, Exercise Bharat Shakti 2024, IAF's Vayu Shakti firepower demonstration), Mahajan Field Firing Range near Bikaner, Chandan range near Jaisalmer, Defence Laboratory Jodhpur, and the Jodhpur air base which hosted a phase of the multinational Exercise Tarang Shakti in 2024.

1. The Defence Research and Development Organisation (DRDO) was established in the year: RAS PRE 2024

(A) 1948 (B) 1958
(C) 1962 (D) 1972

2. The motto of the Defence Research and Development Organisation (DRDO) is: RAS PRE 2018

(A) Satyameva Jayate
(B) Shramev Jayate
(C) Balasya Mulam Vigyanam
(D) Yogah Karmasu Kaushalam

3. Lukoskin, a polyherbal drug developed by DRDO, is used in the treatment of: RAS PRE 2016

(A) Leucoderma (vitiligo)
(B) Malaria
(C) Tuberculosis
(D) Cataract

4. The anti-COVID-19 drug 2-deoxy-D-glucose (2-DG) was developed by which DRDO laboratory? HARD

(A) DFRL, Mysuru
(B) DIPAS, Delhi
(C) DEBEL, Bengaluru
(D) INMAS, Delhi

5. The DRDO laboratory located in Rajasthan, which works on camouflage, desert environment sciences and radiation detection, is:

(A) Defence Research Laboratory, Tezpur
(B) Defence Laboratory, Jodhpur
(C) Defence Institute of High Altitude Research, Leh
(D) Defence Food Research Laboratory, Mysuru

6. Match List-I with List-II and select the correct answer using the codes given below:
List-I (DRDO laboratory) List-II (Area of work)
A. GTRE, Bengaluru 1. Combat vehicles
B. NPOL, Kochi 2. Aero gas turbine engines
C. CVRDE, Avadi 3. Sonar and underwater systems
D. DFRL, Mysuru 4. Food technology HARD

(A) A-2, B-1, C-3, D-4
(B) A-2, B-3, C-1, D-4
(C) A-3, B-2, C-1, D-4
(D) A-2, B-3, C-4, D-1

7. Consider the following statements about DRDO:
I. It functions under the Department of Defence Research and Development in the Ministry of Defence.
II. It began in 1958 with about ten laboratories.
III. Its headquarters is located at Bengaluru.
Which of the statements given above are correct?

(A) I only
(B) I and III only
(C) I and II only
(D) I, II and III

8. The Integrated Guided Missile Development Programme (IGMDP) was launched in 1983 under the leadership of:

(A) A.P.J. Abdul Kalam
(B) Homi J. Bhabha
(C) V.S. Arunachalam
(D) Raja Ramanna

9. Which one of the following missiles was NOT a part of the Integrated Guided Missile Development Programme? RAS PRE 2016

(A) Trishul
(B) Nag
(C) BrahMos
(D) Akash

10. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Missile) List-II (Type)
A. Prithvi 1. Anti-tank guided missile
B. Akash 2. Surface-to-surface ballistic missile
C. Nag 3. Beyond-visual-range air-to-air missile
D. Astra 4. Surface-to-air missile

(A) A-2, B-4, C-3, D-1
(B) A-2, B-4, C-1, D-3
(C) A-4, B-2, C-1, D-3
(D) A-2, B-1, C-4, D-3

11. Astra, developed by DRDO, is India's first indigenous: RAS PRE 2021

(A) Anti-satellite missile
(B) Man-portable air defence system
(C) Submarine-launched ballistic missile
(D) Beyond-visual-range air-to-air missile

12. In defence technology, the term MANPADS refers to: RAS PRE 2023

(A) A man-portable air defence system
(B) A multi-role naval patrol aircraft
(C) A missile guidance radar
(D) A landmine detection device

13. The Agni-V ballistic missile has a range of more than:

(A) 3,000 km
(B) 5,000 km
(C) 8,000 km
(D) 10,000 km

14. 'Mission Divyastra', conducted in March 2024, was the first Indian flight test of a missile fitted with:

(A) A scramjet engine
(B) A hypersonic glide vehicle
(C) Multiple Independently Targetable Re-entry Vehicles
(D) A nuclear-powered propulsion unit

15. Consider the following statements about the BrahMos missile:
I. It is a joint venture between India and Russia.
II. It is a supersonic cruise missile.
III. Its name is derived from the rivers Brahmaputra and Moskva.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

16. The first foreign country to sign a contract to buy the BrahMos missile from India was:

(A) Vietnam
(B) Indonesia
(C) Malaysia
(D) the Philippines

17. DRDO's Hypersonic Technology Demonstrator Vehicle (HSTDV), successfully flight-tested in September 2020, uses which type of propulsion?

(A) Ramjet
(B) Scramjet
(C) Pulse detonation engine
(D) Ion thruster

18. Under 'Mission Shakti', conducted in March 2019, India demonstrated the capability to:

(A) Launch a submarine-based ballistic missile
(B) Intercept an incoming ballistic missile in space
(C) Destroy a satellite in low Earth orbit
(D) Fly a hypersonic glide vehicle

19. Consider the following statements about the Pinaka system:
I. It is a multi-barrel rocket launcher developed by DRDO.
II. It was used during the Kargil conflict of 1999.
III. Its guided variant has a demonstrated range of over 70 km.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

20. 'Pralay', developed by DRDO, is a: HARD

(A) Short-range quasi-ballistic surface-to-surface missile
(B) Nuclear-powered attack submarine
(C) Shoulder-fired anti-tank weapon
(D) Light combat helicopter

21. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Missile) List-II (Approximate range)
A. Prithvi-II 1. 750 km
B. Agni-III 2. 350 km
C. K-15 (Sagarika) 3. 3,000 km
D. Agni-V 4. 5,000 km HARD

(A) A-2, B-1, C-3, D-4
(B) A-2, B-3, C-1, D-4
(C) A-1, B-3, C-2, D-4
(D) A-2, B-3, C-4, D-1

22. Consider the following statements about India's Ballistic Missile Defence programme:
I. Its Phase-I comprises the exo-atmospheric Prithvi Air Defence and the endo-atmospheric Advanced Air Defence interceptors.
II. It has been developed by DRDO.
III. It was developed jointly with the United States.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

23. The Light Combat Aircraft Tejas is designed by the Aeronautical Development Agency and manufactured by:

(A) DRDO
(B) Bharat Electronics Limited
(C) Bharat Dynamics Limited
(D) Hindustan Aeronautics Limited

24. Consider the following statements about the Advanced Medium Combat Aircraft (AMCA) programme:
I. It aims to develop a fifth-generation stealth fighter.
II. It is being led by the Aeronautical Development Agency.
III. The aircraft has already entered squadron service with the Indian Air Force.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

25. The Light Combat Helicopter inducted into the Indian armed forces in October 2022 is named:

(A) Rudra
(B) Prachand
(C) Dhruv
(D) Vajra

26. TAPAS-BH, a medium-altitude long-endurance unmanned aerial vehicle, has been developed by which DRDO laboratory? HARD

(A) ARDE, Pune
(B) LRDE, Bengaluru
(C) ADE, Bengaluru
(D) IRDE, Dehradun

27. Which one of the following pairs of an aerial platform and its description is NOT correctly matched? HARD

(A) Netra — Airborne early warning and control system
(B) TAPAS — Medium-altitude long-endurance UAV
(C) Nishant — Unmanned aerial vehicle
(D) Prachand — Basic trainer aircraft

28. INS Vikrant, commissioned in September 2022, is significant as:

(A) India's first indigenously designed and built aircraft carrier
(B) India's first nuclear-powered submarine
(C) India's first stealth destroyer
(D) India's largest amphibious transport dock

29. Consider the following statements:
I. INS Arihant is India's first indigenously built nuclear-powered ballistic missile submarine.
II. INS Arighaat was commissioned in 2024.
III. The Kalvari-class submarines are nuclear-powered.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

30. INS Nistar, commissioned in July 2025, is India's first indigenously designed and built: HARD

(A) Nuclear submarine
(B) Stealth frigate
(C) Diving support vessel
(D) Landing platform dock

31. INS Himgiri and INS Udaygiri, commissioned together in August 2025, belong to which class of stealth frigates?

(A) Shivalik class (Project 17)
(B) Visakhapatnam class (Project 15B)
(C) Talwar class
(D) Nilgiri class (Project 17A)

32. Which one of the following pairs of an Indian land-warfare system and its category is NOT correctly matched?

(A) Arjun — Main battle tank
(B) Zorawar — Light tank
(C) Dhanush — Anti-tank guided missile
(D) ATAGS — Towed artillery gun system

33. Match List-I with List-II and select the correct answer using the codes given below:
List-I (System) List-II (Developing DRDO laboratory)
A. Arjun main battle tank 1. ARDE, Pune
B. Pinaka rocket system 2. CVRDE, Avadi
C. Varunastra heavyweight torpedo 3. NPOL, Kochi
D. Sonar systems for warships 4. NSTL, Visakhapatnam HARD

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-2, B-1, C-4, D-3
(D) A-2, B-4, C-1, D-3

34. The Advanced Towed Artillery Gun System (ATAGS) developed by DRDO is of which calibre?

(A) 105 mm
(B) 155 mm
(C) 130 mm
(D) 120 mm

35. Arrange the following events in Indian defence history in the correct chronological order:
I. Establishment of DRDO
II. Launch of the Integrated Guided Missile Development Programme
III. Pokhran-II nuclear tests
IV. Commissioning of the indigenous aircraft carrier INS Vikrant

(A) II, I, III, IV
(B) I, II, III, IV
(C) I, III, II, IV
(D) I, II, IV, III

36. The AK-203 assault rifles for the Indian Army are being manufactured in India at:

(A) Ishapore, West Bengal
(B) Tiruchirappalli, Tamil Nadu
(C) Jabalpur, Madhya Pradesh
(D) Korwa, Amethi, Uttar Pradesh

37. India's first nuclear test, code-named 'Smiling Buddha', was conducted in 1974 at Pokhran, which lies in which district of Rajasthan?

(A) Jaisalmer
(B) Barmer
(C) Bikaner
(D) Jodhpur

38. National Technology Day is observed on 11 May every year to commemorate:

(A) The launch of Aryabhata
(B) The first flight of Tejas
(C) The Pokhran-II nuclear tests of 1998
(D) The founding of DRDO

39. Consider the following statements about India's nuclear doctrine adopted in 2003:
I. It is based on a policy of No First Use.
II. It promises massive retaliation to a nuclear first strike.
III. It permits the first use of nuclear weapons against non-nuclear weapon states.
Which of the statements given above are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

40. Consider the following statements:
I. India joined the Missile Technology Control Regime in 2016.
II. India is a member of the Wassenaar Arrangement and the Australia Group.
III. India is a member of the Nuclear Suppliers Group.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

41. The Ordnance Factory Board was dissolved in 2021 and its units were reorganised into how many new defence public sector undertakings? HARD

(A) 4
(B) 5
(C) 7
(D) 9

42. The two defence industrial corridors announced by the Government of India are located in:

(A) Maharashtra and Gujarat
(B) Karnataka and Telangana
(C) Odisha and Andhra Pradesh
(D) Uttar Pradesh and Tamil Nadu

43. Which one of the following pairs relating to defence indigenisation is NOT correctly matched?

(A) iDEX — Engaging start-ups and MSMEs in defence innovation
(B) SRIJAN portal — Defence indigenisation
(C) Positive Indigenisation Lists — Items barred from import beyond a notified date
(D) Agnipath — Corporatisation of the ordnance factories

44. Consider the following statements about the Agnipath scheme:
I. It was announced in June 2022.
II. Agniveers are enrolled for a tenure of four years.
III. Up to 25 per cent of each batch may be retained in regular service.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

45. The Chief of Defence Staff, a post created in 2019, also functions as the head of which body?

(A) Department of Military Affairs
(B) National Security Council Secretariat
(C) Integrated Defence Staff Headquarters
(D) Strategic Forces Command

46. Which one of the following statements about India's tri-service defence structures is NOT correct?

(A) The Andaman and Nicobar Command, created in 2001, is India's first tri-service command
(B) The Strategic Forces Command was created in 2003
(C) The Chief of Defence Staff heads the Department of Military Affairs
(D) The Integrated Defence Staff was created in 2019

47. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Military exercise) List-II (Partner country)
A. Yudh Abhyas 1. France
B. Varuna 2. Russia
C. Indra 3. United States of America
D. Surya Kiran 4. Nepal

(A) A-3, B-1, C-2, D-4
(B) A-1, B-3, C-2, D-4
(C) A-3, B-2, C-1, D-4
(D) A-3, B-1, C-4, D-2

48. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Military exercise) List-II (Partner country)
A. Mitra Shakti 1. Bangladesh
B. Sampriti 2. Sri Lanka
C. Dustlik 3. Uzbekistan
D. Ekuverin 4. Maldives

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-2, B-1, C-3, D-4
(D) A-2, B-3, C-1, D-4

49. Which one of the following pairs of a joint military exercise and its partner is NOT correctly matched?

(A) Malabar — a naval exercise of India, the USA, Japan and Australia
(B) Garuda — India and France
(C) Nomadic Elephant — India and Mongolia
(D) Al Nagah — India and Egypt

50. Which one of the following pairs of a central armed police force and its year of raising is NOT correctly matched?

(A) National Security Guard — 1974
(B) Central Industrial Security Force — 1969
(C) Indo-Tibetan Border Police — 1962
(D) Border Security Force — 1965

ST12

Environment, Ecology, EIA, Biodiversity and Sustainable Development

70 questions · 30 fact-sheet points

Must-know facts

1. Ecology hierarchy: individual - population - community - ecosystem - landscape - biome - biosphere; term 'ecosystem' coined by A.G. Tansley (1935).
2. 10 per cent law of energy transfer between trophic levels - Raymond Lindeman (1942); pyramid of energy is always upright.
3. Gaseous cycles: carbon, nitrogen, oxygen (reservoir = atmosphere). Sedimentary cycles: phosphorus, sulphur (reservoir = rocks); phosphorus cycle has no gaseous phase.
4. N-fixers: Rhizobium (symbiotic, legumes), Frankia (Casuarina), Azotobacter (free-living aerobic), Clostridium (anaerobic), Anabaena/Nostoc (cyanobacteria, Azolla).
5. Ecotone = transition zone between two ecosystems; edge effect = higher species density at the ecotone. Lichens are pioneers of xerarch (bare rock) succession.
6. Biodiversity hotspot criteria (Norman Myers, 1988): at least 1,500 endemic vascular plants AND at least 70% of primary vegetation lost. Four hotspots reach India: Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland (Nicobar).
7. Alpha diversity = within a community/habitat; Beta = turnover between communities along a gradient; Gamma = total diversity of a landscape or region.
8. IUCN (founded 1948, HQ Gland, Switzerland) Red List has 9 categories; 'threatened' = Critically Endangered + Endangered + Vulnerable.
9. Biosphere reserve zones: core (legally protected) - buffer - transition. India's first biosphere reserve: Nilgiri (1986). Rajasthan has none.
10. Project Tiger 1973 (nine reserves incl. Ranthambore); NTCA statutory in 2006 under the Wild Life (Protection) Act, 1972. Project Elephant 1992; Project Snow Leopard 2009; Project Dolphin 2020.
11. Rajasthan tiger reserves: Ranthambore (Sawai Madhopur), Sariska (Alwar), Mukundra Hills (Kota, 2013), Ramgarh Vishdhari (Bundi, 2022 - India's 52nd), Dholpur-Karauli (2023 - India's 54th).
12. Ramsar sites of Rajasthan (five): Keoladeo Ghana NP (1981), Sambhar Lake (1990), Menar (2025), Khichan (2025) and Siliserh Lake, Alwar (2025).
13. Wild Life (Protection) Act, 1972 (Act 53 of 1972): 2022 amendment (in force 1 April 2023) cut schedules from six to four - Sch. I & II animals, Sch. III plants, Sch. IV CITES specimens.
14. Biological Diversity Act 2002: three-tier - National Biodiversity Authority (Chennai), State Biodiversity Boards, Biodiversity Management Committees (People's Biodiversity Registers). 2023 amendment exempts registered AYUSH practitioners and decriminalises offences.
15. Forest (Conservation) Act 1980 renamed Van (Sanrakshan Evam Samvardhan) Adhiniyam 1980 by the 2023 amendment; exempts land within 100 km of international borders for strategic linear projects and 10 ha for security infrastructure.
16. Forest Rights Act 2006 - nodal Ministry of Tribal Affairs; Gram Sabha is the authority; cut-off date 13 December 2005.
17. Water Act 1974 created CPCB/SPCBs; Air Act 1981; Environment (Protection) Act 1986 (umbrella Act, under Article 253, following Stockholm 1972, after the 1984 Bhopal disaster); Environment Audit Rules notified 29 August 2025.
18. National Green Tribunal Act 2010 - NGT set up 18 October 2010, principal bench New Delhi, zonal benches Bhopal, Pune, Kolkata, Chennai; disposal within 6 months; India was the third country with a dedicated green court.

19. National AQI (2014): 6 categories - Good 0-50, Satisfactory 51-100, Moderate 101-200, Poor 201-300, Very Poor 301-400, Severe 401-500; 8 pollutants (PM_{10}, $PM_{2.5}$, NO_2, SO_2, CO, O_3, NH_3, Pb).
20. Acid rain = SO_2 + NO_x converting to H_2SO_4 and HNO_3; rain with pH below 5.6 is acidic. NCAP launched January 2019; revised target 40% cut in particulate concentration by 2025-26 over 2017.
21. Greenhouse contribution order: CO_2 (~60%) > CH_4 (~20%) > CFCs (~14%) > N_2O (~6%). Highest 100-year GWP: SF_6.
22. BOD measured over 5 days at 20 degrees C; COD always exceeds BOD. Mercury-Minamata, Cadmium-Itai-itai, Nitrate-blue baby syndrome, Fluoride-fluorosis.
23. Single-use plastic ban effective 1 July 2022; carry bag thickness 120 microns from 31 December 2022; E-Waste (Management) Rules 2022 in force 1 April 2023; EPR = Extended Producer Responsibility.
24. Ozone: Vienna Convention 1985, Montreal Protocol 1987 (World Ozone Day 16 September), Kigali Amendment 2016 phases down HFCs (in force 1 January 2019; India ratified 2021).
25. Climate diplomacy: IPCC 1988 (WMO+UNEP); UNFCCC Rio 1992; Kyoto 1997; Paris 2015 (well below 2 C, pursue 1.5 C); COP26 Glasgow - Panchamrit and net-zero 2070; COP28 Dubai - first Global Stocktake; COP29 Baku - NCQG USD 300 bn/yr by 2035; COP30 Belem 2025 - 'global mutirao'; COP31 Antalya (Turkiye) 2026.
26. India's NDC: 45% emissions-intensity cut and 50% non-fossil capacity by 2030; new NDC (2031-35) approved 25 March 2026 - 47% intensity cut and 60% non-fossil capacity by 2035, net-zero 2070.
27. NAPCC (2008) - eight missions: Solar, Enhanced Energy Efficiency, Sustainable Habitat, Water, Sustaining the Himalayan Ecosystem, Green India, Sustainable Agriculture, Strategic Knowledge for Climate Change.
28. ISA launched 2015 (India-France), HQ Gurugram - first treaty-based international organisation headquartered in India; CDRI launched 2019, HQ New Delhi; Mission LiFE launched October 2022 at Kevadia.
29. EIA Notification 2006 (under EP Act 1986): stages Screening - Scoping - Public Consultation - Appraisal; Category A cleared by MoEFCC/EAC, Category B by SEIAA/SEAC (B1 needs EIA, B2 does not); public hearing conducted by the SPCB.
30. Disaster Management Act 2005: NDMA chaired by PM, SDMA by CM, DDMA by District Collector; NEC headed by Union Home Secretary; NDRF and NIDM created under it; Sendai Framework 2015-2030 has 4 priorities and 7 global targets.

1. The '10 per cent law' of energy transfer between successive trophic levels in an ecosystem was proposed by:

(A) Raymond Lindeman (B) Charles Elton
(C) Arthur Tansley (D) Eugene Odum

2. Which one of the following ecological pyramids is always upright?

(A) Pyramid of numbers (B) Pyramid of energy
(C) Pyramid of biomass (D) Pyramid of standing crop

3. Consider the following statements about food chains:
I. The grazing food chain begins with green plants.
II. The detritus food chain begins with dead organic matter.
III. In terrestrial ecosystems, a much larger fraction of energy flows through the detritus food chain than through the grazing food chain.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

4. Which one of the following pairs of an ecological category and its example is NOT correctly matched?

(A) Keystone species — Sea otter
(B) Indicator species — Lichen
(C) Flagship species — Bengal tiger
(D) Invasive species — Great Indian Bustard

5. The zone of transition between two adjacent ecosystems, which usually supports greater species richness than either of them, is called:

(A) Ecotone
(B) Ecotype
(C) Ecad
(D) Ecocline

6. Which one of the following biogeochemical cycles is a purely sedimentary cycle with no significant gaseous phase?

(A) Nitrogen cycle
(B) Phosphorus cycle
(C) Carbon cycle
(D) Oxygen cycle

7. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Organism) List-II (Mode of nitrogen fixation)

A. Rhizobium 1. Symbiotic with roots of legumes

B. Frankia 2. Symbiotic with Casuarina

C. Azotobacter 3. Free-living aerobic bacterium

D. Anabaena 4. Cyanobacterium associated with Azolla

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-3, B-2, C-1, D-4

8. In primary ecological succession on a bare rock surface, the pioneer community is normally formed by:

(A) Mosses
(B) Grasses
(C) Shrubs
(D) Lichens

9. The diversity of species found within a single community or habitat is termed: RAS PRE 2018

(A) Beta diversity
(B) Alpha diversity
(C) Gamma diversity
(D) Delta diversity

10. Consider the following statements:

I. Beta diversity refers to the change in species composition between communities along an environmental gradient.

II. Gamma diversity refers to the total species diversity of a large geographical region or landscape.

III. Alpha diversity of a habitat is always greater than the gamma diversity of the region containing it.

Which of the statements given above are correct? RAS PRE 2016

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

11. Which one of the following biodiversity hotspots does NOT extend into Indian territory? RAS PRE 2013

(A) Himalaya
(B) Indo-Burma
(C) Sundaland
(D) Cerrado

12. The concept of 'biodiversity hotspots' was first proposed in 1988 by: HARD

(A) Norman Myers
(B) Robert Whittaker
(C) Eugene Odum
(D) Edward O. Wilson

13. Which one of the following is NOT a criterion for designating an area as a biodiversity hotspot? HARD

(A) It must contain at least 500 endemic vertebrate species
(B) It must contain at least 1,500 species of endemic vascular plants
(C) It must have lost at least 70 per cent of its original primary vegetation
(D) It is identified on the basis of both endemism and degree of threat

14. In the IUCN Red List, the three categories that together constitute 'threatened' species are:

(A) Extinct, Extinct in the Wild and Critically Endangered
(B) Critically Endangered, Endangered and Vulnerable
(C) Endangered, Vulnerable and Near Threatened
(D) Vulnerable, Near Threatened and Least Concern

15. Consider the following statements:
I. National parks and biosphere reserves are examples of in-situ conservation.
II. Cryopreservation of gametes is a form of ex-situ conservation.
III. Botanical gardens and seed banks are examples of ex-situ conservation.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

16. The three zones into which a biosphere reserve is conventionally divided are:

(A) Core, buffer and transition
(B) Core, sanctuary and buffer
(C) Nucleus, mantle and periphery
(D) Reserve, restricted and open

17. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Biosphere reserve) List-II (State)
A. Nokrek 1. Meghalaya
B. Simlipal 2. Odisha
C. Dihang-Dibang 3. Arunachal Pradesh
D. Agasthyamalai 4. Kerala and Tamil Nadu

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

18. Consider the following statements about Project Tiger:
I. It was launched in 1973.
II. The National Tiger Conservation Authority was given statutory status in 2006.
III. Ranthambore was one of the nine tiger reserves created at the launch of the project.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

19. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Tiger reserve of Rajasthan) List-II (District)
A. Ranthambore 1. Sawai Madhopur
B. Sariska 2. Alwar
C. Mukundra Hills 3. Kota
D. Ramgarh Vishdhari 4. Bundi RAS PRE 2024

(A) A-1, B-2, C-4, D-3
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-3, D-4
(D) A-1, B-3, C-2, D-4

20. Which of the following pairs of Rajasthan wetlands were the state's first two Ramsar sites? RAS PRE 2016, 2023

(A) Menar and Khichan
(B) Sambhar and Menar
(C) Keoladeo and Sambhar
(D) Keoladeo and Khichan

21. The National Tiger Conservation Authority is a statutory body constituted under:

(A) The Environment (Protection) Act, 1986
(B) The Biological Diversity Act, 2002
(C) The Wild Life (Protection) Act, 1972
(D) The Forest (Conservation) Act, 1980

22. After the Wild Life (Protection) Amendment Act, 2022, the number of schedules in the Wild Life (Protection) Act, 1972 was reduced to:

(A) Two
(B) Four
(C) Five
(D) Six

23. Consider the following statements about the schedules of the Wild Life (Protection) Act, 1972 after the 2022 amendment:
I. Schedules I and II list animal species.
II. Schedule III lists plant species.
III. Schedule IV lists specimens included in the appendices of CITES.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

24. The National Biodiversity Authority constituted under the Biological Diversity Act, 2002 has its headquarters at:

(A) Chennai
(B) New Delhi
(C) Hyderabad
(D) Bengaluru

25. Consider the following statements about the Biological Diversity (Amendment) Act, 2023:
I. It exempts registered AYUSH practitioners from sharing benefits for the use of codified traditional knowledge.
II. It decriminalises offences under the Act and replaces imprisonment with monetary penalties.
III. It abolishes Biodiversity Management Committees at the local level.
Which of the statements given above are correct? HARD

(A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

26. Under the Biological Diversity Act, 2002, the body constituted at the local body level is the:

(A) State Biodiversity Board
(B) Biodiversity Heritage Committee
(C) Biodiversity Management Committee
(D) People's Biodiversity Council

27. The Forest (Conservation) Amendment Act, 2023 renamed the parent Act of 1980 as:

(A) Bharatiya Van Adhiniyam, 1980
(B) Rashtriya Van Adhiniyam, 1980
(C) Van Samvardhan Adhiniyam, 1980
(D) Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980

28. Consider the following statements about the Forest (Conservation) Amendment Act, 2023:
I. Forest land within 100 km of an international border proposed for a strategic linear project of national importance is exempted from prior central approval.
II. Up to 10 hectares of forest land proposed for security-related infrastructure is exempted.
III. The Act was passed to give statutory backing to the Compensatory Afforestation Fund.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

29. The nodal ministry for implementation of the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 is the:

(A) Ministry of Environment, Forest and Climate Change
(B) Ministry of Tribal Affairs
(C) Ministry of Rural Development
(D) Ministry of Panchayati Raj

30. The Environment (Protection) Act, 1986 was enacted by Parliament under Article 253 of the Constitution to implement the decisions taken at the:

(A) Rio Earth Summit, 1992
(B) Nairobi Conference, 1982
(C) United Nations Conference on the Human Environment, Stockholm, 1972
(D) Vienna Convention, 1985

31. The Central Pollution Control Board was constituted under which one of the following Acts?

(A) The Environment (Protection) Act, 1986
(B) The Air (Prevention and Control of Pollution) Act, 1981
(C) The Factories Act, 1948
(D) The Water (Prevention and Control of Pollution) Act, 1974

32. The National Green Tribunal, established under the NGT Act, 2010, has its principal bench at:

(A) New Delhi
(B) Bhopal
(C) Pune
(D) Kolkata

33. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Legislation) List-II (Year of enactment)
A. Water (Prevention and Control of Pollution) Act 1. 1974
B. Air (Prevention and Control of Pollution) Act 2. 1981
C. Environment (Protection) Act 3. 1986
D. National Green Tribunal Act 4. 2010

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-3, B-2, C-1, D-4
(D) A-1, B-2, C-3, D-4

34. Acid rain is caused mainly by the atmospheric emission of which pair of pollutants? RAS PRE 2016, 2018

(A) Sulphur dioxide and oxides of nitrogen
(B) Carbon dioxide and methane
(C) Carbon monoxide and ozone
(D) Chlorofluorocarbons and nitrous oxide

35. The National Air Quality Index of India, launched in 2014, classifies air quality into how many categories?

(A) Five
(B) Six
(C) Four
(D) Seven

36. Match List-I with List-II and select the correct answer using the codes given below:
List-I (AQI value range) List-II (Category)
A. 0-50 1. Good
B. 51-100 2. Satisfactory
C. 201-300 3. Poor
D. 401-500 4. Severe HARD

(A) A-1, B-2, C-4, D-3
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-3, D-4
(D) A-1, B-3, C-2, D-4

37. Consider the following statements about the National Clean Air Programme:
I. It was launched in January 2019.
II. Its revised target is a 40 per cent reduction in particulate matter concentration by 2025-26 over the 2017 level.
III. It is implemented by the Ministry of Road Transport and Highways.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I, II and III
(D) I and II only

38. Which one of the following pollutants is NOT included in the computation of India's National Air Quality Index? HARD

(A) Methane
(B) Ammonia
(C) Ground-level ozone
(D) Lead

39. A high biochemical oxygen demand (BOD) value of a water sample indicates:

(A) Low content of biodegradable organic matter
(B) High content of biodegradable organic matter
(C) High dissolved oxygen content
(D) Complete absence of microorganisms

40. Consider the following statements about eutrophication:
I. It is caused by enrichment of a water body with nutrients, chiefly nitrates and phosphates.
II. Algal blooms raise the dissolved oxygen level of the water body.
III. Eutrophication can cause large-scale fish kills.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

41. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Pollutant) List-II (Disease)
A. Mercury 1. Minamata disease
B. Cadmium 2. Itai-itai disease
C. Fluoride 3. Skeletal fluorosis
D. Nitrate 4. Blue baby syndrome

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-3, C-2, D-4
(D) A-1, B-2, C-3, D-4

42. The ban on manufacture, sale and use of identified single-use plastic items in India came into effect from: HARD

(A) 1 July 2022
(B) 2 October 2019
(C) 1 April 2021
(D) 1 January 2023

43. Which one of the following pairs of a programme and its year of launch is NOT correctly matched?

(A) Namami Gange — 2014
(B) National Action Plan on Climate Change — 2005
(C) National Clean Air Programme — 2019
(D) Swachh Bharat Mission — 2014

44. Which one of the following ambient noise standards for day time, prescribed under the Noise Pollution (Regulation and Control) Rules, 2000, is NOT correctly stated? HARD

(A) Industrial area — 75 dB(A) Leq
(B) Commercial area — 65 dB(A) Leq
(C) Silence zone — 60 dB(A) Leq
(D) Residential area — 55 dB(A) Leq

45. Which one of the following gases makes the largest contribution to the enhanced greenhouse effect? RAS PRE 2018

(A) Methane
(B) Nitrous oxide
(C) Carbon dioxide
(D) Chlorofluorocarbons

46. Arrange the following greenhouse gases in descending order of their contribution to the enhanced greenhouse effect:
I. Methane
II. Carbon dioxide
III. Nitrous oxide
IV. Chlorofluorocarbons RAS PRE 2016 HARD

(A) II, I, III, IV
(B) I, II, IV, III
(C) II, IV, I, III
(D) II, I, IV, III

47. The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 by:

(A) The World Meteorological Organization and UNEP
(B) UNESCO and FAO
(C) UNDP and the World Bank
(D) The UNFCCC and UNEP

48. The United Nations Framework Convention on Climate Change was adopted at the Earth Summit held in 1992 at:

(A) Stockholm
(B) Rio de Janeiro
(C) Kyoto
(D) Johannesburg

49. Arrange the following in correct chronological order:
I. Montreal Protocol
II. Stockholm Conference on the Human Environment
III. Paris Agreement
IV. Kyoto Protocol

(A) I, II, IV, III
(B) II, I, III, IV
(C) II, I, IV, III
(D) I, II, III, IV

50. The Paris Agreement, 2015 commits the parties to holding the increase in global average temperature to:

(A) Below 3 degrees C above pre-industrial levels
(B) Below 2.5 degrees C above pre-industrial levels
(C) Exactly 2 degrees C above 1990 levels
(D) Well below 2 degrees C, and to pursue efforts to limit it to 1.5 degrees C, above pre-industrial levels

51. The 30th Conference of the Parties (COP30) to the UNFCCC was held in November 2025 at:

(A) Belem, Brazil
(B) Baku, Azerbaijan
(C) Dubai, United Arab Emirates
(D) Antalya, Turkiye

52. Consider the following statements about recent climate conferences:
I. COP28 at Dubai in 2023 concluded the first Global Stocktake under the Paris Agreement.
II. COP29 at Baku in 2024 agreed a New Collective Quantified Goal on climate finance of at least USD 300 billion a year by 2035.
III. COP30 at Belem adopted a binding roadmap for transitioning away from fossil fuels.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

53. India announced its target of achieving net-zero emissions by 2070, as part of the 'Panchamrit', at:

(A) COP21, Paris
(B) COP24, Katowice
(C) COP26, Glasgow
(D) COP28, Dubai

54. Consider the following statements about India's Nationally Determined Contribution for 2031-2035:
I. India will reduce the emissions intensity of its GDP by 47 per cent by 2035 from the 2005 level.
II. India will achieve about 60 per cent of cumulative electric power installed capacity from non-fossil fuel sources by 2035.
III. India's declared net-zero target year remains 2070.
Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

55. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Mission under the NAPCC) List-II (Nodal ministry)
A. National Solar Mission 1. Ministry of New and Renewable Energy
B. National Water Mission 2. Ministry of Jal Shakti
C. National Mission for a Green India 3. Ministry of Environment, Forest and Climate Change
D. National Mission for Enhanced Energy Efficiency 4. Ministry of Power

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-1, B-3, C-2, D-4

56. Which one of the following is NOT one of the eight missions under the National Action Plan on Climate Change?

(A) National Water Mission
(B) National Clean Air Mission
(C) National Mission on Sustainable Habitat
(D) National Mission for Sustaining the Himalayan Ecosystem

57. The International Solar Alliance, jointly launched by India and France in 2015, has its headquarters at:

(A) New Delhi
(B) Abu Dhabi
(C) Gurugram, Haryana
(D) Paris

58. The Environmental Impact Assessment Notification currently governing environmental clearance in India was issued in:

(A) 1994
(B) 2002
(C) 2006
(D) 2020

59. Consider the following statements about the EIA Notification, 2006:
I. Category 'A' projects are appraised at the central level by an Expert Appraisal Committee.
II. Category 'B1' projects require preparation of an EIA report, whereas 'B2' projects do not.
III. The public hearing is conducted by the Ministry of Environment, Forest and Climate Change.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

60. Under the EIA Notification, 2006, the public hearing for a project is conducted by the:

(A) Expert Appraisal Committee
(B) Ministry of Environment, Forest and Climate Change
(C) National Green Tribunal
(D) State Pollution Control Board

61. Environmental clearance for Category 'B' projects under the EIA Notification, 2006 is granted by the: HARD

(A) State Environment Impact Assessment Authority
(B) Central Pollution Control Board
(C) Ministry of Environment, Forest and Climate Change
(D) District Collector

62. Which one of the following is NOT one of the four stages of the environmental clearance process prescribed under the EIA Notification, 2006?

(A) Screening
(B) Cost-benefit certification
(C) Scoping
(D) Appraisal

63. The Montreal Protocol on Substances that Deplete the Ozone Layer was adopted to give effect to which one of the following framework conventions?

(A) The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes
(B) The Stockholm Convention on Persistent Organic Pollutants
(C) The Vienna Convention for the Protection of the Ozone Layer
(D) The Rotterdam Convention on Prior Informed Consent

64. The Montreal Protocol of 1987 deals with:

(A) Reduction of greenhouse gas emissions
(B) Phasing out of ozone-depleting substances
(C) Transboundary movement of hazardous waste
(D) Conservation of migratory species

65. The classic definition of sustainable development as development that meets the needs of the present without compromising the ability of future generations to meet their own needs was given in:

(A) Agenda 21, 1992
(B) The Limits to Growth, 1972
(C) Our Common Future (Brundtland Report), 1987
(D) The Future We Want, 2012

66. Consider the following statements about the Sustainable Development Goals:
I. There are 17 goals and 169 targets.
II. They were adopted in September 2015 as part of the 2030 Agenda for Sustainable Development.
III. In India, the SDG India Index is published by NITI Aayog.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

67. Consider the following statements about the Disaster Management Act, 2005:
I. It provides for the constitution of the National Institute of Disaster Management.
II. It provides for the constitution of the National Disaster Response Force.
III. The National Executive Committee under the Act is chaired by the Union Home Secretary.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

68. India's first National Disaster Management Plan, aligned with the Sendai Framework, was released in:

(A) 2016
(B) 2009
(C) 2013
(D) 2019

69. The Sendai Framework for Disaster Risk Reduction, adopted in 2015, covers the period: HARD

(A) 2015-2025
(B) 2016-2030
(C) 2015-2035
(D) 2015-2030

70. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Body under the Disaster Management Act, 2005) List-II (Ex-officio head)

A. National Disaster Management Authority 1. Prime Minister

B. State Disaster Management Authority 2. Chief Minister

C. District Disaster Management Authority 3. District Collector

D. National Executive Committee 4. Union Home Secretary

- (A) A-1, B-2, C-3, D-4
- (B) A-1, B-2, C-4, D-3
- (C) A-2, B-1, C-3, D-4
- (D) A-1, B-3, C-2, D-4

ST13

Agriculture, Horticulture, Forestry, Animal Husbandry (Rajasthan focus); S&T Policy and Indian Scientists

65 questions · 30 fact-sheet points

Must-know facts

1. Soil Health Card Scheme launched by the Prime Minister on 19 February 2015 from Suratgarh, Sri Ganganagar (Rajasthan).
2. Nutrient deficiency symptoms: nitrogen - chlorosis of older leaves; phosphorus - purplish tinge; potassium - scorching of leaf margins; zinc - khaira disease in rice. Calcium, magnesium and sulphur are secondary macronutrients.
3. Seed multiplication chain: breeder seed (golden-yellow tag) - foundation seed (white tag) - certified seed (azure-blue tag); truthfully labelled seed carries an opal-green label.
4. Rajasthan leads India in bajra, rapeseed-mustard, guar, coriander, cumin, fenugreek, isabgol, henna and moth bean, and is the largest wool producer and holder of the largest camel population.
5. HHB-67 Improved is an early-maturing, downy-mildew-resistant bajra hybrid; RGC-936 and RGC-1003 are guar varieties; Pusa Bold and Bio-902 are mustard varieties.
6. ICAR-Directorate of Rapeseed-Mustard Research is at Sewar, Bharatpur (Rajasthan).
7. Desert locust = Schistocerca gregaria. The Locust Warning Organisation is headquartered at Jodhpur under the Directorate of Plant Protection, Quarantine and Storage, Faridabad; the Scheduled Desert Area lies in Rajasthan and Gujarat.
8. PMKSY (2015): 'Har Khet Ko Pani' and 'Per Drop More Crop' (micro-irrigation). Namo Drone Didi provides agricultural drones to women's self-help groups.
9. Khadin (Jaisalmer) is a traditional runoff-harvesting cropping system; other traditional structures are tanka, nadi, johad, baori and toba.
10. Seed spices of Rajasthan: coriander, cumin, fennel, fenugreek, ajwain, dill, celery and nigella; Rajasthan and Gujarat together dominate India's seed-spice output.
11. ICAR-National Research Centre on Seed Spices, Tabiji, Ajmer - established 2000.
12. ICAR-National Research Centre on Camel, Bikaner - established 1984; ICAR-Central Institute for Arid Horticulture, Bikaner - established 1993.
13. ICAR-Central Arid Zone Research Institute (CAZRI), Jodhpur - 1959; ICAR-Central Sheep and Wool Research Institute (CSWRI), Avikanagar, Malpura tehsil, Tonk district - 1962.
14. Arid Forest Research Institute (AFRI), Jodhpur - under ICFRE; Rajasthan Remote Sensing Centre, Jodhpur; Ch. Charan Singh National Institute of Agricultural Marketing (NIAM), Jaipur.
15. CSIR-Central Electronics Engineering Research Institute (CEERI), Pilani is the only CSIR laboratory in Rajasthan; BITS Pilani is a leading private deemed university of the state.
16. Udaipur Solar Observatory, on an island in Fateh Sagar Lake, was founded in 1975 by Dr. Arvind Bhatnagar and is operated by the Physical Research Laboratory, Ahmedabad; it hosts the Multi-Application Solar Telescope (MAST).
17. Agricultural universities: MPUAT Udaipur, SKRAU Bikaner, SKN Agriculture University Jobner, Agriculture University Jodhpur, Agriculture University Kota, RAJUVAS Bikaner; RARI is at Durgapura, Jaipur.
18. Sojat (Pali) is the henna (Lawsonia inermis) capital and holds a GI tag; isabgol is Plantago ovata; ashwagandha is Withania somnifera; safed musli is Chlorophytum borivilianum; guggul is Commiphora wightii.

19. Khejri (Prosopis cineraria) is the state tree, 'kalpavriksha of the desert', yielding sangri pods; Rohida (Tecomella undulata) is the state flower, called Marwar teak; Prosopis juliflora is the invasive 'vilayati babool'.
20. Kinnow belt - Sri Ganganagar and Hanumangarh; olive plantations - Lunkaransar (Bikaner) and Bassi (Jaipur); orange - Jhalawar; date palm - Jaisalmer and Bikaner; pomegranate - Barmer and Jalore.
21. Cattle breeds: Rathi (Bikaner-Sri Ganganagar, best milch), Tharparkar (Jaisalmer-Barmer), Kankrej (Jalore-Sirohi), Nagauri (Nagaur, draught), Malvi (Jhalawar), Hariana (Bharatpur-Alwar).
22. Sheep breeds: Chokla (Shekhawati, finest wool), Magra (Bikaner-Jaisalmer, lustrous carpet wool), Marwari (Jodhpur-Pali), Nali (Sri Ganganagar-Churu), Sonadi (Udaipur-Dungarpur), Pugal, Jaisalmeri, Malpuri.
23. Goat breeds: Sirohi (meat), Jakhrana of Alwar (milk), Marwari, Barbari. Camel breeds: Bikaneri, Jaisalmeri (riding), Mewari (milk), Kachchhi; camel declared state animal (domestic) in 2014 and protected by a 2015 state Act.
24. Livestock disease control: NADCP (2019) targets foot-and-mouth disease and brucellosis; Rashtriya Gokul Mission (2014) develops indigenous bovine breeds; lumpy skin disease is caused by a capripoxvirus.
25. National Forest Policy 1988 targets 33 per cent of geographical area under forest and tree cover; Joint Forest Management circular issued in 1990; Van Dhan Vikas Kendras are run by TRIFED under the Ministry of Tribal Affairs.
26. Science days: National Science Day 28 February (Raman Effect, 1928), National Technology Day 11 May (Pokhran-II, 1998), National Space Day 23 August (Chandrayaan-3, 2023), National Mathematics Day 22 December (Ramanujan).
27. S&T institutions: CSIR founded 1942, DST 1971, DSIR and DBT in the 1980s; STIP 2020 is India's fifth S&T policy; the Anusandhan National Research Foundation was created by an Act of 2023 with the Prime Minister as ex-officio President.
28. Indian Nobel laureates in science: C.V. Raman (Physics 1930), Har Gobind Khorana (Physiology or Medicine 1968), S. Chandrasekhar (Physics 1983), Venkatraman Ramakrishnan (Chemistry 2009).
29. Scientists and contributions: Meghnad Saha - thermal ionisation equation; S.N. Bose - Bose-Einstein statistics; J.C. Bose - crescograph; Homi Bhabha - Bhabha scattering and the three-stage nuclear programme; Vikram Sarabhai - PRL and ISRO; Birbal Sahni - palaeobotany; Salim Ali - ornithology.
30. Daulat Singh Kothari, born at Udaipur, chaired the Education Commission of 1964-66 and was the first Scientific Adviser to the Ministry of Defence.

1. The Soil Health Card Scheme was launched by the Prime Minister in February 2015 from which place in Rajasthan?

(A) Suratgarh, Sri Ganganagar
(B) Jobner, Jaipur
(C) Durgapura, Jaipur
(D) Tabiji, Ajmer

2. Yellowing of older leaves, known as chlorosis, is the classic symptom of the deficiency of which nutrient in crops?

(A) Potassium
(B) Nitrogen
(C) Phosphorus
(D) Sulphur

3. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Nutrient) List-II (Deficiency symptom)
A. Nitrogen 1. General chlorosis of older leaves
B. Phosphorus 2. Purplish tinge on leaves and stems
C. Potassium 3. Scorching of leaf margins
D. Zinc 4. Khaira disease of rice

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-1, B-3, C-2, D-4

4. Which one of the following is NOT a micronutrient for plants?

(A) Zinc
(B) Boron
(C) Molybdenum
(D) Calcium

5. In India's seed certification system, the colour of the tag used for certified seed is: HARD

(A) Golden yellow
(B) White
(C) Opal green
(D) Azure blue

6. Arrange the following classes of seed in the correct order of the seed multiplication chain:
I. Certified seed
II. Breeder seed
III. Foundation seed

(A) II, III, I
(B) III, II, I
(C) I, II, III
(D) II, I, III

7. Consider the following statements:
I. Rajasthan is India's largest producer of bajra.
II. Rajasthan is India's largest producer of rapeseed-mustard.
III. Rajasthan is India's largest producer of sugarcane.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

8. Consider the following statements about guar (cluster bean) in Rajasthan:
I. Rajasthan accounts for the bulk of India's guar production.
II. Guar gum is an important export item used in the oil and gas industry.
III. Guar is grown in Rajasthan as a rabi crop.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

9. HHB-67 Improved, an early-maturing and downy-mildew-resistant variety widely grown in Rajasthan, belongs to which crop?

(A) Mustard
(B) Guar
(C) Barley
(D) Bajra

10. The ICAR Directorate of Rapeseed-Mustard Research is located at:

(A) Bharatpur, Rajasthan
(B) Hisar, Haryana
(C) Indore, Madhya Pradesh
(D) Kanpur, Uttar Pradesh

11. Which one of the following is NOT a kharif crop of Rajasthan?

(A) Bajra
(B) Barley
(C) Moth bean
(D) Guar

12. Inclusion of a legume in a crop rotation chiefly improves soil fertility by:

(A) Increasing the soil pH
(B) Adding phosphorus to the soil
(C) Fixing atmospheric nitrogen through root nodule bacteria
(D) Increasing the clay content of the soil

13. Consider the following statements about dryland farming:
I. It is practised in areas receiving less than about 750 mm of annual rainfall.
II. Conservation of soil moisture is its central objective.
III. Mulching reduces evaporation losses from the soil surface.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

14. Consider the following statements about locusts in Rajasthan:
I. The species responsible for swarming attacks in western Rajasthan is Schistocerca gregaria, the desert locust.
II. The Locust Warning Organisation is headquartered at Jodhpur.
III. The Scheduled Desert Area for locust control lies in Rajasthan and Gujarat.
Which of the statements given above are correct?

(A) I, II and III
(B) I and II only
(C) II and III only
(D) I and III only

15. The headquarters of India's Locust Warning Organisation is at:

(A) Bikaner
(B) Jodhpur
(C) Faridabad
(D) Jaisalmer

16. 'Per Drop More Crop', the micro-irrigation component promoting drip and sprinkler systems, is part of which scheme?

(A) Rashtriya Krishi Vikas Yojana
(B) National Food Security Mission
(C) Pradhan Mantri Krishi Sinchayee Yojana
(D) Paramparagat Krishi Vikas Yojana

17. Consider the following statements about Integrated Pest Management:
I. It gives priority to cultural, mechanical and biological methods over chemical pesticides.
II. Trichogramma is used in it as a biological control agent.
III. Its objective is the complete eradication of the pest species.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

18. Which one of the following is NOT a protected cultivation structure?

(A) Poly-house
(B) Shade net house
(C) Walk-in tunnel
(D) Khadin

19. The 'Namo Drone Didi' scheme provides agricultural drones to:

(A) Women's self-help groups
(B) Krishi Vigyan Kendras
(C) Agricultural universities
(D) Primary agricultural credit societies

20. Sprinkler irrigation is particularly suited to the sandy soils of western Rajasthan mainly because it:

(A) Raises the water table
(B) Increases soil salinity
(C) Requires no external energy
(D) Reduces conveyance and deep percolation losses

21. The ICAR National Research Centre on Seed Spices is located at: RAS PRE 2023

(A) Jodhpur
(B) Tabiji, Ajmer
(C) Bikaner
(D) Kota

22. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Institute) List-II (Location)
A. National Research Centre on Seed Spices 1. Ajmer
B. Central Institute for Arid Horticulture 2. Bikaner
C. Central Arid Zone Research Institute 3. Jodhpur
D. Central Sheep and Wool Research Institute 4. Avikanagar (Tonk) RAS PRE 2018, 2023

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-1, B-2, C-3, D-4
(D) A-3, B-2, C-1, D-4

23. Which one of the following groups consists entirely of seed spices grown in Rajasthan?

(A) Coriander, cumin, fennel and fenugreek
(B) Turmeric, ginger, cardamom and clove
(C) Chilli, pepper, cinnamon and nutmeg
(D) Saffron, vanilla, mace and asafoetida

24. Sojat in Rajasthan, whose produce carries a Geographical Indication tag, is famous for:

(A) Cumin
(B) Isabgol
(C) Kinnow
(D) Henna (mehndi)

25. Isabgol, of which Rajasthan is the leading producer in India, is obtained from the plant:

(A) Plantago ovata
(B) Cassia angustifolia
(C) Chlorophytum borivilianum
(D) Withania somnifera

26. Which one of the following pairs of a horticultural crop and its main growing area in Rajasthan is NOT correctly matched?

(A) Kinnow — Sri Ganganagar
(B) Henna — Pali
(C) Olive — Jhalawar
(D) Isabgol — Jalore

27. Consider the following statements about the Central Institute for Arid Horticulture, Bikaner:
I. It works on arid fruits such as ber, date palm, aonla and pomegranate.
II. It was established in 1993 as the National Research Centre for Arid Horticulture.
III. It functions under the Ministry of Environment, Forest and Climate Change.
Which of the statements given above are correct? RAS PRE 2016

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

28. Date palm cultivation in Rajasthan, promoted through tissue-culture plantations, is concentrated mainly in the districts of: HARD

(A) Kota and Baran
(B) Bharatpur and Dholpur
(C) Udaipur and Banswara
(D) Jaisalmer and Bikaner

29. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Plant) List-II (Botanical name)
A. Henna 1. Lawsonia inermis
B. Ashwagandha 2. Withania somnifera
C. Safed musli 3. Chlorophytum borivilianum
D. Guggul 4. Commiphora wightii

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

30. The state tree of Rajasthan, the mainstay of desert agroforestry and the source of pods called 'sangri', is:

(A) Khejri (Prosopis cineraria)
(B) Rohida (Tecomella undulata)
(C) Kair (Capparis decidua)
(D) Babool (Acacia nilotica)

31. Rohida (Tecomella undulata), the state flower of Rajasthan, is also popularly known as:

(A) Kalpavriksha of the desert
(B) Marwar teak or the teak of the desert
(C) Desert date
(D) Flame of the forest

32. Consider the following statements:
I. The National Forest Policy, 1988 set the goal of bringing one-third of the country's geographical area under forest and tree cover.
II. Joint Forest Management was formally introduced through a Government of India circular in 1990.
III. The National Forest Policy, 1988 gave first priority to maximising revenue from forests.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

33. The Van Dhan Vikas Kendra scheme for value addition to minor forest produce is implemented by:

(A) Indian Council of Forestry Research and Education
(B) NABARD
(C) Forest Survey of India
(D) TRIFED under the Ministry of Tribal Affairs

34. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Cattle breed) List-II (Home tract in Rajasthan)
A. Rathi 1. Bikaner and Sri Ganganagar
B. Tharparkar 2. Jaisalmer and Barmer
C. Kankrej 3. Jalore and Sirohi
D. Nagauri 4. Nagaur

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

35. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Sheep breed) List-II (Home tract)
A. Chokla 1. Sikar and Jhunjhunu
B. Magra 2. Bikaner and Jaisalmer
C. Sonadi 3. Udaipur and Dungarpur
D. Marwari 4. Jodhpur and Pali

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-3, B-2, C-1, D-4

36. The Jakhrana breed of goat, noted for its milk yield, belongs to which district of Rajasthan? HARD

(A) Sirohi (B) Nagaur
(C) Alwar (D) Ajmer

37. Consider the following statements about the camel in Rajasthan:
I. The camel was declared the state animal (domestic) of Rajasthan in 2014.
II. The chinkara is the state animal (wild) of Rajasthan.
III. Rajasthan has the largest camel population among Indian states.
Which of the statements given above are correct?

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

38. The National Research Centre on Camel is located at: RAS PRE 2018

(A) Bikaner (B) Jaisalmer
(C) Jodhpur (D) Barmer

39. Which one of the following livestock diseases is NOT caused by a virus?

(A) Foot-and-mouth disease (B) Brucellosis
(C) Peste des petits ruminants (D) Lumpy skin disease

40. Consider the following statements about lumpy skin disease:
I. It is caused by a capripoxvirus.
II. It affects cattle and buffalo and is spread mainly by biting insects.
III. It is a zoonotic disease that spreads to humans through milk.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

41. The National Animal Disease Control Programme launched in 2019 targets the control and eradication of:

(A) Anthrax and rabies
(B) Bird flu and classical swine fever
(C) Lumpy skin disease and PPR
(D) Foot-and-mouth disease and brucellosis

42. Rashtriya Gokul Mission, launched in 2014, is concerned with:

(A) Promotion of goat rearing
(B) Development and conservation of indigenous bovine breeds
(C) Setting up of poultry hatcheries
(D) Development of marine fisheries

43. Arrange the following research institutes in the correct chronological order of their establishment:
I. National Research Centre on Camel, Bikaner
II. Central Arid Zone Research Institute, Jodhpur
III. National Research Centre on Seed Spices, Ajmer
IV. Central Sheep and Wool Research Institute, Avikanagar HARD

(A) I, II, IV, III
(B) II, I, IV, III
(C) II, IV, I, III
(D) IV, II, I, III

44. The Udaipur Solar Observatory, situated on an island in Fateh Sagar Lake, is at present operated by the: RAS PRE 2018, 2021

(A) Indian Institute of Astrophysics, Bengaluru
(B) Indian Space Research Organisation
(C) Aryabhatta Research Institute of Observational Sciences
(D) Physical Research Laboratory, Ahmedabad

45. The Central Sheep and Wool Research Institute at Avikanagar is situated in which district of Rajasthan?

(A) Tonk
(B) Ajmer
(C) Nagaur
(D) Jaipur

46. The Arid Forest Research Institute (AFRI), functioning under the Indian Council of Forestry Research and Education, is located at:

(A) Bikaner
(B) Jodhpur
(C) Udaipur
(D) Alwar

47. The only laboratory of the Council of Scientific and Industrial Research located in Rajasthan is the:

(A) Central Salt and Marine Chemicals Research Institute
(B) National Physical Laboratory
(C) Central Electronics Engineering Research Institute, Pilani
(D) Central Institute of Mining and Fuel Research

48. The Ch. Charan Singh National Institute of Agricultural Marketing (NIAM) is located at:

(A) Nagpur
(B) Hyderabad
(C) Bhopal
(D) Jaipur

49. Which one of the following research institutes is NOT located in Rajasthan? RAS PRE 2016

(A) Central Institute for Research on Goats
(B) Central Arid Zone Research Institute
(C) National Research Centre on Seed Spices
(D) Arid Forest Research Institute

50. Match List-I with List-II and select the correct answer using the codes given below:

List-I (University) List-II (Headquarters)

A. Maharana Pratap University of Agriculture and Technology 1. Udaipur

B. Swami Keshwanand Rajasthan Agricultural University 2. Bikaner

C. Sri Karan Narendra Agriculture University 3. Jobner

D. Agriculture University 4. Kota

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

51. Consider the following statements:

I. The Department of Science and Technology was established in 1971.

II. The Council of Scientific and Industrial Research was founded in 1942.

III. The Science, Technology and Innovation Policy of 2020 is India's fifth national science and technology policy document.

Which of the statements given above are correct? HARD

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

52. National Science Day is observed in India on 28 February to commemorate:

(A) The discovery of the Raman Effect
(B) The founding of CSIR
(C) The Pokhran-II nuclear tests
(D) The launch of Aryabhata

53. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Day) List-II (Date)

A. National Science Day 1. 28 February

B. National Technology Day 2. 11 May

C. National Space Day 3. 23 August

D. National Mathematics Day 4. 22 December

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-1, B-3, C-2, D-4

54. Consider the following statements about the Anusandhan National Research Foundation:
I. It was established by an Act of Parliament in 2023.
II. The Prime Minister is the ex-officio President of its Governing Board.
III. It replaced the Council of Scientific and Industrial Research.
Which of the statements given above are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II and III

55. The Shanti Swarup Bhatnagar Prize for Science and Technology is awarded annually by the:

(A) Department of Science and Technology
(B) Indian National Science Academy
(C) Council of Scientific and Industrial Research
(D) Indian Science Congress Association

56. The INSPIRE scheme of the Department of Science and Technology stands for:

(A) Indian Scheme for Promotion of Innovation and Research
(B) Institutional Support for Pioneering Research
(C) Integrated Science Programme for Indian Research
(D) Innovation in Science Pursuit for Inspired Research

57. C.V. Raman was awarded the Nobel Prize in Physics in 1930 for his work on:

(A) Scattering of light
(B) Thermal ionisation
(C) Cosmic rays
(D) The structure of the atom

58. Match List-I with List-II and select the correct answer using the codes given below:
List-I (Scientist) List-II (Contribution)
A. Meghnad Saha 1. Thermal ionisation equation
B. Satyendra Nath Bose 2. Bose-Einstein statistics
C. Jagadish Chandra Bose 3. Crescograph
D. Homi J. Bhabha 4. Bhabha scattering

(A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-3, D-4
(C) A-1, B-2, C-4, D-3
(D) A-4, B-2, C-3, D-1

59. Consider the following statements:
I. C.V. Raman received the Nobel Prize in Physics in 1930.
II. Subrahmanyan Chandrasekhar received the Nobel Prize in Physics in 1983.
III. Har Gobind Khorana received the Nobel Prize in Chemistry in 1968.
Which of the statements given above are correct?

(A) I and III only
(B) II and III only
(C) I and II only
(D) I, II and III

60. The Physical Research Laboratory at Ahmedabad, founded in 1947, was established by:

(A) Homi J. Bhabha
(B) Meghnad Saha
(C) Satish Dhawan
(D) Vikram Sarabhai

61. Venkatraman Ramakrishnan was awarded the Nobel Prize in Chemistry in 2009 for studies on the structure and function of the:

(A) Ribosome (B) Mitochondrion
(C) Chloroplast (D) Nucleosome

62. Operation Flood, which made India the world's largest producer of milk, was led by:

(A) M.S. Swaminathan (B) Verghese Kurien
(C) Hiralal Chaudhary (D) Sam Pitroda

63. Which one of the following pairs of a colour revolution and its associated product is NOT correctly matched?

(A) Green Revolution — Foodgrains (B) White Revolution — Milk
(C) Blue Revolution — Oilseeds (D) Golden Revolution — Horticulture

64. Salim Ali, known as the 'Birdman of India', is closely associated with the ornithological survey of which site in Rajasthan?

(A) Sambhar Lake (B) Desert National Park
(C) Ranthambore (D) Keoladeo Ghana, Bharatpur

65. Daulat Singh Kothari, born at Udaipur, chaired the Education Commission of 1964-66 and was the first Scientific Adviser to the: HARD

(A) Ministry of Defence (B) Ministry of Education
(C) Department of Atomic Energy (D) Planning Commission

Answer key & explanations

ST01 — PHYSICS IN EVERYDAY LIFE

ST01·1 **A**
ST01·2 **D**
ST01·3 **C**
ST01·4 **C**
ST01·5 **D**
ST01·6 **D**
ST01·7 **D**
ST01·8 **C**
ST01·9 **C**
ST01·10 **A**
ST01·11 **A**
ST01·12 **B**
ST01·13 **C**
ST01·14 **C**
ST01·15 **D**
ST01·16 **D**
ST01·17 **B**
ST01·18 **D**
ST01·19 **D**
ST01·20 **D**
ST01·21 **A**
ST01·22 **A**
ST01·23 **B**
ST01·24 **A**
ST01·25 **A**
ST01·26 **D**
ST01·27 **B**
ST01·28 **D**
ST01·29 **A**
ST01·30 **D**
ST01·31 **B**
ST01·32 **B**
ST01·33 **C**
ST01·34 **A**
ST01·35 **A**
ST01·36 **B**
ST01·37 **B**
ST01·38 **C**
ST01·39 **D**
ST01·40 **B**
ST01·41 **B**
ST01·42 **B**
ST01·43 **D**
ST01·44 **D**
ST01·45 **A**
ST01·46 **C**
ST01·47 **C**
ST01·48 **B**
ST01·49 **C**
ST01·50 **B**
ST01·51 **B**
ST01·52 **A**
ST01·53 **B**
ST01·54 **C**
ST01·55 **C**

ST02 — CHEMISTRY IN EVERYDAY LIFE

ST02·1 **A**
ST02·2 **B**
ST02·3 **C**
ST02·4 **D**
ST02·5 **A**
ST02·6 **B**
ST02·7 **C**
ST02·8 **D**
ST02·9 **A**
ST02·10 **B**
ST02·11 **C**
ST02·12 **D**
ST02·13 **A**
ST02·14 **C**
ST02·15 **C**
ST02·16 **D**
ST02·17 **A**
ST02·18 **B**
ST02·19 **C**
ST02·20 **B**
ST02·21 **A**
ST02·22 **C**
ST02·23 **C**
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ST02·25 **D**
ST02·26 **D**
ST02·27 **A**
ST02·28 **A**
ST02·29 **B**
ST02·30 **A**
ST02·31 **C**
ST02·32 **B**
ST02·33 **D**
ST02·34 **A**
ST02·35 **B**
ST02·36 **A**
ST02·37 **A**
ST02·38 **C**
ST02·39 **D**
ST02·40 **B**
ST02·41 **B**
ST02·42 **C**
ST02·43 **B**
ST02·44 **D**
ST02·45 **C**
ST02·46 **D**
ST02·47 **D**
ST02·48 **D**
ST02·49 **A**
ST02·50 **C**
ST02·51 **B**
ST02·52 **C**
ST02·53 **B**
ST02·54 **C**
ST02·55 **B**

ST03 — BIOLOGY: CELL, TISSUES AND THE HUMAN BODY

ST03·1 **A**
ST03·2 **C**
ST03·3 **B**
ST03·4 **D**
ST03·5 **B**
ST03·6 **D**
ST03·7 **C**
ST03·8 **C**
ST03·9 **C**
ST03·10 **B**
ST03·11 **C**
ST03·12 **B**
ST03·13 **D**
ST03·14 **B**
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ST03·16 **C**
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ST03·20 **B**
ST03·21 **D**
ST03·22 **B**
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ST03·25 **D**
ST03·26 **A**
ST03·27 **C**
ST03·28 **B**
ST03·29 **C**
ST03·30 **A**
ST03·31 **A**
ST03·32 **D**
ST03·33 **D**
ST03·34 **D**
ST03·35 **B**
ST03·36 **D**
ST03·37 **B**
ST03·38 **C**
ST03·39 **B**
ST03·40 **A**
ST03·41 **A**
ST03·42 **C**
ST03·43 **B**
ST03·44 **D**
ST03·45 **A**
ST03·46 **C**
ST03·47 **D**
ST03·48 **C**
ST03·49 **C**
ST03·50 **B**
ST03·51 **B**
ST03·52 **D**
ST03·53 **C**
ST03·54 **A**
ST03·55 **D**

ST04 — HUMAN HEALTH, DISEASES AND PUBLIC HEALTH PROGRAMMES

ST04·1 **B**
ST04·2 **C**
ST04·3 **B**
ST04·4 **C**
ST04·5 **D**
ST04·6 **A**
ST04·7 **C**
ST04·8 **A**
ST04·9 **A**
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ST04·27 **B**
ST04·28 **C**
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ST04·30 **B**
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ST04·33 **A**
ST04·34 **D**
ST04·35 **A**
ST04·36 **A**
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ST04·49 **A**
ST04·50 **D**
ST04·51 **D**
ST04·52 **C**
ST04·53 **C**
ST04·54 **D**
ST04·55 **D**
ST04·56 **A**
ST04·57 **C**
ST04·58 **D**
ST04·59 **A**
ST04·60 **A**

ST05 — FOOD AND NUTRITION

ST05·1 **C**
ST05·2 **C**
ST05·3 **A**
ST05·4 **C**
ST05·5 **B**
ST05·6 **C**
ST05·7 **C**
ST05·8 **D**
ST05·9 **A**
ST05·10 **A**
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ST05·31 **B**
ST05·32 **A**
ST05·33 **B**
ST05·34 **A**
ST05·35 **C**
ST05·36 **A**
ST05·37 **D**
ST05·38 **A**
ST05·39 **D**
ST05·40 **B**

ST06 — GENETICS: INHERITANCE AND VARIATION

ST06·1 **B**
ST06·2 **C**
ST06·3 **B**
ST06·4 **A**
ST06·5 **C**
ST06·6 **C**
ST06·7 **C**
ST06·8 **B**
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ST06·26 **B**
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ST06·28 **A**
ST06·29 **C**
ST06·30 **B**
ST06·31 **B**
ST06·32 **B**
ST06·33 **C**
ST06·34 **A**
ST06·35 **A**
ST06·36 **A**
ST06·37 **B**
ST06·38 **A**
ST06·39 **B**
ST06·40 **A**
ST06·41 **B**
ST06·42 **B**
ST06·43 **A**
ST06·44 **B**
ST06·45 **A**

ST07 — BIOTECHNOLOGY AND GENETIC ENGINEERING

ST07·1 **A**
ST07·2 **B**
ST07·3 **B**
ST07·4 **A**
ST07·5 **B**
ST07·6 **B**
ST07·7 **C**
ST07·8 **A**
ST07·9 **A**
ST07·10 **D**
ST07·11 **A**
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ST07·35 **C**
ST07·36 **D**
ST07·37 **D**
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ST07·39 **D**
ST07·40 **C**
ST07·41 **C**
ST07·42 **D**
ST07·43 **D**
ST07·44 **D**
ST07·45 **D**

ST08 — NANOTECHNOLOGY AND NEW MATERIALS

ST08·1 **C**
ST08·2 **A**
ST08·3 **A**
ST08·4 **D**
ST08·5 **C**
ST08·6 **A**
ST08·7 **D**
ST08·8 **B**
ST08·9 **B**
ST08·10 **A**
ST08·11 **A**
ST08·12 **D**
ST08·13 **A**
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ST08·26 **B**
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ST08·28 **B**
ST08·29 **D**
ST08·30 **C**
ST08·31 **B**
ST08·32 **C**
ST08·33 **C**
ST08·34 **D**
ST08·35 **A**

ST09 — COMPUTERS, IT AND ICT

ST09·1 **C**
ST09·2 **B**
ST09·3 **A**
ST09·4 **A**
ST09·5 **D**
ST09·6 **D**
ST09·7 **A**
ST09·8 **C**
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ST09·29 **C**
ST09·30 **D**
ST09·31 **A**
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ST09·33 **D**
ST09·34 **B**
ST09·35 **B**
ST09·36 **A**
ST09·37 **B**
ST09·38 **D**
ST09·39 **D**
ST09·40 **A**
ST09·41 **C**
ST09·42 **C**
ST09·43 **C**
ST09·44 **D**
ST09·45 **C**
ST09·46 **B**
ST09·47 **B**
ST09·48 **C**
ST09·49 **D**
ST09·50 **A**
ST09·51 **A**
ST09·52 **B**
ST09·53 **D**
ST09·54 **C**
ST09·55 **A**

ST10 — SPACE TECHNOLOGY

ST10·1 **B**
ST10·2 **C**
ST10·3 **A**
ST10·4 **D**
ST10·5 **D**
ST10·6 **A**
ST10·7 **D**
ST10·8 **D**
ST10·9 **A**
ST10·10 **D**
ST10·11 **B**
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ST10·30 **B**
ST10·31 **C**
ST10·32 **A**
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ST10·34 **A**
ST10·35 **D**
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ST10·37 **A**
ST10·38 **C**
ST10·39 **D**
ST10·40 **C**
ST10·41 **C**
ST10·42 **D**
ST10·43 **C**
ST10·44 **D**
ST10·45 **C**
ST10·46 **D**
ST10·47 **C**
ST10·48 **B**
ST10·49 **A**
ST10·50 **A**
ST10·51 **B**
ST10·52 **D**
ST10·53 **B**
ST10·54 **D**
ST10·55 **A**

ST11 — DEFENCE TECHNOLOGY

ST11·1 **B**
ST11·2 **C**
ST11·3 **A**
ST11·4 **D**
ST11·5 **B**
ST11·6 **B**
ST11·7 **C**
ST11·8 **A**
ST11·9 **C**
ST11·10 **B**
ST11·11 **D**
ST11·12 **A**
ST11·13 **B**
ST11·14 **C**
ST11·15 **A**
ST11·16 **D**
ST11·17 **B**
ST11·18 **C**
ST11·19 **A**
ST11·20 **A**
ST11·21 **B**
ST11·22 **C**
ST11·23 **D**
ST11·24 **A**
ST11·25 **B**
ST11·26 **C**
ST11·27 **D**
ST11·28 **A**
ST11·29 **B**
ST11·30 **C**
ST11·31 **D**
ST11·32 **C**
ST11·33 **C**
ST11·34 **B**
ST11·35 **B**
ST11·36 **D**
ST11·37 **A**
ST11·38 **C**
ST11·39 **A**
ST11·40 **B**
ST11·41 **C**
ST11·42 **D**
ST11·43 **D**
ST11·44 **D**
ST11·45 **A**
ST11·46 **D**
ST11·47 **A**
ST11·48 **C**
ST11·49 **D**
ST11·50 **A**

ST12 — ENVIRONMENT, ECOLOGY, EIA, BIODIVERSITY AND SUSTAINABLE DEVELOPMENT

ST12·1 **A**
ST12·2 **B**
ST12·3 **D**
ST12·4 **D**
ST12·5 **A**
ST12·6 **B**
ST12·7 **C**
ST12·8 **D**
ST12·9 **B**
ST12·10 **C**
ST12·11 **D**
ST12·12 **A**
ST12·13 **A**
ST12·14 **B**
ST12·15 **D**
ST12·16 **A**
ST12·17 **B**
ST12·18 **D**
ST12·19 **C**
ST12·20 **C**
ST12·21 **C**
ST12·22 **B**
ST12·23 **D**
ST12·24 **A**
ST12·25 **B**
ST12·26 **C**
ST12·27 **D**
ST12·28 **A**
ST12·29 **B**
ST12·30 **C**
ST12·31 **D**
ST12·32 **A**
ST12·33 **D**
ST12·34 **A**
ST12·35 **B**
ST12·36 **C**
ST12·37 **D**
ST12·38 **A**
ST12·39 **B**
ST12·40 **C**
ST12·41 **D**
ST12·42 **A**
ST12·43 **B**
ST12·44 **C**
ST12·45 **C**
ST12·46 **D**
ST12·47 **A**
ST12·48 **B**
ST12·49 **C**
ST12·50 **D**
ST12·51 **A**
ST12·52 **B**
ST12·53 **C**
ST12·54 **D**
ST12·55 **C**
ST12·56 **B**
ST12·57 **C**
ST12·58 **C**
ST12·59 **B**
ST12·60 **D**
ST12·61 **A**
ST12·62 **B**
ST12·63 **C**
ST12·64 **B**
ST12·65 **C**
ST12·66 **A**
ST12·67 **D**
ST12·68 **A**
ST12·69 **D**
ST12·70 **A**

ST13 — AGRICULTURE, HORTICULTURE, FORESTRY, ANIMAL HUSBANDRY (RAJASTHAN FOCUS); S&T POLICY AND INDIAN SCIENTISTS

ST13·1 **A**
ST13·2 **B**
ST13·3 **C**
ST13·4 **D**
ST13·5 **D**
ST13·6 **A**
ST13·7 **B**
ST13·8 **C**
ST13·9 **D**
ST13·10 **A**
ST13·11 **B**
ST13·12 **C**
ST13·13 **D**
ST13·14 **A**
ST13·15 **B**
ST13·16 **C**
ST13·17 **B**
ST13·18 **D**
ST13·19 **A**
ST13·20 **D**
ST13·21 **B**
ST13·22 **C**
ST13·23 **A**
ST13·24 **D**
ST13·25 **A**
ST13·26 **C**
ST13·27 **B**
ST13·28 **D**
ST13·29 **A**
ST13·30 **A**
ST13·31 **B**
ST13·32 **C**
ST13·33 **D**
ST13·34 **A**
ST13·35 **B**
ST13·36 **C**
ST13·37 **D**
ST13·38 **A**
ST13·39 **B**
ST13·40 **C**
ST13·41 **D**
ST13·42 **B**
ST13·43 **C**
ST13·44 **D**
ST13·45 **A**
ST13·46 **B**
ST13·47 **C**
ST13·48 **D**
ST13·49 **A**
ST13·50 **B**
ST13·51 **D**
ST13·52 **A**
ST13·53 **A**
ST13·54 **B**
ST13·55 **C**
ST13·56 **D**

ST13·57 **A**	ST13·60 **D**	ST13·63 **C**
ST13·58 **B**	ST13·61 **A**	ST13·64 **D**
ST13·59 **C**	ST13·62 **B**	ST13·65 **A**

ST01 — EXPLANATIONS

1. (A) Candela — The candela (luminous intensity) is one of the seven SI base units, along with metre, kilogram, second, ampere, kelvin and mole. Newton, pascal and joule are derived units.

2. (D) I, II, III and IV — All four measure pressure. One atmosphere equals 1.013 x 10^5 pascal, which is 760 torr, 76 cm of mercury and about 1.013 bar.

3. (C) 746 watt — 1 HP = 746 W. The watt, the SI unit of power, is one joule per second.

4. (C) A-2, B-1, C-3, D-4 — Dioptre measures the power of a lens (the reciprocal of the focal length in metres), poise measures viscosity, becquerel measures radioactivity and tesla measures magnetic flux density.

5. (D) Light year — Time — A light year is the distance travelled by light in one year, that is, a unit of DISTANCE. One parsec equals 3.26 light years.

6. (D) Speed — Speed has magnitude only and is therefore a scalar. Impulse, torque and momentum all have direction and are vectors.

7. (D) I, II and III — All three are correct. The first law states that a body continues in its state of rest or of uniform motion unless acted upon by an external force, and inertia increases with mass.

8. (C) Newton's third law of motion — Hot gases ejected backwards exert an equal and opposite forward reaction on the rocket, which is Newton's third law of motion.

9. (C) conservation of linear momentum — The total linear momentum of the gun-and-bullet system is zero before firing, so the forward momentum of the bullet is balanced by the backward momentum of the gun.

10. (A) I and II only — I and II are correct. Because the earth bulges at the equator, g is minimum at the equator and maximum at the poles, so statement III is wrong.

11. (A) I and II only — I and II are correct. Remote sensing satellites use low polar or sun-synchronous orbits of about 800 to 900 km; the geostationary orbit is used mainly for communication and meteorology.

12. (B) the astronaut and the spacecraft are both in a state of free fall — Gravity is very much present in orbit; the sensation of weightlessness arises because the astronaut and the spacecraft fall around the earth together, so there is no normal reaction on the astronaut.

13. (C) zero, because the force and the displacement are mutually perpendicular — Work equals force multiplied by displacement multiplied by the cosine of the angle between them; here the upward force and the horizontal displacement are perpendicular, so the work done against gravity is zero.

14. (C) 3.6 x 10^6 joule — 1 kWh = 1000 W x 3600 s = 3.6 x 10^6 joule. It is also called the Board of Trade unit and is what a domestic electricity meter records.

15. (D) Pascal's law — Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished in all directions; it is used in hydraulic brakes, hydraulic lifts and hydraulic presses.

16. (D) Pressure cooker — Surface tension — A pressure cooker works because an increase in pressure raises the boiling point of water; surface tension plays no part in it.

17. (B) the increased pressure inside raises the boiling point of water — Trapped steam raises the pressure inside, and since the boiling point of water rises with pressure, water boils above 100 degrees Celsius and food cooks faster. On hills, where atmospheric pressure is low, water boils below 100 degrees Celsius.

18. (D) A-1, B-3, C-2, D-4 — A lactometer is a hydrometer calibrated for milk; a barometer measures atmospheric pressure and an anemometer measures wind speed.

19. (D) Bernoulli's principle — Water between the boats flows faster and therefore exerts less pressure than the water on the outer sides, so the boats are pushed together. The same principle gives an aeroplane wing its lift.

20. (D) The operation of a hydraulic jack — A hydraulic jack works on Pascal's law. The other three are classic consequences of surface tension and the related phenomenon of capillarity.

21. (A) I and II only — I and II are correct. Surface tension DECREASES as temperature rises, one reason why hot water with detergent cleans better than cold water.

22. (A) decreases — Heating weakens the intermolecular attraction in a liquid, so it flows more easily. In contrast, the viscosity of a GAS increases with temperature.

23. (B) a very high specific heat capacity — Water has a specific heat of about 4.2 joule per gram per kelvin, the highest among common liquids, so it absorbs a large quantity of heat for a small rise in temperature.

24. (A) I and II only — I and II are correct. During a change of state the temperature stays constant because the heat supplied is used to break intermolecular bonds; this is also why a burn from steam is worse than one from boiling water.

25. (A) A-1, B-2, C-3, D-4 — Land and sea breezes are convection currents arising from the high specific heat of water; radiation, unlike the other two modes, needs no material medium, which is how heat reaches the earth from the sun.

26. (D) absorption of latent heat when the liquid refrigerant evaporates — The refrigerant evaporates in the coils inside the cabinet and absorbs latent heat from the contents; the compressor then compresses the vapour and the condenser rejects that heat outside.

27. (B) attains its maximum density at 4 degrees Celsius — This anomalous expansion means water at 4 degrees Celsius sinks while colder water and ice remain on top, so aquatic life survives beneath a frozen surface.

28. (D) Mercury is used in thermometers because it has a very high specific heat — Mercury is preferred because it expands uniformly, conducts heat well, is opaque and does not wet glass; its specific heat is in fact low.

29. (A) I and II only — I and II are correct. Loudness depends on AMPLITUDE; frequency determines pitch, while the waveform determines quality or timbre.

30. (D) I, II and III — All three are correct. Ultrasound is also used in sonography, echocardiography, cleaning delicate parts and detecting internal flaws in metal castings; bats and dolphins navigate by echolocation.

31. (B) multiple reflection of sound — Sound from the patient's body is repeatedly reflected inside the tubes of the stethoscope and is thus carried to the doctor's ears with very little loss of intensity.

32. (B) the Doppler effect — The Doppler effect arises from relative motion between source and observer; the same principle is used in radar speed guns and in measuring the red shift of receding galaxies.

33. (C) 17.2 metre — The ear retains a sound sensation for about 0.1 second; with the speed of sound about 344 metre per second the sound must travel 34.4 m to and fro, so the reflector must be at least 17.2 m away.

34. (A) scattering of sunlight by the molecules of the atmosphere — Shorter wavelengths such as blue are scattered far more strongly than longer ones by air molecules. Above the atmosphere there is no scattering, so an astronaut sees a black sky.

35. (A) I, II, III, IV — In increasing order of wavelength the electromagnetic spectrum runs gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves and radio waves; frequency and photon energy decrease in the same direction.

36. (B) IV, III, II, I — The visible spectrum in increasing order of wavelength is violet, indigo, blue, green, yellow, orange, red. Violet is deviated the most by a prism and red the least.

37. (B) infrared rays — A remote control uses an infrared light-emitting diode to send coded pulses to the receiver on the television set. Infrared lies just beyond the red end of the visible spectrum.

38. (C) atmospheric refraction of starlight — Continuously changing atmospheric refraction makes the apparent position and brightness of a point-like star fluctuate; planets are extended sources, so the fluctuations average out.

39. (D) I, II and III — All three depend on total internal reflection. Diamond has a refractive index of about 2.42 and a critical angle of only about 24 degrees, so light is reflected repeatedly inside it before emerging.

40. (B) total internal reflection — Light entering the optically denser core strikes the cladding at an angle greater than the critical angle and is repeatedly totally internally reflected along the length of the fibre.

41. (B) A-2, B-1, C-4, D-3 — In myopia the image forms in front of the retina and needs a diverging (concave) lens; in hypermetropia it forms behind the retina and needs a converging (convex) lens.

42. (B) a virtual, erect and diminished image with a wide field of view — A convex mirror always gives an erect, diminished virtual image and covers a much wider field of view than a plane mirror of the same size.

43. (D) Correction of short-sightedness — Convex lens — Short-sightedness, or myopia, is corrected with a CONCAVE (diverging) lens; a convex lens is used for long-sightedness or hypermetropia.

44. (D) I, II and III — All three are correct. Sunlight entering a raindrop is dispersed, internally reflected once and refracted again as it emerges towards an observer whose back is to the sun.

45. (A) I and II only — I and II are correct. A voltmeter, having very high resistance, is always connected in PARALLEL; an ammeter, having very low resistance, is connected in series. Parallel wiring lets each appliance get the full 220 volt and work independently.

46. (C) a thin wire of low melting point connected in series in the live wire — The fuse wire, usually a tin-lead alloy of low melting point, melts and breaks the circuit when the current exceeds a safe value, protecting appliances against short circuits and overloading.

47. (C) provide a low-resistance path to the ground so that a person touching it is not shocked — If the live wire touches the metal casing, the earth wire carries the leakage current safely to the ground and the resulting large current blows the fuse.

48. (B) I and III only — A transformer needs a continuously changing magnetic flux and therefore works only with alternating current, so statement II is wrong.

49. (C) A-1, B-2, C-3, D-4 — A dynamo or generator works on Faraday's law of electromagnetic induction; Fleming's right-hand rule applies to the generator and the left-hand rule to the motor.

50. (B) solar energy into electrical energy — A photovoltaic cell is a semiconductor device, usually silicon based, that converts sunlight straight into direct-current electricity; a solar cooker, by contrast, converts solar energy into heat.

51. (B) a superconductor — Superconductivity was discovered by Kamerlingh Onnes in mercury in 1911; superconductors also expel magnetic flux, an effect called the Meissner effect that is used in magnetic levitation.

52. (A) A-1, B-2, C-4, D-3 — Roentgen discovered X-rays in 1895 and received the first Nobel Prize in Physics in 1901; Einstein received the 1921 Nobel Prize for explaining the photoelectric effect.

53. (B) nuclear fusion of hydrogen nuclei into helium — In the sun's core, hydrogen nuclei fuse to form helium and release enormous energy; the fission of uranium-235 is what powers a nuclear reactor on earth.

54. (C) Cadmium and boron rods are used as moderators to slow down neutrons — Cadmium and boron rods are CONTROL rods that absorb neutrons to regulate the chain reaction; moderators such as heavy water and graphite slow the neutrons down.

55. (C) gamma radiation — Alpha particles (helium nuclei) are stopped by a sheet of paper, beta particles (electrons) by a few millimetres of aluminium, while gamma rays are high-energy electromagnetic waves needing thick lead or concrete.

ST02 — EXPLANATIONS

1. (A) NaHCO3 — Baking soda is sodium hydrogencarbonate, $NaHCO_3$. On heating it releases carbon dioxide, which makes cakes rise; it is also used as a mild antacid and in soda-acid fire extinguishers.

2. (B) heating gypsum to about 100 degrees Celsius — Gypsum ($CaSO_4.2H_2O$) heated to about 373 K loses three-quarters of its water of crystallisation to give plaster of Paris, $CaSO_4.(1/2)H_2O$, which sets into a hard mass when mixed with water.

3. (C) passing chlorine over dry slaked lime — Chlorine passed over dry calcium hydroxide gives bleaching powder, $CaOCl_2$. It is used for bleaching cotton and paper pulp and for making drinking water safe.

4. (D) concentrated hydrochloric acid and concentrated nitric acid in the ratio 3 : 1 — Aqua regia, literally royal water, is three parts concentrated hydrochloric acid to one part concentrated nitric acid; it dissolves noble metals such as gold and platinum.

5. (A) deuterium oxide — Heavy water is D_2O, in which ordinary hydrogen is replaced by its isotope deuterium. It slows down fast neutrons without absorbing them, which is why it is used as a moderator.

6. (B) acetic acid — Table vinegar contains about 5 to 8 per cent acetic acid (ethanoic acid). The acidic medium checks the growth of spoilage micro-organisms, which is why it is used for pickling.

7. (C) acetylene — Calcium carbide with water gives acetylene (ethyne), which mimics the natural ripening hormone ethylene. Its use on food is banned in India because commercial carbide carries traces of arsenic and phosphorus.

8. (D) 60 carbon atoms — Buckminsterfullerene, C_{60}, has sixty carbon atoms arranged in twenty hexagons and twelve pentagons. Graphene and carbon nanotubes are other modern carbon forms.

9. (A) its layers slide over one another easily and it has free electrons that conduct electricity — In graphite each carbon atom is bonded to only three others, leaving one free electron for conduction, and the hexagonal layers are held by weak forces so they slip over each other.

10. (B) diamond — Diamond, an allotrope of carbon in which every atom is covalently bonded to four others in a rigid three-dimensional network, tops the Mohs scale at 10.

11. (C) fluorine — Fluorine is the most electronegative element and also the most reactive non-metal; electronegativity increases across a period and decreases down a group.

12. (D) argon along with some nitrogen — These chemically inert gases prevent the tungsten filament from burning away and reduce its evaporation, thereby prolonging the life of the bulb.

13. (A) absorbs harmful ultraviolet radiation from the sun — Ozone (O_3) shields living organisms from ultraviolet-B radiation. Chlorofluorocarbons deplete it, and the Montreal Protocol of 1987 was adopted to phase out ozone-depleting substances.

14. (C) Carbon dioxide — Carbon dioxide is heavier than air and does not support combustion, so it blankets the fire and cuts off oxygen; it also leaves no residue on electrical equipment.

15. (C) Nylon — Nylon is a synthetic polyamide made wholly from petrochemicals. Cotton and jute are natural cellulosic fibres, while rayon is regenerated cellulose and is therefore cellulosic.

16. (D) Tin — Tinning gives a chemically inert lining so that acidic food does not react with the copper of the brass. Brass itself is an alloy of copper and zinc.

17. (A) mercury — Any alloy of a metal with mercury is called an amalgam; dental amalgam, for example, contains mercury with silver and tin.

18. (B) chromium and nickel — Typical stainless steel contains about 18 per cent chromium and 8 per cent nickel besides iron and a little carbon; the chromium forms a protective oxide film that resists rusting.

19. (C) copper, zinc and nickel — German silver contains no silver at all; it owes its silvery appearance to nickel. It is also called nickel silver.

20. (B) A-3, B-1, C-2, D-4 — Bronze is the copper-tin alloy of the Bronze Age; duralumin, being light and strong, is used in aircraft bodies.

21. (A) German silver — Copper, silver and nickel — German silver is an alloy of copper, zinc and nickel and contains no silver whatsoever.

22. (C) A-1, B-2, C-3, D-4 — Cinnabar is mercuric sulphide, galena is lead sulphide and zinc blende is zinc sulphide; haematite and magnetite are the chief ores of iron.

23. (C) I and II only — Sodium and potassium are stored under KEROSENE, because they react violently with water and even with moist air. Statements I and II are correct.

24. (D) hydrated ferric oxide — Rust is $Fe_2O_3.xH_2O$ and its formation requires BOTH oxygen and moisture; that is why iron does not rust in dry air or in air-free water.

25. (D) zinc — Zinc protects iron even when the coating is scratched, because zinc is more reactive and corrodes in preference to iron, a form of sacrificial protection.

26. (D) Rusting is a physical change that can be reversed simply by heating the article — Rusting is a chemical change, an oxidation reaction, and cannot be reversed by heating; the metal is actually consumed.

27. (A) Both A and R are true and R is the correct explanation of A — The chromium oxide layer is self-repairing, so even a scratched surface re-passivates. This is why stainless steel cutlery and surgical instruments resist corrosion.

28. (A) Lithium-ion cell — Lithium-ion cells offer high energy density and are rechargeable. Lead-acid batteries are used in vehicles for starting, and a fuel cell converts hydrogen and oxygen directly into electricity, giving only water as a by-product.

29. (B) I and II only — An acidic solution has pH LESS than 7. Toothpastes are mildly basic in order to neutralise the acid produced by bacteria in the mouth.

30. (A) A-1, B-3, C-4, D-2 — Oxalic acid occurs in tomato and spinach, malic acid in apples and hydrochloric acid in the gastric juice of the stomach.

31. (C) II, IV, I, III — Gastric juice is strongly acidic (pH about 1.2 to 3), pure water is neutral at 7, milk of magnesia is a weak base at about 10 and sodium hydroxide solution is strongly basic at about 14.

32. (B) A-1, B-2, C-3, D-4 — Quicklime slaked with water gives slaked lime, and a solution of slaked lime is limewater, which turns milky with carbon dioxide.

33. (D) Hypo — Sodium hydrogencarbonate — Hypo is sodium thiosulphate ($Na_2S_2O_3.5H_2O$), used as a fixer in photography and for removing chlorine from bleached fabric.

34. (A) A-4, B-1, C-2, D-3 — Green vitriol is $FeSO_4.7H_2O$ and white vitriol is $ZnSO_4.7H_2O$, while oil of vitriol is concentrated sulphuric acid. Hypo, the photographic fixer, is sodium thiosulphate.

35. (B) I and II only — Group 17 elements are the HALOGENS; group 1 elements are the alkali metals. Mendeleev's original table was based on atomic mass, which Moseley replaced with atomic number.

36. (A) I and II only — In DIAMOND each carbon atom is bonded to FOUR others in a rigid tetrahedral network, leaving no free electron; in graphite each atom is bonded to three, leaving one free electron for conduction.

37. (A) A-1, B-2, C-3, D-4 — The hydrogenation of vegetable oil over nickel converts liquid oils into solid vanaspati and is a classic example of catalysis in the food industry.

38. (C) I, II and III — All three are correct. Liquid nitrogen boils at about minus 196 degrees Celsius and is used for cryopreservation of semen, blood and tissues.

39. (D) I, II and III — All three are correct. LPG and CNG are both odourless in their pure state, so a strong-smelling mercaptan is deliberately added as a safety measure.

40. (B) Water gas — Carbon monoxide and carbon dioxide — Water gas is a mixture of carbon monoxide and HYDROGEN, obtained by passing steam over red-hot coke; it is also called synthesis gas.

41. (B) hydrogen — Hydrogen has a calorific value of about 150 kilojoule per gram, far higher than any other fuel, and it burns to give only water; storage and transport, however, remain difficult.

42. (C) Bakelite — Bakelite, made from phenol and formaldehyde, is a thermosetting plastic: once moulded it cannot be softened and reshaped by heating. Melamine is another thermosetting plastic.

43. (B) sulphur — Charles Goodyear's process forms sulphur cross-links between the polymer chains, so vulcanised rubber resists heat, wear and organic solvents; it is used for vehicle tyres.

44. (D) I, II and III — All three are correct. Hard water contains calcium and magnesium ions that form an insoluble scum with soap, whereas the calcium and magnesium salts of detergents remain soluble.

45. (C) slow down the rate of setting — Without gypsum, Portland cement would set almost immediately on mixing with water, leaving no time for placing and finishing the concrete.

46. (D) silica — Soda-lime glass is made by fusing silica (sand) with sodium carbonate and calcium carbonate. Borosilicate or Pyrex glass, which contains boron oxide, withstands sudden temperature changes.

47. (D) I, II and III — All three are correct. Excessive use of nitrogenous fertiliser causes nitrate leaching into groundwater and eutrophication of surface water bodies.

48. (D) potassium nitrate, charcoal and sulphur — In gunpowder the potassium nitrate acts as the oxidiser while charcoal and sulphur are the fuel. Dynamite, invented by Alfred Nobel, is nitroglycerine absorbed in kieselguhr.

49. (A) I, II and III — All three are correct. Adulteration of mustard oil with argemone oil causes epidemic dropsy, and metanil yellow is a non-permitted coal-tar dye.

50. (C) a narcotic analgesic — Morphine relieves severe pain by acting on the central nervous system and is habit forming, so it is regulated as a narcotic. Aspirin and paracetamol are non-narcotic analgesics.

51. (B) Alexander Fleming — Fleming observed in 1928 that the mould Penicillium notatum inhibited bacterial growth; he shared the 1945 Nobel Prize with Florey and Chain, who purified the drug.

52. (C) an antiseptic is applied to living tissue while a disinfectant is used on non-living objects — Dettol and tincture of iodine are antiseptics for wounds and skin; phenol and chlorine compounds are disinfectants for floors, drains and instruments. Dilute phenol acts as an antiseptic while concentrated phenol is a disinfectant.

53. (B) Aspirin — Antibiotic — Aspirin is an analgesic and antipyretic, not an antibiotic; it is also used in low doses as a blood thinner.

54. (C) A-2, B-3, C-1, D-4 — Carbon-14, with a half-life of about 5730 years, is used for radiocarbon dating; phosphorus-32 is used as a tracer in agricultural research.

55. (B) Camphor — Camphor, naphthalene, iodine, ammonium chloride and dry ice all sublime. Sublimation is used to separate such solids from non-volatile impurities.

ST03 — EXPLANATIONS

1. (A) Robert Hooke — Robert Hooke coined the word 'cell' in 1665 after observing thin slices of cork under his microscope. Robert Brown discovered the nucleus in 1831.

2. (C) Rudolf Virchow — Rudolf Virchow (1855) extended the cell theory of Schleiden and Schwann by establishing that new cells arise only from pre-existing cells.

3. (B) Mitochondrion — The mitochondrion generates ATP through the Krebs cycle and oxidative phosphorylation. It has a double membrane, an inner membrane folded into cristae, and its own circular DNA.

4. (D) Lysosome — Lysosomes, discovered by Christian de Duve, contain hydrolytic enzymes that can digest the cell itself; hence the name suicide bag.

5. (B) 70S — Prokaryotes have 70S ribosomes (50S + 30S subunits). Eukaryotic cytoplasmic ribosomes are 80S, but mitochondria and chloroplasts also carry 70S ribosomes.

6. (D) Cell wall of plants — The plant cell wall is made chiefly of cellulose. Fungal cell walls contain chitin and bacterial cell walls contain peptidoglycan.

7. (C) Four haploid cells — Meiosis is a reduction division producing four haploid daughter cells from one diploid cell; it is the basis of gamete formation.

8. (C) Pachytene — Crossing over occurs at the pachytene stage of prophase-I, mediated by the enzyme recombinase; chiasmata become visible later at diplotene.

9. (C) 9 kilocalories — Fat yields about 9 kcal per gram, more than double that of carbohydrate or protein (about 4 kcal/g each); alcohol yields about 7 kcal/g.

10. (B) Uracil — RNA contains uracil in place of the thymine of DNA, and its sugar is ribose instead of deoxyribose.

11. (C) Three — Guanine pairs with cytosine through three hydrogen bonds, while adenine pairs with thymine through two; this is why G-C rich DNA has a higher melting temperature.

12. (B) R.H. Whittaker — R.H. Whittaker (1969) proposed the five kingdoms - Monera, Protista, Fungi, Plantae and Animalia - using cell structure, body organisation, nutrition and reproduction as criteria.

13. (D) Genus — In binomial nomenclature, given by Carolus Linnaeus, the first word is the genus (capitalised) and the second the species (in lower case); the name is italicised.

14. (B) Water — Oxygen is released by the photolysis (splitting) of water during the light reaction, as proved by experiments using heavy-oxygen labelled water.

15. (C) Stroma — The light reaction occurs on the thylakoid membranes of the grana, while the Calvin cycle occurs in the stroma, the fluid matrix of the chloroplast.

16. (C) Ethylene — Ethylene is the only gaseous plant hormone; it hastens fruit ripening and senescence. Auxin causes apical dominance and gibberellin causes stem elongation.

17. (B) Ptyalin — Ptyalin, or salivary amylase, converts starch into maltose in the mouth. Pepsin acts on protein in the stomach and trypsin is a pancreatic protease.

18. (D) Gall bladder — Bile is produced in the liver and stored and concentrated in the gall bladder. Bile contains no digestive enzyme; it emulsifies fat and makes the medium alkaline.

19. (B) Right atrium — The sino-atrial (SA) node lies in the wall of the right atrium and initiates each heartbeat; the atrio-ventricular node lies in the inter-atrial septum.

20. (B) William Harvey — William Harvey described the circulation of blood in 1628. Karl Landsteiner discovered the ABO blood groups in 1901.

21. (D) O — Group O red cells carry neither A nor B antigen, so they are not agglutinated by the recipient's anti-A or anti-B antibodies; O-negative is the true universal donor. Group AB is the universal recipient.

22. (B) The Rhesus monkey — Karl Landsteiner and A.S. Wiener identified the antigen in 1940 in the Rhesus monkey (Macaca mulatta), hence 'Rh' factor. About 93-95 per cent of Indians are Rh-positive.

23. (C) 120 days — A human RBC survives about 120 days; it is produced in the red bone marrow and destroyed mainly in the spleen, which is therefore called the graveyard of RBCs.

24. (A) Nephron — Each human kidney contains roughly one million nephrons. The Bowman's capsule together with the glomerulus forms the Malpighian corpuscle.

25. (D) Liver — Urea is formed in the liver by the ornithine (urea) cycle from ammonia and carbon dioxide, and is then excreted by the kidneys; humans are therefore ureotelic.

26. (A) Pituitary gland — The pituitary (hypophysis), lying in the sella turcica at the base of the brain, secretes trophic hormones that regulate other endocrine glands; it is itself controlled by the hypothalamus.

27. (C) Beta cells of the islets of Langerhans — Beta cells of the islets of Langerhans secrete insulin, which lowers blood glucose; alpha cells secrete glucagon, which raises it. Insulin deficiency causes diabetes mellitus.

28. (B) 206 — An adult human has 206 bones; a newborn has about 300, many of which fuse during growth. The axial skeleton has 80 bones and the appendicular 126.

29. (C) Stapes — The stapes (stirrup) of the middle ear is the smallest human bone; the femur (thigh bone) is the longest.

30. (A) Rods — Rod cells contain the pigment rhodopsin (visual purple) and are responsible for scotopic or dim-light vision; cones contain iodopsin and give colour vision.

31. (A) I and II only — The Calvin cycle occurs in the stroma of the chloroplast, not in the mitochondrion. The semi-autonomous nature of mitochondria (own DNA, 70S ribosomes) supports the endosymbiotic theory.

32. (D) I and III only — The SA node lies in the right atrium, so statement II is wrong. The second sound 'dub' is produced by the closure of the semilunar valves.

33. (D) I, II and III — All three are correct. Because AB plasma has no anti-A or anti-B antibody, group AB is the universal recipient, while group O is the universal donor.

34. (D) I, II and III — All three statements are correct. The liver also stores glycogen, iron and fat-soluble vitamins and detoxifies drugs and toxins.

35. (B) II and III only — Meiosis occurs in germ cells, not somatic cells, so statement I is wrong. Mitosis is equational and meiosis-I is reductional.

36. (D) II and III only — Certain RNA molecules called ribozymes also act as biological catalysts, so statement I is wrong. Enzymes are highly specific and are not consumed in the reaction.

37. (B) I and III only — Glucagon raises blood glucose by promoting glycogenolysis; it is antagonistic to insulin. Hence statement II is wrong.

38. (C) I and III only — Balance and coordination are controlled by the cerebellum; the medulla oblongata regulates heartbeat, breathing and blood pressure. Humans also have 12 pairs of cranial nerves.

39. (B) I and III only — Humans are ureotelic (they excrete urea); birds and reptiles are uricotelic. ADH deficiency leads to excessive dilute urine, a condition called diabetes insipidus.

40. (A) I and II only — Oxygen is evolved during the light reaction by the photolysis of water, so statement III is wrong. Haemoglobin, in contrast to chlorophyll, contains iron.

41. (A) A-2, B-4, C-1, D-3 — Ribosomes synthesise protein, lysosomes digest intracellular material, chloroplasts carry out photosynthesis and the Golgi body packages and secretes.

42. (C) A-3, B-1, C-2, D-4 — Thyroxine is from the thyroid, adrenaline from the adrenal medulla, insulin from the pancreatic islets and growth hormone from the anterior pituitary.

43. (B) A-3, B-4, C-2, D-1 — Excess growth hormone before puberty causes gigantism and after puberty acromegaly; hypothyroidism in infancy causes cretinism and in adults myxoedema.

44. (D) A-2, B-4, C-3, D-1 — Abscisic acid is called the stress hormone because it closes stomata and promotes dormancy; gibberellin causes bolting in rosette plants.

45. (A) A-3, B-1, C-4, D-2 — Ptyalin is salivary amylase, pepsin is secreted as pepsinogen by gastric glands, trypsin comes from the pancreas as trypsinogen, and bile is made by the liver.

46. (C) A-3, B-1, C-2, D-4 — The femur is the longest bone and the stapes the smallest; the gluteus maximus of the buttock is the largest muscle and the stapedius of the middle ear the smallest.

47. (D) A-3, B-1, C-2, D-4 — Landsteiner discovered the ABO blood groups (Nobel Prize 1930), Harvey described blood circulation, Brown discovered the nucleus and Golgi the organelle named after him.

48. (C) Ribosome - Synthesis of lipids — Ribosomes are the site of protein synthesis, not lipid synthesis; lipid synthesis occurs on the smooth endoplasmic reticulum.

49. (C) Secretion of insulin — Insulin is secreted by the beta cells of the islets of Langerhans in the pancreas, not by the liver. The liver secretes bile, stores glycogen and forms urea.

50. (B) Parathyroid - Calcitonin — Calcitonin is secreted by the thyroid gland; the parathyroid glands secrete parathormone, which raises blood calcium and is antagonistic to calcitonin.

51. (B) Mature mammalian red blood corpuscles possess a nucleus — Mature mammalian RBCs are anucleate, which increases their oxygen-carrying capacity; camels and llamas are exceptional in having oval RBCs but these too lack a nucleus.

52. (D) Cochlea — The cochlea is the spirally coiled hearing organ of the internal ear. The human brain consists of the forebrain, midbrain and hindbrain.

53. (C) Blood group AB - Anti-A and anti-B antibodies — Group AB plasma contains neither anti-A nor anti-B antibody, which is why AB is the universal recipient.

54. (A) IV, II, I, III — The correct sequence is prophase, metaphase, anaphase, telophase. Chromosomes align at the equator in metaphase and sister chromatids separate in anaphase.

55. (D) II, IV, I, III — Food passes from the oesophagus to the stomach, then to the duodenum, jejunum and ileum of the small intestine. Maximum absorption occurs in the ileum through villi.

ST04 — EXPLANATIONS

1. (B) Female Anopheles mosquito — Only the female Anopheles mosquito transmits Plasmodium, the malaria parasite. Culex transmits filariasis and Japanese encephalitis, and Aedes transmits dengue and chikungunya.

2. (C) Plasmodium falciparum — Plasmodium falciparum causes malignant tertian or cerebral malaria and accounts for almost all malaria deaths. Plasmodium vivax is the commonest species in India.

3. (B) Ronald Ross — Ronald Ross demonstrated the malaria parasite in the stomach of an Anopheles mosquito at Secunderabad in 1897 and received the Nobel Prize in 1902.

4. (C) Aedes aegypti — Aedes aegypti, a daytime biter that breeds in clean stagnant water, transmits dengue, chikungunya, Zika and yellow fever.

5. (D) Leishmania donovani — Kala-azar is caused by the protozoan Leishmania donovani and is transmitted by the bite of the female sandfly Phlebotomus argentipes.

6. (A) Wuchereria bancrofti — Wuchereria bancrofti, a filarial nematode transmitted by Culex mosquitoes, blocks the lymphatic vessels and causes gross swelling of the limbs and genitalia.

7. (C) Pig — The pig is the amplifying host of the Japanese encephalitis virus and the ardeid water bird is the natural reservoir; the vector is the Culex tritaeniorhynchus mosquito.

8. (A) Yersinia pestis — Plague is caused by the bacterium Yersinia pestis and is transmitted by the rat flea, Xenopsylla cheopis. The Surat plague outbreak of 1994 is the best-known recent Indian episode.

9. (A) A larval trombiculid mite — Scrub typhus is caused by Orientia tsutsugamushi and is transmitted by the larval stage (chigger) of the trombiculid mite; an eschar at the bite site is characteristic.

10. (B) Robert Koch — Robert Koch identified Mycobacterium tuberculosis in 1882 and received the Nobel Prize in 1905; World TB Day is observed on 24 March, the day he announced the discovery.

11. (A) Tuberculosis — Bacillus Calmette-Guerin is a live attenuated vaccine against tuberculosis, especially the severe childhood forms such as miliary TB and TB meningitis.

12. (A) Mycobacterium leprae — Mycobacterium leprae was discovered by the Norwegian physician G.H.A. Hansen in 1873. Leprosy is treated with multi-drug therapy under the National Leprosy Eradication Programme.

13. (D) Typhoid — The Widal test detects agglutinating antibodies against Salmonella typhi, the causative organism of typhoid fever, which spreads through contaminated food and water.

14. (B) Vibrio cholerae — Vibrio cholerae, a comma-shaped bacterium identified by Robert Koch, causes profuse watery diarrhoea; oral rehydration solution is the mainstay of treatment.

15. (A) Hepatitis A — Hepatitis A and hepatitis E spread by the faeco-oral route, while hepatitis B, C and D spread through blood and body fluids. Hepatitis E is particularly dangerous in pregnancy.

16. (B) Live attenuated vaccine developed by Albert Sabin — Albert Sabin developed the live attenuated oral polio vaccine; Jonas Salk developed the killed injectable polio vaccine (IPV), which India introduced into routine immunisation in 2015.

17. (C) 2014 — The WHO South-East Asia Region, including India, was certified polio-free in March 2014; India's last case of wild poliovirus was reported from Howrah, West Bengal, in January 2011.

18. (D) Louis Pasteur — Louis Pasteur developed the rabies vaccine in 1885. Rabies is caused by a lyssavirus transmitted through the bite of an infected animal, most often a dog, and is nearly always fatal once symptoms appear.

19. (B) H1N1 — Swine flu is caused by the influenza A H1N1 virus, responsible for the 2009 pandemic; H5N1 is the avian influenza or bird flu virus.

20. (A) The fruit bat — Fruit bats of the genus Pteropus are the natural reservoir of Nipah virus; pigs can act as intermediate hosts. India has recorded outbreaks in West Bengal and Kerala.

21. (B) CD4 helper T lymphocytes — HIV, a retrovirus, infects and depletes CD4 helper T lymphocytes, crippling cell-mediated immunity. Screening is done by ELISA and confirmation by Western blot.

22. (B) Edward Jenner — Edward Jenner used cowpox material to protect against smallpox in 1796. WHO declared smallpox globally eradicated in 1980; India's last case was in 1975.

23. (A) Trisomy of chromosome 21 — Down syndrome results from an extra copy of chromosome 21, giving 47 chromosomes; it causes intellectual disability, a characteristic facial appearance and often congenital heart defects.

24. (C) X-linked recessive — Haemophilia, like colour blindness, is X-linked recessive; it appears far more often in males, who have only one X chromosome. It is called the royal disease.

25. (D) 45, XO — Turner syndrome is 45, XO - a female with a single X chromosome, showing short stature and sterility. Klinefelter syndrome is 47, XXY and affects males.

26. (C) Vitamin C — Scurvy results from a deficiency of vitamin C (ascorbic acid), impairing collagen synthesis; it causes bleeding gums and delayed wound healing.

27. (B) Iodine — Iodine deficiency prevents adequate thyroxine synthesis, causing compensatory enlargement of the thyroid; universal salt iodisation is the principal control measure.

28. (C) Rs 5 lakh — PM-JAY, launched on 23 September 2018 from Ranchi, provides Rs 5 lakh per family per year for secondary and tertiary care hospitalisation, with no cap on family size.

29. (B) Ayushman Arogya Mandir — In November 2023 the Health and Wellness Centres were renamed Ayushman Arogya Mandir with the tagline 'Arogyam Parmam Dhanam'; they deliver comprehensive primary health care.

30. (B) 2014 — Mission Indradhanush was launched on 25 December 2014 to raise full immunisation coverage of children and pregnant women to at least 90 per cent; Intensified Mission Indradhanush followed in October 2017.

31. (A) 2025 — India aims to eliminate TB by 2025, five years ahead of the global SDG target of 2030. The RNTCP was renamed the National TB Elimination Programme in 2020.

32. (C) 14 — The ABHA number is a unique 14-digit identifier that links an individual's digital health records. ABDM was launched nationwide on 27 September 2021.

33. (A) I and II only — Plasmodium falciparum causes malignant tertian (cerebral) malaria, the most fatal form; benign tertian malaria is caused by Plasmodium vivax. Hence statement III is wrong.

34. (D) I, II and III — All three statements are correct. Aedes aegypti breeds in clean stagnant water in and around houses, which is why source reduction is the key control measure.

35. (A) I and II only — BCG is a live attenuated vaccine, so statement III is wrong. Active immunity is slow to develop but long-lasting, whereas passive immunity acts at once but is short-lived.

36. (A) I and II only — PM-JAY is fully government-funded and beneficiaries pay no premium, so statement III is wrong. Beneficiaries were identified on the basis of deprivation criteria of SECC 2011.

37. (D) I, II and III — All three are correct. ABDM was piloted from 15 August 2020 in six union territories as the National Digital Health Mission before its national launch on 27 September 2021.

38. (C) I and III only — India's TB elimination target is 2025, five years ahead of the SDG target of 2030, so statement II is wrong. Ni-kshay Poshan Yojana began in April 2018.

39. (A) I and II only — Thalassaemia is an autosomal recessive disorder of haemoglobin synthesis, not a sex-linked one. Sickle cell anaemia is likewise autosomal recessive.

40. (D) I, II and III — All three statements are correct. The Red Line campaign marks prescription-only antibiotics in India to discourage self-medication.

41. (B) II and III only — An Rh-negative recipient develops anti-Rh antibodies after the first exposure, so a later transfusion of Rh-positive blood causes haemolysis; statement I is therefore wrong.

42. (A) I and II only — Only malaria, kala-azar and lymphatic filariasis are targeted for elimination; dengue, chikungunya and Japanese encephalitis are outbreak-prone diseases managed through control measures.

43. (D) I, II and III — All three are correct. Pulse Polio Immunisation began in India in 1995 and injectable polio vaccine was added to routine immunisation in 2015.

44. (A) I and II only — Rickets in children is caused by deficiency of vitamin D, which impairs calcium absorption; deficiency of vitamin C causes scurvy.

45. (A) A-3, B-4, C-2, D-1 — Malaria is caused by Plasmodium, kala-azar by Leishmania donovani, filariasis by Wuchereria bancrofti and typhoid by Salmonella typhi.

46. (D) A-3, B-1, C-4, D-2 — Anopheles transmits malaria, Aedes aegypti dengue, Culex lymphatic filariasis and the sandfly kala-azar. Disease-vector matching is a recurring RPSC device.

47. (B) A-3, B-4, C-1, D-2 — Tuberculosis is bacterial, dengue viral, ringworm fungal and amoebic dysentery protozoal (Entamoeba histolytica).

48. (C) A-3, B-4, C-1, D-2 — Vitamin A deficiency causes night blindness, vitamin C deficiency scurvy, vitamin D deficiency rickets and iodine deficiency goitre.

49. (A) A-3, B-4, C-2, D-1 — NRHM was launched in 2005, Mission Indradhanush in December 2014, PM-JAY in September 2018 and ABDM nationally in September 2021.

50. (D) A-3, B-4, C-1, D-2 — Down syndrome is trisomy 21, Turner syndrome 45 XO, Klinefelter syndrome 47 XXY and haemophilia an X-linked recessive gene disorder.

51. (D) A-3, B-4, C-2, D-1 — Koch identified the tubercle bacillus, Ross the malaria parasite in the mosquito, Jenner introduced smallpox vaccination and Fleming discovered penicillin.

52. (C) Kala-azar - Tsetse fly — Kala-azar is transmitted by the sandfly Phlebotomus argentipes; the tsetse fly transmits African sleeping sickness caused by Trypanosoma.

53. (C) Typhoid — Typhoid is a bacterial disease caused by Salmonella typhi. Measles, mumps and chickenpox are all caused by viruses.

54. (D) Pradhan Mantri Bhartiya Janaushadhi Pariyojana — The Janaushadhi Pariyojana, which supplies generic medicines through dedicated stores, is a separate scheme of the Department of Pharmaceuticals. Ayushman Arogya Mandir is the new name of the Health and Wellness Centre.

55. (D) Tetanus — Tetanus is caused by Clostridium tetani, whose spores enter the body through wounds contaminated with soil; it is not transmitted through water.

56. (A) BCG - Hepatitis B — BCG protects against tuberculosis, not hepatitis B; hepatitis B has its own separate vaccine, which is part of India's Universal Immunisation Programme.

57. (C) Tuberculosis — The programme covers malaria, dengue, chikungunya, Japanese encephalitis, kala-azar and lymphatic filariasis. Tuberculosis is dealt with under the National TB Elimination Programme.

58. (D) Anaemia Mukt Bharat - 2012 — Anaemia Mukt Bharat was launched in 2018 on a 6x6x6 strategy covering six beneficiary groups, six interventions and six institutional mechanisms.

59. (A) II, IV, I, III — NRHM was launched in 2005, Mission Indradhanush in December 2014, PM-JAY in September 2018 and the Ayushman Bharat Digital Mission nationally in September 2021.

60. (A) Both A and R are true and R is the correct explanation of A — Falciparum infects erythrocytes of every age and the parasitised cells adhere to capillary walls, producing cerebral malaria; this is why it is called malignant malaria.

ST05 — EXPLANATIONS

1. (C) Thiamine — Vitamin B1 is thiamine; its deficiency causes beriberi. Riboflavin is B2, niacin is B3 and pyridoxine is B6.

2. (C) Pellagra — Nicotinic acid (niacin, Vitamin B3) deficiency causes pellagra, marked by the three Ds - dermatitis, diarrhoea and dementia. Tryptophan in the diet can be converted to niacin.

3. (A) A-2, B-4, C-1, D-3 — Riboflavin = B2, cyanocobalamin = B12, ascorbic acid = C and tocopherol = E. Vitamin B12 is the only vitamin that contains a metal, cobalt.

4. (C) A-4, B-1, C-2, D-3 — Vitamin A deficiency causes night blindness and xerophthalmia, Vitamin D causes rickets in children, Vitamin C causes scurvy and Vitamin K causes delayed clotting because prothrombin synthesis fails.

5. (B) A, D, E, K — Vitamins A, D, E and K are fat-soluble; they are stored in the liver and adipose tissue and can cause hypervitaminosis. All B-complex vitamins and Vitamin C are water-soluble.

6. (C) Cobalt — Cyanocobalamin (Vitamin B12) has a cobalt atom at the centre of its corrin ring. Iron is central to haemoglobin and magnesium to chlorophyll.

7. (C) 9 kilocalories — Fat yields about 9 kcal/g, more than twice that of carbohydrate or protein (4 kcal/g each). Alcohol yields about 7 kcal/g.

8. (D) Pyridoxine - Pellagra — Pyridoxine (B6) deficiency causes anaemia, dermatitis and convulsions; pellagra is caused by deficiency of niacin (B3), not pyridoxine.

9. (A) I and II only — Kwashiorkor typically appears in children aged 1-3 years after weaning, not below six months; marasmus is the form seen in infants under one year. Oedema distinguishes kwashiorkor from marasmus.

10. (A) A-4, B-1, C-2, D-3 — Marasmus is severe protein-energy malnutrition with extreme wasting and no oedema. Stunting reflects chronic undernutrition, wasting reflects acute undernutrition and underweight is a composite indicator.

11. (C) Iodine — Iodine deficiency impairs thyroxine synthesis, causing compensatory enlargement of the thyroid gland (goitre); in the foetus and infant it causes cretinism.

12. (B) 30 ppm at manufacture and 15 ppm at consumer level — Iodine is volatile and is lost in storage, so the standard is 30 ppm at the point of manufacture and at least 15 ppm when it reaches the consumer.

13. (D) I, II and III — All three are correct. Fluorosis is a serious problem in several districts of western Rajasthan, where groundwater fluoride exceeds the permissible limit of 1.5 mg/L.

14. (A) Vitamin A — Bitot's spots are foamy white patches caused by Vitamin A deficiency; if untreated it progresses to xerophthalmia and keratomalacia, a major preventable cause of childhood blindness.

15. (D) Ascorbic acid — Ascorbic acid (Vitamin C) is required for hydroxylation of proline in collagen synthesis; its deficiency causes scurvy. Humans and guinea pigs cannot synthesise Vitamin C.

16. (B) Osteomalacia — The same deficiency causes rickets in growing children (soft, bowed bones) and osteomalacia in adults (softening of already formed bone). Vitamin D is synthesised in skin from 7-dehydrocholesterol using UV-B.

17. (A) I and II only — Vitamin K is fat-soluble, not water-soluble, so statement III is wrong. It is required for gamma-carboxylation of clotting factors including prothrombin, and menaquinone (K2) is produced by intestinal bacteria.

18. (D) I, II and III — All three are correct. For Asian Indians the cut-offs are lowered, with 23 and above taken as overweight and 25 and above as obese, because of higher body fat at a given BMI.

19. (B) Nine — Nine amino acids - histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine - cannot be synthesised by the human body and must come from food. Of the 20 standard amino acids the rest are non-essential.

20. (C) Fortified with micronutrients — The '+F' logo identifies fortified foods. FSSAI has notified fortification standards for salt, wheat flour, rice, milk and edible oil.

21. (C) A-2, B-1, C-3, D-4 — Ordinary iodised salt carries iodine only; double fortified salt carries iodine and iron. Rice and wheat flour are fortified with iron, folic acid and Vitamin B12, and milk and edible oil with Vitamins A and D.

22. (D) I, II and III — All three are correct. FSSAI, headquartered in New Delhi, also runs RUCO for used cooking oil, the Clean Street Food Hub scheme and the BHOG initiative for places of worship.

23. (D) Milk - Brick powder — Brick powder is an adulterant of chilli powder, not milk. Milk is commonly adulterated with water, starch, urea and detergent.

24. (A) Argemone oil — Argemone mexicana seed oil contains the alkaloids sanguinarine and dihydrosanguinarine, which cause epidemic dropsy. Argemone seeds resemble mustard seeds, making the adulteration easy.

25. (C) Kesari dal (Lathyrus sativus) — Kesari dal contains the neurotoxin beta-ODAP (BOAA); when it forms a large part of the diet or adulterates arhar dal it causes neurolathyrism.

26. (B) 1975 — ICDS was launched on 2 October 1975 and delivers six services through anganwadi centres - supplementary nutrition, pre-school non-formal education, nutrition and health education, immunisation, health check-up and referral services.

27. (B) Jhunjhunu — The Prime Minister launched POSHAN Abhiyaan from Jhunjhunu in Rajasthan on International Women's Day, 8 March 2018. It targets a 2% annual reduction in stunting and underweight and a 3% annual reduction in anaemia.

28. (D) I, II and III — All three are correct. The Mid-Day Meal Scheme was launched nationally on 15 August 1995 and was renamed PM POSHAN in September 2021, extending coverage to pre-primary Bal Vatika children.

29. (B) 5 kg — Priority households get 5 kg per person per month while Antyodaya Anna Yojana households get 35 kg per household per month. The Act covers up to 75% of the rural and 50% of the urban population.

30. (D) 6x6x6 — Anaemia Mukt Bharat uses a 6x6x6 strategy - six target beneficiary groups, six interventions and six institutional mechanisms. India's anaemia burden remains high: NFHS-5 recorded 67.1% among children aged 6-59 months.

31. (B) Pearl millet — Dhanashakti is an iron and zinc dense pearl millet (bajra) variety. Biofortification raises nutrient content through plant breeding rather than by adding nutrients during processing.

32. (A) I and II only — Golden Rice carries genes for the biosynthesis of beta-carotene, the precursor of Vitamin A, giving the grain its golden colour. It is not an iron-biofortified rice, so statement III is wrong.

33. (B) 72 degrees C for 15 seconds — HTST or flash pasteurisation uses 72 C for 15 seconds followed by rapid cooling. LTLT holding pasteurisation uses 63 C for 30 minutes and UHT uses 135-140 C for 2-3 seconds.

34. (A) A-2, B-1, C-4, D-3 — Lyophilisation is freeze drying; irradiation uses Cobalt-60 gamma rays and carries the Radura logo; canning was invented by Nicolas Appert; blanching inactivates enzymes before freezing.

35. (C) Tocopherol — Tocopherol (Vitamin E) is fat-soluble. Thiamine, niacin and folic acid all belong to the water-soluble B-complex group, which is not stored in the body and must be supplied regularly.

36. (A) Probiotics are live beneficial microorganisms while prebiotics are non-digestible food components that promote their growth — Probiotics are live microorganisms such as Lactobacillus and Bifidobacterium; prebiotics are non-digestible substrates such as inulin and fructo-oligosaccharides. A product containing both is called a synbiotic.

37. (D) Glycogen — Glycogen is the storage carbohydrate of animals and is readily digested. Cellulose, hemicellulose, pectin, gums and lignin resist human digestive enzymes and provide roughage.

38. (A) I, II, III, IV — ICDS 1975, National Food Security Act 2013, POSHAN Abhiyaan March 2018 and PM POSHAN (renaming of the Mid-Day Meal Scheme) September 2021.

39. (D) I, II and III — All three are correct. Rajasthan is India's largest producer of bajra (pearl millet), and the Indian Institute of Millets Research, Hyderabad, is the nodal institute for millets.

40. (B) 2 per cent by weight — India capped industrially produced trans fats in oils, fats and foods at 2% by weight with effect from January 2022, ahead of the WHO's global target date, under the 'Trans Fat Free India' campaign.

ST06 — EXPLANATIONS

1. (B) Pisum sativum — Mendel used the garden pea, Pisum sativum, because it is naturally self-pollinating, has clear contrasting characters and a short life cycle. He studied seven pairs of contrasting characters.

2. (C) Seven — Mendel studied seven pairs of contrasting characters, including seed shape, seed colour, flower colour, pod shape, pod colour, flower position and stem height.

3. (B) Law of segregation — The law of segregation, also called the law of purity of gametes, holds universally because alleles simply separate during gamete formation. Dominance has exceptions (incomplete dominance, codominance) and independent assortment fails for linked genes.

4. (A) A-3, B-4, C-2, D-1 — The monohybrid F2 is 3:1 phenotypically and 1:2:1 genotypically, the dihybrid F2 is 9:3:3:1, and a dihybrid test cross gives 1:1:1:1. In incomplete dominance the F2 phenotypic ratio also becomes 1:2:1.

5. (C) 9:3:3:1 — The 9:3:3:1 ratio in F2 demonstrates the law of independent assortment. A dihybrid test cross gives 1:1:1:1.

6. (C) Homozygous recessive — A test cross uses the homozygous recessive parent so that the phenotypes of the offspring directly reveal the gametes and hence the genotype of the unknown parent.

7. (C) A-2, B-3, C-4, D-1 — Pink flowers in Antirrhinum and Mirabilis jalapa show incomplete dominance, the AB blood group shows codominance, phenylketonuria shows pleiotropy and human skin colour is polygenic.

8. (B) Codominance — In group AB both I(A) and I(B) alleles are fully and simultaneously expressed on the red cell surface, which is codominance. The ABO system also illustrates multiple allelism.

9. (D) I, II and III — All three are correct: two heterozygous group A parents (I(A)i x I(A)i) can produce an ii child of group O. The ABO gene lies on chromosome 9.

10. (A) A or B only — The father can give either I(A) or I(B) and the mother can give only i, so the children will be I(A)i (group A) or I(B)i (group B). Neither AB nor O is possible.

11. (C) An Rh-negative mother carries an Rh-positive foetus — An Rh-negative mother sensitised to the Rh antigen of an Rh-positive foetus produces anti-Rh antibodies that destroy foetal red cells, usually in the second and later pregnancies. It is prevented by giving anti-D immunoglobulin.

12. (B) Pleiotropy — Pleiotropy is one gene with many effects, as in phenylketonuria and sickle cell anaemia. Polygenic inheritance is the opposite - many genes acting on one trait, as in human skin colour and height.

13. (D) Polygenic inheritance — Both are quantitative traits governed by several genes with additive effects, giving continuous variation in a population. Kolreuter's study of wheat kernel colour was an early demonstration.

14. (C) T. H. Morgan — T. H. Morgan and his students used the fruit fly to establish the chromosomal theory of linkage; his student Alfred Sturtevant used recombination frequency to construct the first genetic map.

15. (B) A-3, B-4, C-1, D-2 — Humans follow XX-XY and grasshoppers XX-XO (both male heterogametic); birds follow ZZ-ZW with the female heterogametic; honeybees follow haplodiploidy, drones developing parthenogenetically from unfertilised eggs.

16. (C) Domestic fowl — Birds, including the domestic fowl, follow the ZZ-ZW system in which the female is ZW and therefore heterogametic. Humans, Drosophila and grasshoppers all have male heterogamety.

17. (A) The mother contributes the Y chromosome to a male child — A mother is XX and can contribute only an X chromosome; the Y always comes from the father. The SRY gene on the short arm of the Y chromosome triggers testis development.

18. (A) I and II only — A father passes his Y chromosome, not his X, to his sons, so statement III is wrong; he passes the defective X to all his daughters, who become carriers. About 8% of men and 0.4% of women are affected.

19. (A) Factor VIII — Haemophilia A, the classical form, is due to deficiency of clotting factor VIII; haemophilia B or Christmas disease is due to factor IX deficiency. Both are X-linked recessive.

20. (C) One-half — The carrier mother passes either the normal or the defective X with equal probability; sons receiving the defective X are haemophilic, so half the sons are affected while all daughters are phenotypically normal (half of them carriers).

21. (A) Sickle cell anaemia — Sickle cell anaemia is an autosomal recessive disorder of the beta-globin gene on chromosome 11. The other three are X-linked recessive and therefore far commoner in males.

22. (C) Trisomy of chromosome 21 — Down syndrome is trisomy 21, giving a chromosome number of 47. It arises from non-disjunction during gamete formation and its incidence rises sharply with maternal age.

23. (C) 47 — An extra copy of chromosome 21 raises the count from 46 to 47. The karyotype is written 47, +21.

24. (B) I, II and III — All three statements are correct. Congenital heart disease and intellectual disability are also usual features; prenatal detection is possible by karyotyping of amniotic fluid cells.

25. (B) A-2, B-1, C-4, D-3 — Klinefelter syndrome is 47, XXY (a sterile male with gynaecomastia), Turner syndrome 45, X0 (a sterile female with a webbed neck), Down syndrome trisomy 21 and Edwards syndrome trisomy 18.

26. (B) One — The number of Barr bodies equals the number of X chromosomes minus one. A Klinefelter male has two X chromosomes and therefore one Barr body; a normal male has none and a Turner female has none.

27. (A) I and II only — It is a single base substitution (GAG to GTG), a missense point mutation, not a frameshift mutation, so statement III is wrong. Heterozygous carriers show the sickle cell trait and are relatively protected against falciparum malaria.

28. (A) Klinefelter syndrome - 45, X0 — Klinefelter syndrome is 47, XXY; the karyotype 45, X0 is Turner syndrome. The other three pairs are correctly matched.

29. (C) A-4, B-1, C-2, D-3 — Watson and Crick proposed the double helix in 1953 using Rosalind Franklin's X-ray data; Griffith discovered transformation in 1928; Hershey and Chase used bacteriophage in 1952; de Vries proposed the mutation theory.

30. (B) I and III only — Adenine and thymine are joined by two hydrogen bonds, while guanine and cytosine are joined by three, so statement II is wrong. One turn is 3.4 nm long with ten base pairs, each 0.34 nm apart.

31. (B) I, II and III — All three are correct. Meselson and Stahl used the heavy isotope nitrogen-15 with density gradient centrifugation; the Okazaki fragments of the lagging strand are finally sealed by DNA ligase.

32. (B) I, II, III, IV — Griffith 1928, Avery-MacLeod-McCarty 1944, Hershey-Chase 1952 and Meselson-Stahl 1958. Together they established DNA as the genetic material and its mode of replication.

33. (C) Three — UAA, UAG and UGA are stop codons; the remaining 61 are sense codons. AUG is both the initiation codon and the codon for methionine.

34. (A) Tryptophan — Tryptophan (UGG) and methionine (AUG) are each specified by a single codon; leucine, serine and arginine have six codons each, illustrating the degeneracy of the code.

35. (A) It is ambiguous — The genetic code is unambiguous - one codon specifies only one amino acid. It is degenerate because one amino acid may have several codons, and it is nearly universal.

36. (A) Frameshift mutation — Insertions or deletions that are not multiples of three shift the reading frame and usually produce a completely altered, non-functional protein. Acridine dyes are classic frameshift mutagens.

37. (B) Oenothera lamarckiana — De Vries worked on the evening primrose, Oenothera lamarckiana, and argued that evolution proceeds by sudden large changes he called saltations, unlike Darwin's gradual variations.

38. (A) I and II only — Genetic drift and natural selection disturb the equilibrium rather than maintain it, so statement III is wrong. Equilibrium needs a large randomly mating population with no mutation, migration, drift or selection.

39. (B) Directional selection — Soot-darkened tree trunks favoured the dark form over the light form, shifting the population mean in one direction - the definition of directional selection. It is the best documented case of natural selection in action.

40. (A) Both A and R are true and R is the correct explanation of A — Common ancestry with adaptation to different functions produces homologous organs and divergent evolution. Analogous organs such as the wings of a bird and an insect indicate convergent evolution.

41. (B) A-2, B-1, C-3, D-4 — Bateson coined 'genetics', Johannsen coined 'gene', Jeffreys developed DNA fingerprinting in 1984 and Khorana shared the 1968 Nobel Prize for work on the genetic code.

42. (B) Repetitive DNA sequences such as VNTRs — Variable Number Tandem Repeats and short tandem repeats differ in copy number between individuals, giving each person a unique pattern. Alec Jeffreys developed the technique in 1984; in India, Lalji Singh of CCMB Hyderabad pioneered it.

43. (A) I and II only — The project found only about 20,000 to 25,000 protein-coding genes, far fewer than the earlier estimate of one lakh, so statement III is wrong. Craig Venter's Celera Genomics ran a parallel private effort.

44. (B) It is present in one copy in normal females — Normal females are XX and carry no Y chromosome at all. The Y is much smaller than the X, carries the SRY gene and is present singly in a normal male.

45. (A) Amniocentesis — Amniocentesis samples foetal cells from the amniotic fluid for karyotyping. In India its misuse for sex determination is prohibited under the Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994.

ST07 — EXPLANATIONS

1. (A) The bacterial strain — In EcoRI, 'E' is for the genus Escherichia, 'co' for the species coli, 'R' for the strain RY13 and the Roman numeral I for the order in which it was isolated from that strain.

2. (B) I and II only — Joining fragments is the work of DNA ligase, not of restriction enzymes, so statement III is wrong. Enzymes that cut in a staggered way generate sticky ends, which is what makes recombination possible.

3. (B) Ability to code for a human protein — A vector must be able to replicate (ori), allow selection of transformed cells (marker) and have unique restriction sites for inserting foreign DNA. Coding for a human protein is a property of the inserted gene, not of the vector.

4. (A) Agrobacterium tumefaciens — Agrobacterium tumefaciens, which causes crown gall disease, naturally transfers a segment of its tumour-inducing (Ti) plasmid into plant cells; it is disarmed and used as the standard vector for dicot transformation.

5. (B) A-2, B-1, C-4, D-3 — Ligase joins fragments, PCR amplifies, electrophoresis separates fragments by size and the biolistic gene gun bombards plant cells with DNA-coated gold or tungsten particles.

6. (B) I, II and III — All three are correct. Taq polymerase survives the repeated denaturation step at about 95 C, which is what made automated cycling possible; about 30 cycles amplify the target roughly a billion-fold.

7. (C) II, I, III — Each cycle proceeds as denaturation (about 95 C), annealing of primers (about 50-55 C) and extension (about 72 C); around 30 cycles amplify the target about a billion-fold.

8. (A) Anode, because DNA is negatively charged — The phosphate backbone gives DNA a net negative charge, so it migrates to the positively charged anode; smaller fragments travel farther. Bands are stained with ethidium bromide and viewed under UV light.

9. (A) Insulin — Recombinant human insulin, marketed as Humulin, was the first rDNA product approved for human use. The A chain of 21 and the B chain of 30 amino acids were made separately in E. coli and joined by disulphide bonds.

10. (D) A-3, B-1, C-4, D-2 — Cas9 is an endonuclease guided to its target by a guide RNA and is used for genome editing; hybridoma technology yields monoclonal antibodies; nuclear transfer produced Dolly; RNA interference silences messenger RNA.

11. (A) CRISPR-Cas9 — FELUDA was developed by CSIR-Institute of Genomics and Integrative Biology, New Delhi, using a Cas9 protein to detect the viral genome, and was named after Satyajit Ray's fictional detective.

12. (A) DRR Dhan 100 (Kamala) and Pusa DST Rice 1 — DRR Dhan 100 'Kamala' was developed by ICAR-Indian Institute of Rice Research and Pusa DST Rice 1 by ICAR-IARI, both using site-directed nuclease editing that introduces no foreign gene and is therefore exempt from GMO rules.

13. (A) Bacillus thuringiensis — Bacillus thuringiensis produces crystal (cry) protein toxins. Genes cry1Ac and cry2Ab control cotton bollworms, while cry1Ab controls the corn borer.

14. (B) I, II and III — All three are correct. Because the toxin needs an alkaline gut to be activated, it does not harm the bacterium itself and is not toxic to humans, whose stomach is acidic.

15. (A) Bt cotton — Bt cotton (Bollgard-I) was approved by GEAC in March 2002 and Bollgard-II in 2006. Bt brinjal is under moratorium and GM mustard DMH-11 has not been released commercially.

16. (C) 2010 — GEAC recommended approval in October 2009, but after nationwide public consultations the then Union Environment Minister Jairam Ramesh announced an indefinite moratorium in February 2010. Bangladesh has since commercialised Bt brinjal.

17. (C) I and II only — GEAC recommended environmental release of DMH-11 in October 2022, but the Supreme Court gave a split verdict in July 2024 and referred the matter to a larger bench; it is not in commercial cultivation, so statement III is wrong.

18. (D) Flavr Savr tomato — The Flavr Savr tomato, engineered for delayed softening by suppressing the polygalacturonase enzyme, was approved in the United States in 1994.

19. (C) I, II and III — All three are correct. Dolly was born on 5 July 1996 to the team of Ian Wilmut and Keith Campbell, proving that the nucleus of a differentiated cell can be reprogrammed to form a whole organism.

20. (D) Totipotency — Totipotency is the basis of plant tissue culture; G. Haberlandt, who first proposed it, is called the Father of plant tissue culture. Animal embryonic stem cells are only pluripotent.

21. (C) I, II and III — All three are correct. Murashige and Skoog medium is the standard culture medium, and micropropagation produces genetically identical plants called somaclones.

22. (B) Callus culture - Production of transgenic animals — Callus culture is a plant tissue culture technique used to regenerate plantlets; transgenic animals are produced by microinjection of DNA into a fertilised egg or by nuclear transfer.

23. (D) A-2, B-1, C-4, D-3 — Rhizobium fixes nitrogen in legume nodules, Azotobacter is a free-living aerobic fixer, Anabaena azollae lives inside the Azolla fern used in paddy fields, and mycorrhizal fungi mainly improve phosphorus absorption.

24. (D) Digests hydrocarbons of oil spills and was the first patented living organism — Chakrabarty engineered Pseudomonas putida to break down several hydrocarbons in oil spills; the US Supreme Court in Diamond v. Chakrabarty (1980) allowed the first patent on a living organism.

25. (D) A-2, B-4, C-1, D-3 — Aspergillus niger yields citric acid, Penicillium chrysogenum penicillin, Trichoderma polysporum the immunosuppressant cyclosporin A and Monascus purpureus the cholesterol-lowering statins.

26. (B) Acetobacter aceti - Citric acid — Acetobacter aceti produces acetic acid (vinegar); citric acid is produced by the fungus Aspergillus niger.

27. (A) Edible food crops such as sugarcane, corn and vegetable oils — First-generation biofuels come from edible feedstocks and therefore compete with food; second generation uses non-food lignocellulosic residues, third generation uses algae and fourth generation uses engineered organisms.

28. (D) Third-generation biofuels — Algae, with their high lipid content and very high productivity per unit area, are the feedstock of third-generation biofuels; they need neither cropland nor fresh water in most systems.

29. (D) Genetically or metabolically engineered algae and microbes aimed at capturing carbon dioxide — Fourth-generation biofuels use genetically engineered algae and micro-organisms designed to fix and store more carbon dioxide than the fuel releases, making the fuel potentially carbon-negative.

30. (C) I, II and III — All three are correct. India reached the 20% ethanol blending level in 2025, five years ahead of the originally notified target year of 2030.

31. (D) Adult somatic cells reprogrammed to an embryonic-like state — Shinya Yamanaka reprogrammed adult fibroblasts into pluripotent cells using four transcription factors, sharing the 2012 Nobel Prize with John Gurdon. iPSCs avoid the ethical problems of embryonic stem cells.

32. (D) Adenosine deaminase deficiency — A four-year-old girl with severe combined immunodeficiency caused by adenosine deaminase (ADA) deficiency received lymphocytes carrying a functional ADA gene. The effect was not permanent as mature lymphocytes are short-lived.

33. (D) Covaxin — Covaxin (BBV152) was developed by Bharat Biotech with ICMR-National Institute of Virology, Pune, using an inactivated whole virus with an Algel-IMDG adjuvant.

34. (D) Recombinant viral vector — Covishield is the Oxford-AstraZeneca ChAdOx1 nCoV-19 vaccine, which uses a replication-deficient chimpanzee adenovirus vector carrying the SARS-CoV-2 spike protein gene.

35. (C) I, II and III — All three are correct. ZyCoV-D was developed by Zydus and received emergency use authorisation in 2021 for persons aged 12 years and above.

36. (D) A-2, B-4, C-1, D-3 — Covaxin is an inactivated whole-virion vaccine, Covishield an adenoviral vector vaccine, ZyCoV-D a DNA plasmid vaccine and GEMCOVAC-19 India's first indigenous mRNA vaccine.

37. (D) Hybridoma technology — Hybridoma technology fuses an antibody-producing B lymphocyte with a myeloma cell to give an immortal clone secreting a single type of antibody. Georges Kohler and Cesar Milstein shared the 1984 Nobel Prize for it.

38. (C) I and II only — GEAC works under the Ministry of Environment, Forest and Climate Change, not the Department of Biotechnology, so statement III is wrong. RCGM and RDAC are the DBT-level bodies.

39. (D) Chennai — The National Biodiversity Authority is located at Chennai. The three-tier structure comprises the NBA, State Biodiversity Boards and Biodiversity Management Committees at the local body level, which maintain People's Biodiversity Registers.

40. (C) It abolished the National Biodiversity Authority — The 2023 amendment retained the National Biodiversity Authority. Its main changes were exemptions for AYUSH practitioners and cultivated medicinal plants and the conversion of criminal offences into civil penalties.

41. (C) Protection of Plant Varieties and Farmers' Rights Act, 2001 - Regulation of biofuel blending — The Protection of Plant Varieties and Farmers' Rights Act, 2001 is India's sui generis law for protecting plant varieties and, uniquely, farmers' rights; it has nothing to do with biofuel blending.

42. (D) Economy, Environment and Employment — BioE3 stands for Biotechnology for Economy, Environment and Employment. It promotes high-performance biomanufacturing, biofoundries and bio-AI hubs across themes such as smart proteins, precision biotherapeutics and carbon capture.

43. (D) II, I, III, IV — Dolly was born in 1996, Bt cotton was approved in 2002, the CRISPR Nobel Prize was awarded in 2020 and GEAC recommended release of DMH-11 in 2022.

44. (D) Department of Biotechnology — The Biotechnology Industry Research Assistance Council was set up in 2012 as a not-for-profit enterprise of the Department of Biotechnology, which itself was created in 1986 under the Ministry of Science and Technology.

45. (D) Silencing a specific messenger RNA so that the protein is not translated — Double-stranded RNA triggers the degradation of the complementary mRNA, silencing the gene at the post-transcriptional level. Andrew Fire and Craig Mello received the 2006 Nobel Prize for discovering the mechanism.

ST08 — EXPLANATIONS

1. (C) 10^-9 metre — One nanometre is one-billionth of a metre, i.e. 10^-9 m, which equals 10 angstrom. A human hair is about 80,000-100,000 nm thick.

2. (A) 1-100 nm — The internationally accepted definition of the nanoscale is 1-100 nm. Below this range one deals with individual atoms and molecules; above it, bulk properties dominate.

3. (A) Richard Feynman — Richard Feynman delivered it at the American Physical Society meeting at Caltech on 29 December 1959, envisaging manipulation of matter atom by atom.

4. (D) Norio Taniguchi — Norio Taniguchi of Tokyo University of Science coined the term in 1974. K. Eric Drexler later popularised it through 'Engines of Creation' (1986).

5. (C) Quantum dot — A quantum dot confines carriers in all three dimensions and is therefore described as a zero-dimensional structure or an 'artificial atom'. A quantum well confines them in one direction only.

6. (A) I and II only — Statements I and II are correct. III is wrong — quantum confinement makes emission strongly size-dependent, smaller dots emitting bluer light.

7. (D) 20 hexagons and 12 pentagons — C60 is a truncated icosahedron of 60 carbon atoms made up of 20 hexagons and 12 pentagons, resembling a football. Kroto, Curl and Smalley won the 1996 Nobel Prize in Chemistry for it.

8. (B) Sumio Iijima — Sumio Iijima of NEC, Japan reported multi-walled carbon nanotubes in 1991. CNTs are rolled-up graphene sheets, far stronger than steel yet much lighter.

9. (B) Graphene — They isolated graphene at the University of Manchester in 2004 using the adhesive-tape (mechanical exfoliation) method. Graphene is a single atomic layer of sp2 carbon.

10. (A) A-2, B-4, C-3, D-1 — The four-generation scheme is due to M.C. Roco: passive nanostructures (coatings, nanoparticles), active nanostructures (targeted drugs, actuators), systems of nanosystems (nanorobotics) and molecular nanosystems.

11. (A) Nitinol — Nitinol is a nickel-titanium shape-memory alloy, not a form of carbon. Graphene, fullerene, carbon nanotube, graphite and diamond are all carbon allotropes.

12. (D) Binnig and Rohrer — Gerd Binnig and Heinrich Rohrer built the STM at IBM Zurich and shared the 1986 Nobel Prize in Physics. The STM works on quantum tunnelling of electrons between a sharp tip and a conducting surface.

13. (A) I and II only — Top-down methods break bulk material down (ball milling, lithography, etching); bottom-up methods build up from atoms and molecules (sol-gel, chemical vapour deposition, self-assembly). Hence III is wrong.

14. (A) Department of Science and Technology — The Nano Mission was launched in May 2007 by the Department of Science and Technology with an allocation of about Rs 1,000 crore. It concluded on 31 March 2017 and continues as the National Programme on Nano Science and Technology.

15. (A) A-2, B-4, C-1, D-3 — INST is at Mohali, ARCI at Hyderabad, CeNS at Bengaluru and the National Centre for Nanosciences and Nanotechnology is at the University of Mumbai.

16. (D) A-1, B-2, C-3, D-4 — TEM transmits electrons through an ultrathin sample; AFM senses tip-sample force; STM measures tunnelling current and works only on conducting surfaces; XRD determines crystal structure.

17. (C) Dendrimers — Dendrimers have a central core with regularly branched generations and numerous surface groups, allowing drugs to be carried inside cavities or attached to the surface.

18. (B) Generation of electricity by nuclear fission — Nuclear fission power depends on the splitting of heavy nuclei, not on nanoscale engineering. The other three are established nanotechnology applications.

19. (B) Gold and silver — The glass contains gold and silver nanoparticles whose surface plasmon resonance changes the colour with the direction of light. Michael Faraday studied colloidal gold in 1857.

20. (D) I, II and III — All three are correct. These two effects — an enormous specific surface area and quantum confinement — are why nanomaterials behave differently from the same substance in bulk.

21. (A) II, I, IV, III — Feynman's lecture 1959, fullerene 1985, carbon nanotubes 1991, graphene 2004.

22. (B) Quantum dots — Moungi Bawendi, Louis Brus and Alexei Ekimov shared the 2023 Chemistry Nobel for quantum dots, whose properties are governed by size rather than composition.

23. (D) A single layer of carbon atoms arranged in a honeycomb lattice — Graphene is a one-atom-thick, two-dimensional honeycomb sheet of sp2 carbon. The cage of sixty atoms is fullerene, the rolled sheet is a nanotube and the sp3 network is diamond.

24. (C) A-1, B-2, C-3, D-4 — A quantum dot confines carriers in all three dimensions (0-D), a quantum wire in two (1-D free motion), a quantum well in one (2-D free motion) and a bulk crystal not at all (3-D).

25. (C) A-2, B-1, C-4, D-3 — Nitinol is a nickel-titanium shape-memory alloy, Kevlar an aramid fibre, quartz a piezoelectric crystal and aerogel the ultra-light 'frozen smoke' insulator.

26. (B) Meissner effect — The Meissner effect, along with zero electrical resistance, defines superconductivity. Superconductivity was discovered by Heike Kamerlingh Onnes in mercury in 1911.

27. (D) I, II and III — All three are correct. Metamaterials are periodic arrays of sub-wavelength elements whose collective response can bend electromagnetic waves in ways no natural material does.

28. (B) An extremely light thermal insulator — Aerogel is a gel in which the liquid has been replaced by gas; it is more than 90 per cent air, among the lightest solids known, and an outstanding thermal insulator used in spacecraft and clothing.

29. (D) Nitinol — a ceramic-matrix composite — Nitinol is an alloy of nickel and titanium, not a composite. A composite always has a distinct matrix and a reinforcing phase.

30. (C) Ministry of Electronics and Information Technology — The India Semiconductor Mission is a specialised business division set up under the Ministry of Electronics and Information Technology to run the Semicon India Programme.

31. (B) Rs 76,000 crore — The programme was notified with an outlay of Rs 76,000 crore. As per a PIB release of 1 April 2026, ten projects with investment commitments of about Rs 1.6 lakh crore have been approved and ISM 2.0 was announced in the Union Budget 2026-27.

32. (C) Dholera, Gujarat — The Tata-PSMC fab is at Dholera in Gujarat. Tata's Jagiroad (Assam) plant is an assembly and test (OSAT) unit, while Micron and Kaynes have units at Sanand.

33. (C) Both I and II — Both are correct. A group-15 dopant supplies an extra electron (n-type); a group-13 dopant creates a hole (p-type). Their junction forms the p-n diode, the building block of all electronics.

34. (D) I, II and III — All three are correct. Rare earths are 17 elements in all. India launched the National Critical Mineral Mission in January 2025 to secure such minerals.

35. (A) They are always electrical insulators — CNTs are not insulators. Depending on chirality they behave as metals (armchair type) or as semiconductors (zigzag and chiral types).

ST09 — EXPLANATIONS

1. (C) Charles Babbage — Charles Babbage designed the Difference Engine (1822) and the Analytical Engine (1837), which already had the elements of a modern computer. Ada Lovelace, who wrote algorithms for it, is called the first programmer.

2. (B) The first general-purpose electronic digital computer — ENIAC used about 18,000 vacuum tubes and belongs to the first generation. UNIVAC-I was the first commercial computer and EDVAC introduced the stored-programme concept.

3. (A) A-3, B-4, C-1, D-2 — Vacuum tube, transistor, integrated circuit and microprocessor mark the first four generations. The fifth generation is associated with artificial intelligence and ULSI.

4. (A) I and II only — I and II are correct. Registers, located inside the CPU, are the fastest — not the slowest — storage in the memory hierarchy.

5. (D) 8 bits — A byte is 8 bits and can represent 256 different values. A nibble is 4 bits and 1 kilobyte is 1024 bytes.

6. (D) Nibble — Four bits make a nibble, which is exactly one hexadecimal digit. Two nibbles make one byte.

7. (A) II, IV, I, III — Kilobyte < Megabyte < Gigabyte < Terabyte, each step being a factor of 1024. Petabyte, exabyte, zettabyte and yottabyte follow.

8. (C) 16 — Hexadecimal is base 16 and uses the digits 0-9 followed by A, B, C, D, E and F. Binary is base 2 and octal base 8.

9. (D) 11001 — 25 = 16 + 8 + 1, so the bits for 16, 8 and 1 are set, giving 11001.

10. (C) Plotter — A plotter is an output device that draws vector graphics such as maps and engineering drawings. The other three feed data into the computer.

11. (B) RAM — RAM is volatile working memory. ROM and all secondary storage devices retain data without power and are therefore non-volatile.

12. (D) EPROM — EPROM (Erasable Programmable Read Only Memory) has a quartz window through which ultraviolet light erases the stored data. EEPROM is erased electrically and PROM can be written only once.

13. (B) I and II only — I and II are correct. Instructions must first be brought into main memory before the CPU can execute them, so III is wrong.

14. (A) Both A and R are true and R is the correct explanation of A — Both statements are true and the non-volatility of ROM is precisely why its contents survive a power cut. The BIOS is stored in such memory.

15. (C) Oracle — Oracle is a relational database management system. Ubuntu is a Linux distribution, and Android and macOS are operating systems.

16. (B) FORTRAN — FORTRAN (Formula Translation, 1957) was designed for scientific and engineering computation. COBOL, which followed, was meant for business data processing.

17. (D) Compiler — A compiler translates the whole programme in one pass and lists all errors; an interpreter translates and executes one line at a time and stops at the first error.

18. (C) E. F. Codd — E.F. Codd of IBM proposed the relational model in 1970, in which data are held in tables of rows (tuples) and columns (attributes) and queried using SQL.

19. (A) A-2, B-3, C-4, D-1 — The mesh topology, in which every node links to every other, is the most fault-tolerant but the costliest in cabling; the star is the commonest in offices.

20. (C) I and III only — I and III are correct. The TCP/IP model has four layers — network access, internet, transport and application — so II is wrong.

21. (B) A-2, B-1, C-4, D-3 — SMTP sends mail (port 25), FTP transfers files (ports 20/21), DNS resolves names (port 53) and DHCP allots IP addresses dynamically.

22. (D) 32 bits — IPv4 uses 32 bits written as four decimal octets separated by dots. IPv6, introduced to overcome address exhaustion, uses 128 bits.

23. (B) Tim Berners-Lee — Tim Berners-Lee proposed the Web at CERN and created HTML, HTTP and the first browser. Vint Cerf and Robert Kahn designed the TCP/IP protocols.

24. (D) I, II and III — All three are correct. The third model, Platform as a Service, gives developers a ready environment to build and deploy applications.

25. (C) Blockchain — a centralised database controlled by a single trusted authority — Blockchain is precisely the opposite of a centralised database: it is a decentralised, distributed and tamper-evident ledger maintained across many nodes.

26. (A) I and II only — I and II are correct. Bitcoin was released in 2009 by the pseudonymous 'Satoshi Nakamoto'; the RBI's own digital currency, the Digital Rupee (CBDC), began pilot runs only in 2022.

27. (B) John McCarthy — John McCarthy, who also created the LISP language, coined the term. Alan Turing had earlier proposed the Turing Test in his 1950 paper on machine intelligence.

28. (C) I and III only — I and III are correct. Clustering works on unlabelled data and is therefore unsupervised learning, so II is wrong.

29. (C) Transformer — The transformer architecture, based on the self-attention mechanism, underlies today's large language models. Convolutional networks are mainly used for images.

30. (D) Ministry of Electronics and Information Technology — The mission rests on seven pillars including IndiaAI Compute Capacity, Datasets Platform, FutureSkills, Startup Financing and Safe and Trusted AI. NITI Aayog had earlier issued the National Strategy for AI (2018).

31. (A) Rs 6,004 crore — As per the Department of Science and Technology, the mission was approved on 19 April 2023 with an outlay of Rs 6,003.65 crore for 2023-24 to 2030-31 and targets quantum computers of 50-1000 physical qubits in eight years.

32. (A) A-2, B-3, C-4, D-1 — As notified by DST, the four T-Hubs are at IISc Bengaluru (computing), IIT Madras (communication), IIT Bombay (sensing and metrology) and IIT Delhi (materials and devices).

33. (D) I, II and III — All three are correct. Superposition and entanglement are the two properties that give quantum computers their potential advantage over classical machines.

34. (B) October 2022 — The Prime Minister launched 5G services on 1 October 2022 at the India Mobile Congress in New Delhi, following the 5G spectrum auction held earlier that year.

35. (B) I and III only — I and III are correct. The Bharat 6G Alliance (2023) is an industry-academia body promoted under the Department of Telecommunications, not the Ministry of Defence.

36. (A) Vijay Bhatkar — Dr Vijay P. Bhatkar led the PARAM project and is known as the father of Indian supercomputing. PARAM 8000 was built after India was denied access to foreign supercomputers.

37. (B) IIT (BHU) Varanasi — PARAM Shivay was commissioned at IIT (BHU) Varanasi in 2019. The mission is implemented by C-DAC Pune and IISc Bengaluru under MeitY and DST.

38. (D) The three PARAM Rudra supercomputers were dedicated to the nation in 2019 — The three PARAM Rudra systems, at Pune, Delhi and Kolkata, were dedicated in September 2024, not 2019. PARAM Shivay of 2019 was the first system under the mission.

39. (D) I, II and III — All three are correct. The WannaCry attack of 2017 is the best-known ransomware episode, affecting computers in more than 150 countries.

40. (A) UNCITRAL Model Law on Electronic Commerce, 1996 — The Act was framed on the UNCITRAL Model Law on Electronic Commerce adopted by the UN General Assembly in 1997 and came into force on 17 October 2000. It was substantially amended in 2008.

41. (C) Shreya Singhal v. Union of India — In Shreya Singhal v. Union of India (2015) the Court held Section 66A void for vagueness and for violating the freedom of speech under Article 19(1)(a). The Puttaswamy judgment (2017) recognised the right to privacy.

42. (C) Ministry of Electronics and Information Technology — CERT-In was set up in 2004 under MeitY and derives statutory backing from Section 70B of the IT Act. Critical information infrastructure is protected separately by NCIIPC under NTRO.

43. (C) I and II only — I and II are correct; the maximum penalty is Rs 250 crore, for failure to take reasonable security safeguards, so III is wrong. The DPDP Rules were notified on 14 November 2025.

44. (D) 1 July 2015 — Digital India was launched on 1 July 2015 with three vision areas — digital infrastructure as a utility, governance and services on demand, and digital empowerment of citizens — and nine pillars.

45. (C) 12 digits — Aadhaar is a 12-digit random number. UIDAI was set up in 2009 and given statutory status by the Aadhaar Act, 2016; the first Aadhaar was issued at Tembhli, Maharashtra in September 2010.

46. (B) National Payments Corporation of India — NPCI, an umbrella organisation for retail payments set up under the aegis of the RBI and the Indian Banks' Association, launched UPI in 2016. NPCI also runs RuPay, IMPS, NACH and FASTag.

47. (B) A-2, B-1, C-3, D-4 — DigiLocker (2015) stores documents in the cloud, UMANG (2017) aggregates services in one app, ONDC (2021) opens up e-commerce and SWAYAM hosts free online courses.

48. (C) Kerala — Kerala was declared India's first digital state in February 2016 by the President of India, on the strength of near-total mobile connectivity and internet literacy. RPSC asked this item twice in the same 2016 paper.

49. (D) One number, one card, one identity — Jan Aadhaar, launched on 18 December 2019 under the Planning Department, gives each family a single identity number for delivery of state schemes and subsumed the earlier Bhamashah Yojana.

50. (A) Electronic delivery of government and private services to citizens — e-Mitra, run by the Department of Information Technology and Communication, delivers more than 600 services through over 80,000 kiosks across the state, including bill payment, certificates and applications.

51. (A) A-2, B-1, C-3, D-4 — Raj Kaj handles employee service matters and e-office, RAJ NIVESH is the single-window investment portal, iStart supports startups and Rajasthan SSO gives one login for all state services.

52. (B) The adult woman of the household — Bhamashah, relaunched in 2014, made the adult woman of the family the head for direct benefit transfer, linking bank accounts with Aadhaar. It was later subsumed into the Jan Aadhaar Yojana.

53. (D) Ctrl + V — Save — Ctrl + V is the paste command; the shortcut for save is Ctrl + S.

54. (C) .mp3 — Image file — .mp3 is a compressed audio format. Common image extensions are .jpg, .png, .gif and .bmp.

55. (A) II, III, IV, I — ARPANET 1969, the World Wide Web 1989, commercial internet in India on 15 August 1995 through VSNL, and Digital India on 1 July 2015.

ST10 — EXPLANATIONS

1. (B) 1969 — ISRO was formed on 15 August 1969, superseding INCOSPAR (1962). The Department of Space and the Space Commission were created in 1972.

2. (C) Vikram Sarabhai — Vikram Sarabhai founded INCOSPAR in 1962 and led the space effort until his death in 1971. Satish Dhawan headed ISRO from 1972 to 1984.

3. (A) Kapustin Yar, USSR — The 360 kg satellite was launched on 19 April 1975 by a Soviet Kosmos-3M rocket from Kapustin Yar. It was named after the ancient Indian astronomer.

4. (D) Rohini RS-1 — Rohini RS-1 was orbited by SLV-3 on 18 July 1980; the SLV-3 project director was A.P.J. Abdul Kalam.

5. (D) Nike-Apache — An American-supplied Nike-Apache was launched from Thumba on 21 November 1963. Thumba was chosen because it lies close to the geomagnetic equator.

6. (A) A-2, B-1, C-3, D-4 — VSSC (launch vehicles) is at Thiruvananthapuram, SAC (payloads and applications) at Ahmedabad, NRSC (remote sensing data) at Hyderabad and URSC (satellite building) at Bengaluru.

7. (D) A-2, B-1, C-3, D-4 — LPSC designs liquid stages, NRSC handles earth-observation data, SAC builds payloads, and the ISRO Propulsion Complex at Mahendragiri (Tamil Nadu) tests liquid and cryogenic engines.

8. (D) Hassan — The Master Control Facility is at Hassan in Karnataka, with a second facility at Bhopal. Byalalu hosts the Indian Deep Space Network antenna.

9. (A) Andhra Pradesh — Sriharikota is a barrier island on Pulicat Lake in Andhra Pradesh. The centre was renamed after Satish Dhawan in 2002.

10. (D) I, II and III — All three are correct. PSLV-D1 (1993) failed; 'CA' stands for core alone. PSLV is often called ISRO's workhorse launch vehicle.

11. (B) 104 — PSLV-C37 placed 104 satellites in orbit on 15 February 2017, with Cartosat-2 series as the primary payload. The record was surpassed by SpaceX's Transporter-1 (143 satellites) in January 2021.

12. (C) PSLV-C55 — PSLV-C55 flew a dedicated commercial mission arranged by NewSpace India Limited on 22 April 2023, carrying TeLEOS-2 and Lumelite-4.

13. (B) A-1, B-2, C-3, D-4 — Chandrayaan-1 flew on PSLV-C11 (2008), Mangalyaan on PSLV-C25 (2013), Chandrayaan-3 on LVM3-M4 (2023) and Aditya-L1 on PSLV-C57 (2023).

14. (A) Water molecules on the lunar surface — NASA's Moon Mineralogy Mapper carried on Chandrayaan-1 confirmed hydroxyl and water molecules in 2009. Its Moon Impact Probe struck the surface near the south pole on 14 November 2008.

15. (D) 23 August 2023 — India became the fourth country to soft-land on the Moon and the first to land near the lunar south pole. 23 August is now observed as National Space Day.

16. (C) Shiv Shakti Point — The Chandrayaan-3 landing site was named Shiv Shakti Point; the Chandrayaan-2 crash site was named Tiranga Point and the Chandrayaan-1 impact site is Jawahar Sthal.

17. (C) I and III only — The Vikram lander of Chandrayaan-2 crashed on 7 September 2019, but the orbiter remains operational and later supported Chandrayaan-3.

18. (A) 24 September 2014 — Launched on 5 November 2013 by PSLV-C25, MOM reached Mars orbit on 24 September 2014, making India the first country to succeed on its maiden attempt and the fourth agency to reach Mars.

19. (B) Sun-Earth L1 — Aditya-L1 was inserted into a halo orbit around the Sun-Earth L1 point, about 1.5 million km from Earth, on 6 January 2024. Its primary payload VELC was built by the Indian Institute of Astrophysics.

20. (C) X-ray polarisation of cosmic sources — XPoSat is the world's second dedicated X-ray polarimetry mission after NASA's IXPE. Its payloads are POLIX and XSPECT and it was launched by PSLV-C58.

21. (B) A-3, B-1, C-2, D-4 — AstroSat (2015) is India's first multi-wavelength observatory, Aditya-L1 studies the Sun, XPoSat measures X-ray polarisation and NISAR is a synthetic aperture radar mission.

22. (D) Kalpana-1 — Satellite-based navigation — Kalpana-1 (2002), originally METSAT, was India's first dedicated meteorological satellite, later renamed after Kalpana Chawla. Navigation is provided by the IRNSS/NavIC series.

23. (A) 7 — NavIC uses seven satellites in geostationary and geosynchronous orbits and covers India plus about 1500 km beyond its boundaries. GAGAN, developed with the Airports Authority of India, augments GPS for aviation.

24. (C) I and II only — NavIC is a regional, not a global, navigation system. NVS-01 and NVS-02 are second-generation NavIC satellites.

25. (B) Liquid hydrogen and liquid oxygen — Cryogenic stages burn liquid hydrogen with liquid oxygen at extremely low temperatures, giving the highest specific impulse. Kerosene-LOX is used in semi-cryogenic engines such as the SCE-200 under development.

26. (A) I and II only — GSLV-D3 (2010) failed; the first successful flight of an indigenous cryogenic stage was GSLV-D5 in January 2014.

27. (B) I and III only — SSLV-D1 (August 2022) failed to achieve a stable orbit; SSLV-D2 (February 2023) and SSLV-D3 (August 2024) succeeded. SSLV technology has since been transferred to HAL.

28. (C) Chitradurga, Karnataka — The LEX series of landing experiments was carried out at the Aeronautical Test Range, Chitradurga, Karnataka. The earlier RLV-TD hypersonic experiment (HEX-01) flew in May 2016.

29. (D) China — SpaDeX was launched by PSLV-C60 on 30 December 2024 with two spacecraft, SDX01 (Chaser) and SDX02 (Target); docking was achieved on 16 January 2025.

30. (B) NASA — NISAR carries a dual-frequency synthetic aperture radar — NASA's L-band and ISRO's S-band — and was launched by GSLV-F16 on 30 July 2025 into a sun-synchronous orbit. (Status as of 2026.)

31. (C) I and II only — As of 2026 no Indian crewed flight has been made; uncrewed missions, the first of which carries the humanoid Vyommitra, precede a crewed flight targeted for 2027.

32. (A) Rakesh Sharma — Rakesh Sharma flew aboard Soyuz T-11 to the Salyut-7 station in 1984. Shukla flew on the Axiom-4 mission in June 2025 and was the first Indian aboard the ISS.

33. (B) 2028 — The first module is targeted for 2028 and the completed station for 2035, with an Indian crewed landing on the Moon envisaged by 2040. (Status as of 2026.)

34. (A) I and II only — Chandrayaan-4 was approved on 18 September 2024 with an outlay of about Rs 2,104 crore as an uncrewed sample-return mission. A crewed Indian Moon landing is a separate goal targeted around 2040.

35. (D) 2028 — The mission, with an outlay of about Rs 1,236 crore, is planned around the favourable Earth-Venus alignment of March 2028 and will study the Venusian surface and atmosphere. (Status as of 2026.)

36. (B) Indian Institute of Space Science and Technology — Bengaluru — IIST, India's first space university, is at Valiamala near Thiruvananthapuram, not Bengaluru.

37. (A) Fateh Sagar Lake — Founded in 1976 by Dr Arvind Bhatnagar, it has been run by the Physical Research Laboratory, Ahmedabad, since 1981. The surrounding water reduces ground-heating turbulence and improves image quality.

38. (C) 50 cm — MAST, a 50 cm aperture solar telescope, became operational in 2015. The observatory functions under the Physical Research Laboratory of the Department of Space.

39. (D) I, II and III — All three are correct. The exact geostationary altitude is 35,786 km; sun-synchronous orbits have an inclination of about 98 degrees.

40. (C) 11.2 km/s — Escape velocity from Earth is 11.2 km/s; 7.9 km/s is the orbital velocity for a low Earth orbit and 2.38 km/s is the Moon's escape velocity.

41. (C) Shedding spent stages reduces the mass that must be accelerated further — Discarding empty tanks and engines lowers the vehicle's mass, so the remaining propellant produces a much greater velocity increment.

42. (D) I, II, III, IV — Chandrayaan-1 (2008), Mangalyaan (2013), AstroSat (2015) and Aditya-L1 (2023).

43. (C) I, II, III, IV — SLV-3 first flew in 1979, ASLV in 1987, PSLV in 1993 and GSLV in 2001.

44. (D) IN-SPACe — IN-SPACe (Indian National Space Promotion and Authorisation Centre) is headquartered at Ahmedabad. NewSpace India Limited is the commercial PSU arm and ISRO is to concentrate on research and development.

45. (C) A-2, B-1, C-3, D-4 — Vikram-S (Mission Prarambh, November 2022) was India's first privately built rocket; Agnikul's Agnibaan used the world's first single-piece 3D-printed semi-cryogenic engine.

46. (D) I, II and III — NETRA stands for Network for Space Object Tracking and Analysis; IS4OM is the ISRO System for Safe and Sustainable Space Operations Management.

47. (C) A-3, B-1, C-2, D-4 — Cartosat-3 provides a panchromatic resolution of about 0.25 m; RISAT carries synthetic aperture radar for cloud-independent imaging.

48. (B) The heaviest satellite launched from Indian soil at the time of its launch — The multi-band communication satellite for the Indian Navy weighed about 4,410 kg and was launched from Sriharikota on 2 November 2025. It was surpassed by BlueBird Block-2, carried by LVM3-M6 on 24 December 2025. (As of September 2026.)

49. (A) SLV-3 was India's first vehicle to use a cryogenic upper stage — SLV-3 was an all-solid four-stage vehicle. India's first flight with an indigenous cryogenic stage was GSLV-D5 in January 2014.

50. (A) I, II and III — Bhaskara-I flew in 1979 and APPLE was launched by an Ariane rocket from Kourou in 1981.

51. (B) A-1, B-3, C-2, D-4 — Kalpana-1, launched in 2002 as METSAT, was renamed in memory of Kalpana Chawla. EDUSAT (2004) was the first satellite exclusively for education.

52. (D) Antrix Corporation was set up in 2019 as the commercial arm of ISRO — Antrix Corporation was incorporated in 1992. NewSpace India Limited (NSIL) is the arm created in 2019.

53. (B) I, II and III — JWST was launched in December 2021 on an Ariane-5, Artemis-I flew uncrewed in November-December 2022 and Chang'e-6 brought back far-side samples in June 2024.

54. (D) CNES — Canada — CNES is the French national space agency; Canada's agency is the Canadian Space Agency (CSA).

55. (A) Both A and R are true and R is the correct explanation of A — Constant local solar time gives comparable illumination and shadow conditions between passes, which is essential for change detection in remote sensing.

ST11 — EXPLANATIONS

1. (B) 1958 — DRDO was formed in 1958 by amalgamating the Technical Development Establishment, the Directorate of Technical Development & Production and the Defence Science Organisation. Its headquarters is DRDO Bhawan, New Delhi.

2. (C) Balasya Mulam Vigyanam — 'Balasya Mulam Vigyanam' (बलस्य मूलं विज्ञानम्) means 'Strength's origin is in Science' and is drawn from Kautilya's Arthashastra.

3. (A) Leucoderma (vitiligo) — Lukoskin was developed from Himalayan herbs by the Defence Institute of Bio-Energy Research (DIBER), Haldwani, and the technology was transferred to Aimil Pharmaceuticals for manufacture.

4. (D) INMAS, Delhi — The Institute of Nuclear Medicine and Allied Sciences developed 2-DG with Dr Reddy's Laboratories; it received emergency-use approval in May 2021.

5. (B) Defence Laboratory, Jodhpur — The Defence Laboratory, Jodhpur (DLJ) specialises in desert warfare support, camouflage materials and nuclear-radiation detection systems.

6. (B) A-2, B-3, C-1, D-4 — GTRE developed the Kaveri engine, NPOL builds sonars, CVRDE developed the Arjun tank and DFRL produces combat rations and preserved foods.

7. (C) I and II only — DRDO's headquarters is DRDO Bhawan in New Delhi. The organisation has since grown to about 52 laboratories.

8. (A) A.P.J. Abdul Kalam — The programme covered five missiles — Prithvi, Agni, Trishul, Akash and Nag — and was formally concluded in 2008.

9. (C) BrahMos — BrahMos is a later India-Russia joint venture set up in 1998. IGMDP comprised Prithvi, Agni, Trishul, Akash and Nag.

10. (B) A-2, B-4, C-1, D-3 — Prithvi is a short-range surface-to-surface ballistic missile, Akash a medium-range surface-to-air missile, Nag a fire-and-forget anti-tank missile and Astra a beyond-visual-range air-to-air missile.

11. (D) Beyond-visual-range air-to-air missile — Astra Mk-1 has a range of over 100 km and is integrated on the Su-30MKI and Tejas; the Mk-2 variant extends the range further.

12. (A) A man-portable air defence system — DRDO's indigenous MANPADS project is VSHORADS (Very Short Range Air Defence System), a shoulder-fired missile effective against low-flying aircraft, helicopters and drones at ranges of about 6 km.

13. (B) 5,000 km — Agni-V is a three-stage, solid-fuelled, canisterised missile with a range of over 5,000 km. Agni-IV has a range of about 4,000 km.

14. (C) Multiple Independently Targetable Re-entry Vehicles — Agni-5 was tested with MIRV technology on 11 March 2024, allowing a single missile to deliver several warheads to different targets.

15. (A) I, II and III — BrahMos Aerospace was set up in 1998; the missile flies at about Mach 2.8-3 and can be launched from land, sea, submarine and air platforms.

16. (D) the Philippines — The contract with the Philippines was signed in January 2022 and the first battery was delivered in April 2024; it was India's largest defence export order at the time.

17. (B) Scramjet — A scramjet is a supersonic-combustion ramjet in which airflow through the combustor remains supersonic. The HSTDV achieved about Mach 6.

18. (C) Destroy a satellite in low Earth orbit — The DRDO anti-satellite test destroyed a live satellite at about 283 km altitude, making India the fourth country after the USA, Russia and China to demonstrate ASAT capability.

19. (A) I, II and III — Pinaka was developed by the Armament Research and Development Establishment, Pune. The Mk-I has a range of about 38 km and the guided variant exceeds 70 km.

20. (A) Short-range quasi-ballistic surface-to-surface missile — Pralay is a solid-fuelled, conventionally armed quasi-ballistic missile with a range of about 150-500 km, cleared for induction with the armed forces.

21. (B) A-2, B-3, C-1, D-4 — K-15 Sagarika and the longer-ranged K-4 (about 3,500 km) are submarine-launched missiles carried by Arihant-class boats and form the sea leg of India's nuclear triad.

22. (C) I and II only — The BMD programme is indigenous to DRDO. Phase-II envisages the AD-1 and AD-2 interceptors for longer-range threats.

23. (D) Hindustan Aeronautics Limited — Tejas received Initial Operational Clearance in 2013 and Final Operational Clearance in 2019. The Mk-1A variant is powered by the GE F404 engine.

24. (A) I and II only — The Cabinet Committee on Security approved AMCA development in March 2024; as of 2026 the prototype has not yet flown.

25. (B) Prachand — Built by HAL, Prachand can operate at altitudes above 5,000 m, making it suitable for Siachen-level operations. Its first squadron was raised at Jodhpur.

26. (C) ADE, Bengaluru — The Aeronautical Development Establishment developed TAPAS (earlier called Rustom-II) as well as the Nishant and Lakshya systems.

27. (D) Prachand — Basic trainer aircraft — Prachand is HAL's Light Combat Helicopter. HAL's indigenous basic trainer aircraft is the HTT-40.

28. (A) India's first indigenously designed and built aircraft carrier — Built at Cochin Shipyard, it displaces about 45,000 tonnes. INS Vikramaditya, acquired from Russia, was commissioned in 2013.

29. (B) I and II only — Kalvari-class (Scorpene) boats are conventional diesel-electric submarines built at Mazagon Dock under Project 75 with France's Naval Group; INS Vagsheer, the sixth, was commissioned in January 2025.

30. (C) Diving support vessel — Built by Hindustan Shipyard Limited, Visakhapatnam, it supports deep-sea diving and submarine rescue operations. (Status as of 2026.)

31. (D) Nilgiri class (Project 17A) — Project 17A frigates are built at Mazagon Dock and Garden Reach Shipbuilders. INS Nilgiri, the lead ship of the class, was commissioned in January 2025. (Status as of 2026.)

32. (C) Dhanush — Anti-tank guided missile — Dhanush is a 155 mm/45 calibre indigenous towed howitzer derived from the Bofors gun, often called the 'Desi Bofors'. Nag is India's anti-tank guided missile.

33. (C) A-2, B-1, C-4, D-3 — CVRDE Avadi developed the Arjun tank, ARDE Pune the Pinaka, NSTL Visakhapatnam the Varunastra torpedo and NPOL Kochi India's sonar systems.

34. (B) 155 mm — ATAGS is a 155 mm/52 calibre towed howitzer developed by ARDE, Pune, with Bharat Forge and Tata Advanced Systems as production partners; it has demonstrated ranges of about 48 km.

35. (B) I, II, III, IV — DRDO was set up in 1958, IGMDP launched in 1983, Pokhran-II conducted in 1998 and INS Vikrant commissioned in 2022.

36. (D) Korwa, Amethi, Uttar Pradesh — They are produced by Indo-Russian Rifles Private Limited, an India-Russia joint venture. The AK-203 replaces the indigenous INSAS rifle.

37. (A) Jaisalmer — The test was conducted on 18 May 1974. The Pokhran range in Jaisalmer district was used again on 11 and 13 May 1998 for Operation Shakti.

38. (C) The Pokhran-II nuclear tests of 1998 — On 11 May 1998 India conducted nuclear tests at Pokhran under Operation Shakti; the maiden flight of the indigenous Hansa-3 aircraft and a Trishul missile test on the same day are also commemorated.

39. (A) I and II only — India has pledged non-use of nuclear weapons against non-nuclear weapon states, maintains credible minimum deterrence and observes a unilateral moratorium on further testing.

40. (B) I and II only — India joined MTCR (2016), the Wassenaar Arrangement (2017) and the Australia Group (2018), but is not a member of the NSG and has signed neither the NPT nor the CTBT.

41. (C) 7 — The seven are Munitions India, Armoured Vehicles Nigam (AVANI), Advanced Weapons and Equipment India, Troop Comforts, Yantra India, India Optel and Gliders India; the change took effect on 1 October 2021.

42. (D) Uttar Pradesh and Tamil Nadu — Announced in the 2018-19 Union Budget, the Uttar Pradesh corridor has nodes at Aligarh, Agra, Jhansi, Kanpur, Chitrakoot and Lucknow, and the Tamil Nadu corridor at Chennai, Coimbatore, Hosur, Salem and Tiruchirappalli.

43. (D) Agnipath — Corporatisation of the ordnance factories — Agnipath is the short-service military recruitment scheme launched in 2022; the ordnance factories were corporatised into seven DPSUs in 2021.

44. (D) I, II and III — The scheme applies to all three services and was introduced to lower the average age profile of the armed forces.

45. (A) Department of Military Affairs — General Bipin Rawat became the first CDS on 1 January 2020. The CDS is also the Permanent Chairman of the Chiefs of Staff Committee.

46. (D) The Integrated Defence Staff was created in 2019 — The Integrated Defence Staff was set up in 2001 on the recommendations of the Kargil Review Committee; the post of CDS and the Department of Military Affairs were created in 2019.

47. (A) A-3, B-1, C-2, D-4 — Yudh Abhyas is the India-US army exercise, Varuna the India-France naval exercise, Indra the India-Russia exercise and Surya Kiran the India-Nepal army exercise.

48. (C) A-2, B-1, C-3, D-4 — Mitra Shakti is with Sri Lanka, Sampriti with Bangladesh, Dustlik with Uzbekistan and Ekuverin with the Maldives.

49. (D) Al Nagah — India and Egypt — Al Nagah is an India-Oman army exercise. The India-Egypt army exercise is named 'Cyclone'.

50. (A) National Security Guard — 1974 — The National Security Guard was raised in 1984. The Assam Rifles, raised in 1835, is the oldest paramilitary force in India.

ST12 — EXPLANATIONS

1. (A) Raymond Lindeman — Lindeman (1942) showed that only about 10 per cent of the energy at one trophic level is transferred to the next. Charles Elton gave the concept of the ecological pyramid.

2. (B) Pyramid of energy — Because energy is lost as heat at every transfer, the pyramid of energy can never be inverted. Pyramids of number and biomass may be inverted, as in a single-tree ecosystem or in the sea.

3. (D) I, II and III — All three are correct. In terrestrial ecosystems most net primary production is not grazed but enters the detritus pathway, whereas in many aquatic systems the grazing food chain dominates.

4. (D) Invasive species — Great Indian Bustard — The Great Indian Bustard (Godawan) is a critically endangered native bird and Rajasthan's state bird, not an invasive species. Lichens are classic indicators of sulphur dioxide pollution.

5. (A) Ecotone — The greater species density and richness at an ecotone is called the edge effect, and species largely confined to it are edge species.

6. (B) Phosphorus cycle — The reservoir of phosphorus is rock, and it has no appreciable gaseous component. Nitrogen, carbon and oxygen cycles are gaseous cycles with the atmosphere as reservoir.

7. (C) A-1, B-2, C-3, D-4 — Rhizobium nodulates legume roots, Frankia nodulates Casuarina, Azotobacter is free-living and aerobic, and Anabaena azollae lives in the leaf cavities of the water fern Azolla.

8. (D) Lichens — Lichens secrete acids that weather rock and initiate soil formation; mosses follow. Both xerarch and hydrarch successions ultimately converge on a mesic climax community.

9. (B) Alpha diversity — Alpha diversity is diversity within a habitat; beta diversity is the change in composition between habitats along a gradient; gamma diversity is the total diversity of a landscape or region.

10. (C) I and II only — Statements I and II are the standard Whittaker definitions. Statement III is wrong: gamma diversity, being the regional total, is equal to or greater than the alpha diversity of any one habitat within it.

11. (D) Cerrado — The Cerrado is a South American hotspot. Four hotspots reach India: Himalaya, Indo-Burma, Western Ghats-Sri Lanka and Sundaland (through the Nicobar Islands).

12. (A) Norman Myers — Norman Myers proposed the hotspot concept in 1988; Conservation International later refined it. Thirty-six hotspots are recognised worldwide.

13. (A) It must contain at least 500 endemic vertebrate species — The two quantitative criteria relate to endemic vascular plants and to habitat loss; there is no vertebrate-endemism threshold.

14. (B) Critically Endangered, Endangered and Vulnerable — The IUCN Red List has nine categories in all; the threatened group comprises Critically Endangered, Endangered and Vulnerable. IUCN was founded in 1948 and is headquartered at Gland, Switzerland.

15. (D) I, II and III — All three are correct. In-situ conservation protects species in their natural habitat; ex-situ conservation maintains them outside it, as in zoos, botanical gardens, gene banks and cryopreservation facilities.

16. (A) Core, buffer and transition — Only the core zone is legally protected and free of human interference; limited activity is allowed in the buffer, and settled human use in the transition zone. India's first biosphere reserve was the Nilgiri (1986).

17. (B) A-1, B-2, C-3, D-4 — Rajasthan has no biosphere reserve. India's first was the Nilgiri Biosphere Reserve, notified in 1986.

18. (D) I, II and III — All three are correct. NTCA was constituted under the Wild Life (Protection) Act, 1972 through the 2006 amendment; the 2022 all-India estimate placed the tiger population at about 3,682.

19. (C) A-1, B-2, C-3, D-4 — Ramgarh Vishdhari (Bundi) was notified in 2022 as India's 52nd tiger reserve; Dholpur-Karauli, approved in 2023, is Rajasthan's fifth and India's 54th tiger reserve.

20. (C) Keoladeo and Sambhar — Keoladeo Ghana National Park (Bharatpur) was designated in 1981 and Sambhar Lake in 1990. Khichan and Menar were added on 5 June 2025, taking Rajasthan's Ramsar tally to five. (Siliserh Lake, Alwar, was added in 2025.)

21. (C) The Wild Life (Protection) Act, 1972 — NTCA was created by the 2006 amendment to the Wild Life (Protection) Act, 1972 and is chaired by the Union Minister for Environment, Forest and Climate Change.

22. (B) Four — The 2022 amendment, in force from 1 April 2023, reduced the schedules from six to four and rationalised the species lists.

23. (D) I, II and III — All three are correct. Schedule I species get the highest protection; Schedule IV was introduced to implement India's CITES obligations, with registration of scheduled live specimens through the PARIVESH portal.

24. (A) Chennai — The Act creates a three-tier structure: the NBA at Chennai, State Biodiversity Boards, and Biodiversity Management Committees at the local body level.

25. (B) I and II only — Statement III is wrong: Biodiversity Management Committees continue and remain responsible for People's Biodiversity Registers. The parent Act of 2002 gives effect to the Convention on Biological Diversity, 1992.

26. (C) Biodiversity Management Committee — Biodiversity Management Committees are constituted by every local body and prepare People's Biodiversity Registers documenting local biological resources and associated traditional knowledge.

27. (D) Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 — The 2023 amendment renamed the Forest (Conservation) Act, 1980 as the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 and narrowed the categories of land to which it applies.

28. (A) I and II only — Statement III is wrong: the Compensatory Afforestation Fund has its own statute, the CAF Act, 2016. The 2023 amendment created the border-area and security-infrastructure exemptions and also exempted zoos, safaris and eco-tourism facilities.

29. (B) Ministry of Tribal Affairs — The Gram Sabha is the authority that initiates determination of forest rights; the cut-off date for claims is 13 December 2005.

30. (C) United Nations Conference on the Human Environment, Stockholm, 1972 — The Act is India's umbrella environmental legislation and was passed in the aftermath of the 1984 Bhopal gas disaster. Article 253 empowers Parliament to legislate to implement international agreements.

31. (D) The Water (Prevention and Control of Pollution) Act, 1974 — CPCB was set up in 1974 under the Water Act and was later entrusted with functions under the Air Act, 1981 as well. The Water Act was amended in 2024 to decriminalise several minor offences.

32. (A) New Delhi — The NGT was set up on 18 October 2010 with zonal benches at Bhopal, Pune, Kolkata and Chennai. India became the third country, after Australia and New Zealand, to set up a specialised environmental court.

33. (D) A-1, B-2, C-3, D-4 — These four Acts, along with the Wild Life (Protection) Act, 1972 and the Forest (Conservation) Act, 1980, form the core of India's environmental law.

34. (A) Sulphur dioxide and oxides of nitrogen — Sulphur dioxide and oxides of nitrogen are oxidised in the atmosphere to sulphuric and nitric acid. Rain is regarded as acid rain when its pH falls below 5.6.

35. (B) Six — The six categories are Good, Satisfactory, Moderate, Poor, Very Poor and Severe. The index is computed for eight pollutants: PM10, PM2.5, NO2, SO2, CO, ozone, ammonia and lead.

36. (C) A-1, B-2, C-3, D-4 — The intervening bands are Moderate (101-200) and Very Poor (301-400). The AQI is computed by CPCB and reported as the worst sub-index among the eight pollutants.

37. (D) I and II only — Statement III is wrong: NCAP is implemented by the Ministry of Environment, Forest and Climate Change through CPCB in identified non-attainment cities.

38. (A) Methane — The eight AQI pollutants are PM10, PM2.5, NO2, SO2, CO, ozone, ammonia and lead. Methane, though a potent greenhouse gas, is not an AQI pollutant.

39. (B) High content of biodegradable organic matter — BOD is measured over five days at 20 degrees C and reflects the oxygen consumed by microbes decomposing organic matter. Chemical oxygen demand is always higher than BOD for the same sample.

40. (C) I and III only — Statement II is wrong: decomposition of the algal bloom consumes oxygen and creates hypoxic conditions, which is precisely why fish die.

41. (D) A-1, B-2, C-3, D-4 — Blue baby syndrome (methaemoglobinaemia) results from high nitrate in drinking water. Fluorosis is a serious problem in several districts of western Rajasthan.

42. (A) 1 July 2022 — The ban was imposed under the Plastic Waste Management (Amendment) Rules, 2021. The minimum thickness of plastic carry bags was raised to 120 microns from 31 December 2022.

43. (B) National Action Plan on Climate Change — 2005 — The National Action Plan on Climate Change was released in 2008, not 2005. Namami Gange is implemented by the National Mission for Clean Ganga, and the National Ganga Council is chaired by the Prime Minister.

44. (C) Silence zone — 60 dB(A) Leq — The day-time limit for a silence zone is 50 dB(A) Leq and the night-time limit 40 dB(A) Leq. Silence zones extend 100 metres around hospitals, educational institutions and courts.

45. (C) Carbon dioxide — Carbon dioxide accounts for roughly 60 per cent of the enhanced greenhouse effect, followed by methane (about 20 per cent), CFCs (about 14 per cent) and nitrous oxide (about 6 per cent).

46. (D) II, I, IV, III — The conventional order is carbon dioxide, methane, chlorofluorocarbons and nitrous oxide. By global warming potential per molecule, however, the order is reversed, with sulphur hexafluoride highest of all.

47. (A) The World Meteorological Organization and UNEP — The IPCC assesses published science and does not conduct its own research. Its Sixth Assessment Report cycle concluded with the Synthesis Report in March 2023.

48. (B) Rio de Janeiro — The Rio Earth Summit of 1992 also produced Agenda 21 and the Convention on Biological Diversity; the UNFCCC entered into force in 1994.

49. (C) II, I, IV, III — Stockholm 1972, Montreal Protocol 1987, Kyoto Protocol 1997 and the Paris Agreement 2015. UNEP was created in the wake of the Stockholm Conference and is headquartered at Nairobi.

50. (D) Well below 2 degrees C, and to pursue efforts to limit it to 1.5 degrees C, above pre-industrial levels — The Paris Agreement was adopted at COP21 and entered into force on 4 November 2016. It works through nationally determined contributions and a Global Stocktake every five years.

51. (A) Belem, Brazil — COP30 at Belem adopted the 'global mutirao' decision. COP29 was held at Baku in 2024 and COP31 is scheduled for Antalya, Turkiye, in November 2026 (position as of September 2026).

52. (B) I and II only — Statement III is wrong: no fossil-fuel roadmap was included in the COP30 decision text, though the presidency announced that voluntary roadmaps would be brought to COP31. Position as of September 2026.

53. (C) COP26, Glasgow — The Panchamrit announced at Glasgow in 2021 also included 500 GW of non-fossil capacity by 2030, meeting half of energy requirements from renewables, and cutting projected emissions by one billion tonnes by 2030.

54. (D) I, II and III — All three are correct. The Union Cabinet approved this NDC on 25 March 2026; it also targets an additional carbon sink of 3.5 to 4.0 billion tonnes of CO2 equivalent by 2035. Position as of September 2026.

55. (C) A-1, B-2, C-3, D-4 — The National Action Plan on Climate Change, released in 2008, has eight missions; the other four are Sustainable Habitat, Sustaining the Himalayan Ecosystem, Sustainable Agriculture and Strategic Knowledge for Climate Change.

56. (B) National Clean Air Mission — Clean air is addressed by the National Clean Air Programme of 2019, which is not a NAPCC mission. States prepare State Action Plans on Climate Change aligned with the NAPCC.

57. (C) Gurugram, Haryana — The ISA was launched at COP21 in Paris and its secretariat is at Gwal Pahari, Gurugram. It is the first treaty-based international organisation headquartered in India.

58. (C) 2006 — The EIA Notification of 14 September 2006, issued under the Environment (Protection) Act, 1986, replaced the 1994 notification. A draft notification of 2020 was published but has not replaced it.

59. (B) I and II only — Statement III is wrong: the public hearing is conducted by the concerned State Pollution Control Board or Pollution Control Committee.

60. (D) State Pollution Control Board — The public consultation stage comprises a public hearing at the site and written responses from other concerned persons. Certain categories, such as modernisation of irrigation projects, are exempted from public consultation.

61. (A) State Environment Impact Assessment Authority — SEIAA acts on the recommendation of the State Expert Appraisal Committee. Where no SEIAA has been constituted, the project is treated as Category A and appraised at the central level.

62. (B) Cost-benefit certification — The four stages are screening, scoping, public consultation and appraisal. Screening decides whether a Category B project needs a full EIA; scoping fixes the terms of reference.

63. (C) The Vienna Convention for the Protection of the Ozone Layer — The Vienna Convention (1985) is the framework treaty; the Montreal Protocol (1987) is the binding instrument adopted under it and is the only UN treaty ratified by every member state. The Kigali Amendment (2016) later added the phase-down of hydrofluorocarbons, which do not deplete ozone but have a very high global warming potential; India ratified it in 2021.

64. (B) Phasing out of ozone-depleting substances — It operationalises the Vienna Convention of 1985. World Ozone Day is observed on 16 September, the date the Protocol was signed. Hazardous waste movement is governed by the Basel Convention.

65. (C) Our Common Future (Brundtland Report), 1987 — The Brundtland Commission was formally the World Commission on Environment and Development. 'The Future We Want' was the outcome document of the Rio+20 Conference of 2012.

66. (A) I, II and III — All three are correct. The SDGs succeeded the eight Millennium Development Goals; SDG 13 relates to climate action and SDG 15 to life on land.

67. (D) I, II and III — All three are correct. The Act also created the NDMA at the national level, SDMAs in the states and DDMAs in the districts.

68. (A) 2016 — The plan was released in 2016 and revised in 2019. It covers all phases of disaster management: prevention, mitigation, response and recovery.

69. (D) 2015-2030 — Adopted at the third World Conference on Disaster Risk Reduction at Sendai, Japan, it has four priorities for action and seven global targets, and succeeded the Hyogo Framework of 2005-2015.

70. (A) A-1, B-2, C-3, D-4 — In a district with an elected zila parishad, its chairperson is co-chairperson of the DDMA. The NDMA has up to nine members besides the Prime Minister.

ST13 — EXPLANATIONS

1. (A) Suratgarh, Sri Ganganagar — The scheme was launched on 19 February 2015 from Suratgarh. A soil health card reports twelve parameters, including the primary nutrients N, P and K and the micronutrients.

2. (B) Nitrogen — Nitrogen is mobile in the plant and is withdrawn from older leaves first, so deficiency shows there. Potassium deficiency shows as scorching of leaf margins.

3. (C) A-1, B-2, C-3, D-4 — Zinc deficiency is widespread in the calcareous and sandy soils of Rajasthan. Neem-coated urea was made mandatory in 2015 to improve nitrogen use efficiency.

4. (D) Calcium — Calcium, along with magnesium and sulphur, is a secondary macronutrient. The plant micronutrients are iron, manganese, zinc, copper, boron, molybdenum, chlorine and nickel.

5. (D) Azure blue — Breeder seed carries a golden-yellow tag, foundation seed a white tag and certified seed an azure-blue tag; truthfully labelled seed carries an opal-green label.

6. (A) II, III, I — Breeder seed is produced under the direct supervision of the plant breeder, is multiplied into foundation seed, and that in turn into certified seed for distribution to farmers.

7. (B) I and II only — Uttar Pradesh leads in sugarcane. Rajasthan also leads in guar, coriander, cumin, fenugreek, isabgol and moth bean.

8. (C) I and II only — Statement III is wrong: guar is a kharif crop sown with the onset of the monsoon. Sri Ganganagar, Hanumangarh, Bikaner, Churu and Nagaur are the main guar districts.

9. (D) Bajra — HHB-67 Improved matures in about 62-65 days, which allows it to escape late-season drought in the arid west of the state.

10. (A) Bharatpur, Rajasthan — It is located at Sewar, Bharatpur, and began in 1993 as the National Research Centre on Rapeseed-Mustard. Rajasthan is India's leading rapeseed-mustard producing state.

11. (B) Barley — Barley is a rabi crop, as are wheat, gram and mustard. Bajra, moth bean, guar, jowar, maize and cotton are kharif crops in the state.

12. (C) Fixing atmospheric nitrogen through root nodule bacteria — Rhizobium in the root nodules of legumes fixes atmospheric nitrogen. Guar, moth bean, moong and gram are the important legumes of Rajasthan's rotations.

13. (D) I, II and III — All three are correct. Most of western Rajasthan falls in the dryland and desert category, where khadin cultivation, mulching and drought-escaping short-duration varieties are standard practice.

14. (A) I, II and III — All three are correct. The Locust Warning Organisation works under the Directorate of Plant Protection, Quarantine and Storage at Faridabad; a major upsurge affected Rajasthan in 2019-20.

15. (B) Jodhpur — Circle offices are spread across the desert districts of Rajasthan and Gujarat. The parent Directorate of Plant Protection, Quarantine and Storage is at Faridabad.

16. (C) Pradhan Mantri Krishi Sinchayee Yojana — PMKSY was launched in 2015 with the motto 'Har Khet Ko Pani'. Sprinkler irrigation is especially widespread in Rajasthan because it suits sandy, highly permeable soils.

17. (B) I and II only — Statement III is wrong: IPM seeks to keep pest populations below the economic threshold level, not to eradicate them. Trichogramma is an egg parasitoid used against lepidopteran pests.

18. (D) Khadin — Khadin is a traditional runoff-harvesting cropping system of the Jaisalmer region, in which a bund across a slope impounds runoff for growing a rabi crop on stored moisture.

19. (A) Women's self-help groups — The scheme equips selected women's self-help groups with drones for spraying fertilisers and pesticides, creating a rental service and a livelihood.

20. (D) Reduces conveyance and deep percolation losses — In highly permeable sandy soils, flood irrigation loses much water to deep percolation; sprinklers apply water at a controlled rate close to the infiltration capacity.

21. (B) Tabiji, Ajmer — The centre was established at Tabiji, Ajmer in 2000. Rajasthan and Gujarat together account for the bulk of India's seed-spice production.

22. (C) A-1, B-2, C-3, D-4 — The National Research Centre on Camel is also at Bikaner. CAZRI, established at Jodhpur in 1959, is the country's premier arid-zone research institute.

23. (A) Coriander, cumin, fennel and fenugreek — Seed spices are those whose dried seeds or fruits are used; ajwain, dill, celery and nigella also belong to this group and are worked on at the National Research Centre on Seed Spices, Ajmer.

24. (D) Henna (mehndi) — Sojat lies in Pali district and is the country's main henna processing centre. Henna is obtained from Lawsonia inermis.

25. (A) Plantago ovata — The husk of the seed of Plantago ovata is psyllium, used as a laxative and largely exported. Jalore, Barmer, Sirohi and Nagaur are the main producing districts.

26. (C) Olive — Jhalawar — Jhalawar is famous for oranges. Olive cultivation in Rajasthan is concentrated at Lunkaransar in Bikaner and Bassi in Jaipur, where the state set up its olive plantation and refinery project.

27. (B) I and II only — Statement III is wrong: like all ICAR institutes it functions under the Department of Agricultural Research and Education in the Ministry of Agriculture and Farmers Welfare.

28. (D) Jaisalmer and Bikaner — The hot, dry climate of the extreme west suits the date palm. The Central Institute for Arid Horticulture at Bikaner has been the lead institute for this programme.

29. (A) A-1, B-2, C-3, D-4 — Ashwagandha grown around Nagaur is known locally as asgandh. Guggul is a threatened resin-yielding shrub of the Aravalli and desert tracts.

30. (A) Khejri (Prosopis cineraria) — Khejri is called the kalpavriksha of the desert. The Bishnoi sacrifice at Khejarli in 1730, led by Amrita Devi, is commemorated by the Amrita Devi Bishnoi Wildlife Protection Award.

31. (B) Marwar teak or the teak of the desert — Rohida yields a hard, durable timber. The kalpavriksha of the desert is the khejri, which is the state tree.

32. (C) I and II only — Statement III is wrong: the 1988 policy shifted the emphasis from revenue to ecological stability and the needs of forest-dependent communities. The policy target is 60 per cent cover in hills and 20 per cent in the plains.

33. (D) TRIFED under the Ministry of Tribal Affairs — The scheme, launched in 2018, organises tribal gatherers into self-help groups for primary processing of minor forest produce, complementing the minimum support price scheme for MFP.

34. (A) A-1, B-2, C-3, D-4 — Rathi is the best milch breed of the state, Tharparkar is dual-purpose and Kankrej and Nagauri are prized draught breeds. Malvi is found in Jhalawar and Hariana in the Bharatpur-Alwar tract.

35. (B) A-1, B-2, C-3, D-4 — Chokla, of the Shekhawati tract, yields the finest wool among Indian sheep breeds; Magra wool is lustrous and prized for carpets. Rajasthan is India's largest wool producer.

36. (C) Alwar — Jakhrana is a village in Alwar district. The Sirohi breed is valued mainly for meat, and Marwari and Barbari are the other important goat breeds of the state.

37. (D) I, II and III — All three are correct. The Rajasthan Camel (Prohibition of Slaughter and Regulation of Temporary Migration or Export) Act was passed in 2015; Bikaneri, Jaisalmeri, Mewari and Kachchhi are the recognised breeds.

38. (A) Bikaner — Established at Bikaner in 1984, it works on camel breeding, camel milk and camel-derived products. A camel milk dairy also operates at Bikaner.

39. (B) Brucellosis — Brucellosis is a bacterial and zoonotic disease caused by Brucella species. Foot-and-mouth disease, PPR and lumpy skin disease are all viral.

40. (C) I and II only — Statement III is wrong: lumpy skin disease does not infect humans. The 2022 outbreak caused heavy cattle mortality in Rajasthan and neighbouring states, and the indigenous Lumpi-ProVacInd vaccine was developed in response.

41. (D) Foot-and-mouth disease and brucellosis — The programme provides for 100 per cent central funding of vaccination of cattle, buffalo, sheep, goats and pigs against foot-and-mouth disease and of female calves against brucellosis.

42. (B) Development and conservation of indigenous bovine breeds — The mission supports Gokul Grams, breed improvement through artificial insemination and embryo transfer, and conservation of registered indigenous breeds such as Rathi, Tharparkar, Gir and Kankrej.

43. (C) II, IV, I, III — CAZRI Jodhpur 1959, CSWRI Avikanagar 1962, NRC on Camel Bikaner 1984 and NRC on Seed Spices Ajmer 2000.

44. (D) Physical Research Laboratory, Ahmedabad — The observatory was founded in 1975 by Dr. Arvind Bhatnagar; the island location gives steady daytime seeing. It hosts the Multi-Application Solar Telescope (MAST).

45. (A) Tonk — Avikanagar lies in Malpura tehsil of Tonk district; the institute was established in 1962 and has developed sheep strains such as Avikalin and Bharat Merino.

46. (B) Jodhpur — AFRI at Jodhpur was established in 1988 and works on afforestation of arid and semi-arid tracts of Rajasthan, Gujarat and Dadra and Nagar Haveli.

47. (C) Central Electronics Engineering Research Institute, Pilani — CSIR-CEERI at Pilani was set up in 1953 and works on electronics, semiconductor devices and microwave tubes. CSIR itself was founded in 1942.

48. (D) Jaipur — NIAM at Jaipur, set up in 1988, is an autonomous institute of the Ministry of Agriculture and Farmers Welfare for training and research in agricultural marketing.

49. (A) Central Institute for Research on Goats — The Central Institute for Research on Goats is at Makhdoom in Mathura district, Uttar Pradesh. The other three are at Jodhpur, Ajmer and Jodhpur respectively.

50. (B) A-1, B-2, C-3, D-4 — Rajasthan also has an Agriculture University at Jodhpur and the Rajasthan University of Veterinary and Animal Sciences at Bikaner. The Rajasthan Agricultural Research Institute is at Durgapura, Jaipur.

51. (D) I, II and III — All three are correct. The earlier documents were the Scientific Policy Resolution 1958, the Technology Policy Statement 1983, the Science and Technology Policy 2003 and the Science, Technology and Innovation Policy 2013.

52. (A) The discovery of the Raman Effect — C.V. Raman announced the discovery of the effect on 28 February 1928; the day has been observed since 1986.

53. (A) A-1, B-2, C-3, D-4 — National Technology Day marks the Pokhran-II tests of 1998, National Space Day the Chandrayaan-3 landing of 23 August 2023, and National Mathematics Day the birth anniversary of Srinivasa Ramanujan.

54. (B) I and II only — Statement III is wrong: the ANRF subsumed the Science and Engineering Research Board, while CSIR continues to function as before.

55. (C) Council of Scientific and Industrial Research — Instituted in 1958 and named after the first Director-General of CSIR, it is given in seven disciplines to scientists below the age of 45.

56. (D) Innovation in Science Pursuit for Inspired Research — Launched in 2008, INSPIRE attracts young talent to science through awards, scholarships and fellowships. Vigyan Jyoti, launched in 2019-20, encourages girls to take up STEM careers.

57. (A) Scattering of light — He was the first Asian to win a Nobel Prize in the sciences and was awarded the Bharat Ratna in 1954.

58. (B) A-1, B-2, C-3, D-4 — Homi Bhabha founded the Tata Institute of Fundamental Research in 1945 and framed India's three-stage nuclear programme.

59. (C) I and II only — Statement III is wrong: Khorana's Nobel Prize of 1968 was in Physiology or Medicine, for interpreting the genetic code and its role in protein synthesis.

60. (D) Vikram Sarabhai — Vikram Sarabhai, the father of the Indian space programme, also chaired INCOSPAR, set up in 1962, which evolved into ISRO in 1969. The Udaipur Solar Observatory is today a unit of PRL.

61. (A) Ribosome — He shared the prize with Thomas Steitz and Ada Yonath. He was later President of the Royal Society.

62. (B) Verghese Kurien — Verghese Kurien, the 'Milkman of India', built the Amul cooperative model at Anand and headed the National Dairy Development Board. Operation Flood began in 1970.

63. (C) Blue Revolution — Oilseeds — The Blue Revolution relates to fisheries and aquaculture; oilseeds are associated with the Yellow Revolution. M.S. Swaminathan is regarded as the father of the Green Revolution in India.

64. (D) Keoladeo Ghana, Bharatpur — His long study of the Bharatpur wetland was central to its protection; he wrote 'The Book of Indian Birds'. The Salim Ali Centre for Ornithology and Natural History is at Coimbatore.

65. (A) Ministry of Defence — A physicist trained under Rutherford, Kothari later headed the University Grants Commission; the Kothari Commission's report shaped the National Policy on Education of 1968.